

POs

Program Outcomes (POs):

- **PO1 – Engineering knowledge**
Apply knowledge of mathematics, natural science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
- **PO2 – Problem analysis**
Identify, formulate, research literature and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.
- **PO3 – Design/development of solutions**
Design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal and environmental considerations.
- **PO4 – Investigation**
Conduct investigations of complex problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.
- **PO5 – Modern tool usage**
Create, select and apply appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, to complex engineering problems, with an understanding of the limitations.
- **PO6 – The engineer and society**
Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solutions to complex engineering problems.
- **PO7 – Environment and sustainability**
Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex engineering problems in societal and environmental contexts.
- **PO8 – Ethics**
Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.
- **PO9 – Individual and teamwork**
Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.
- **PO10 – Communication**
Communicate effectively on complex engineering activities with the engineering

community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations and give and receive clear instructions.

- **PO11 – Project management and finance**

Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work as a member and leader in a team, to manage projects and in multi-disciplinary environments.

- **PO12 – Life-long learning**

Recognize the need for, and have the preparation and ability to engage in, independent and life-long learning in the broadest context of technological change.

LOs

Learning Outcomes (LOs):

1. **Relate** the fundamentals of materials science and technology behind industrial processes.
2. **Determine** the ability to ensure suitability and enhancement of industrial management.
3. **Identify** good Health, Safety and Environment (HSE) practices and HSE management systems.
4. **Describe** various managerial practices within the organization.

Learning Summary

Internship Learning Summary

Internship Role: Trainee Material Engineer

Company: Consulting Engineering and Contractors (Pvt) Ltd

Employer: Sri Lanka Road Development Authority (RDA)

Project: Road Construction Project (awarded in four lots through bidding)

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4. Initial Site Testing & Investigation

- Road Investigation
 - DCP Test & DCP Test Summary

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- Identifying Managerial Issues In Preparing DCP Test Reports & Summary
- Develop Automated DCP Report generate workbook
- Soil Sample Collection

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Week6

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2nd six week

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4. Plasticity Index (PI) 

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 - ii. Fine
 - c. SB
 - i. Type I
 - ii. Type II
 - d. EMB
 - i. Type I
 - ii. Type II

- e. Shoulder Material
- f. River-sand, Quarry dust, M-sand (based on project specification and project happening area)

Week10

2. Quality Testing

For soil, ABC and Concrete agg. (course)

- a. SA
- b. PI
- c. Proctor
- d. CBR

Week11

For ABC

- e. FI 
- f. AIV 
- g. LAAV 

Week12

- h. TFV
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For Concrete Aggregate

- j. Specific Gravity

3rd six week

Week13

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Week14

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- Request for Inspection (RFI) Process
- Submission to Resident Engineer (RE) - DPR + RFI
 - Consulting Site Engineer Inspect And Hand Over RFI With Signature
 - RFI Management & QA Documentation
 - Automation With Excel WorkBook - RFI Management & QA Documentation

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 - b. With Subbase
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4th six week

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- Core Cutting for Asphalt Quality Verification

Week24

5. Make research on how to reduce asphalt thickness

Learning prompt version

Internship Learning Summary

Internship Role: Trainee Material Engineer

Company: Consulting Engineering and Contractors (Pvt) Ltd. (Contractor)

Employer: Sri Lanka Road Development Authority (RDA)

Consultant: Euro group for engineering consultancy

Project: Road Construction Project (4 LOTS)

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---Stockpile Management

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-----Concrete Aggregate

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-----Specific Gravity

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according to that we can break main learning outcomes which i got actually.

Project And Company

About The Project

Project Name:

Inclusive Connectivity & Development Project (ICDP)

Purpose:

The project aims to rehabilitate and improve the rural road network in the Homagama area, which will enhance road infrastructure, connectivity, and overall socio-economic conditions in the region. The project also includes the maintenance of these roads to ensure their sustainability.

How the Project Was Received:

The project is funded by the World Bank and received by the Road Development Authority (RDA) of Sri Lanka. The consultant for the project is Euro Group for Engineering Consultants, and the contractor is Consulting Engineers & Contractors (Pvt) Ltd. The contractor company was awarded a total of 4 lots from 5, each covering specific road sections within the Homagama Divisional Secretariat area. These lots cover various roads throughout the Western Province of Sri Lanka.

Procurement Process for Rural Road Rehabilitation in Homagama

1 Invitation for Bids (IFB) or Request for Proposals (RFP)

- RDA (Road Development Authority) issues an Invitation for Bids (IFB) or Request for Proposals (RFP).
- This is a public announcement that details the project scope, specifications, deadlines, and eligibility criteria.
- The project is funded by the World Bank, so the bidding process must comply with international standards and procedures set by the World Bank.

2 Bid Submission

- Interested contractors or companies prepare their proposals or bids. The bid usually includes:
 - Technical documents (project plan, approach, qualifications, etc.)

- Financial documents (cost breakdown, schedule, etc.)
- These bids are submitted within the deadline specified in the IFB/RFP.

3 Bid Evaluation

- After all bids are submitted, the RDA evaluates them based on predefined criteria, including:
 - Technical Competence (experience, expertise, etc.)
 - Financial Proposal (cost-effectiveness, budget alignment)
 - Past Performance (track record of similar projects)
 - Compliance with World Bank guidelines (for international funding)

4 Awarding of Contract

- After evaluation, the RDA selects the best-qualified bidder and awards the contract.
- In this case, the contractor selected for the Rehabilitation/Improvement and Maintenance of Rural Roads project is **Consulting Engineers & Contractors (Pvt) Ltd.**
- The World Bank's approval is also required for the final selection to ensure transparency and compliance with international standards.

5 Contract Signing

- After the bid is awarded, the RDA and the selected contractor sign a formal contract.
- The contract outlines terms, scope, deliverables, timeline, and payment structure — ensuring both parties understand their roles and responsibilities.

6 Implementation

- The contractor begins work according to the project's requirements and timeline.

- The consultant, **Euro Group For Engineering Consultants**, oversees the project — ensuring it stays on track, meets quality standards, and remains within budget.

Supervision and Monitoring

- Regular monitoring and progress reviews are conducted by the RDA, consultant, and contractor.
- World Bank auditors may also participate in monitoring the financials and compliance with project guidelines.

Contract Document in Road Projects

Contract documents are the **legal foundation** of any construction project. In road projects, especially those funded by international agencies like the World Bank, these documents are essential to ensure transparency, accountability, and proper execution in line with ICTAD guidelines.

They are signed between the **Road Development Authority (RDA)** and the selected contractor — here, **Consulting Engineers & Contractors (Pvt) Ltd** — under the Inclusive Connectivity & Development Project (ICDP).

Key Components of Contract Documents

1. Conditions of Contract (SCA/4)

Governs the legal obligations, rights, and responsibilities of all parties involved — including the client (RDA), contractor, and consultant. It ensures all legal terms are clearly defined.

2. Drawings and Specifications

Provide detailed engineering designs, construction methods, material standards, and execution procedures to be followed during the project.

3. Bill of Quantities (BOQ)

Lists all work items, estimated quantities, unit rates, and total costs. The BOQ is used for tendering, budgeting, and progress payments.

4. Schedules (e.g., SCA/8)

Include time schedules, rate schedules for labor and materials, and other predefined operational timelines for various stages of work.

Purpose and Importance

- **Define scope, timeline, and cost** of the project with precision.
- **Prevent disputes** by ensuring all terms and expectations are clearly outlined and mutually agreed upon.
- **Ensure alignment** with ICTAD standards and World Bank guidelines on quality, payments, and penalties.
- **Support project monitoring** by consultants (e.g., Euro Group for Engineering Consultants) and enable effective implementation.

Additional Provisions in the Signed Contract

Project Timeline

Specifies start and end dates, major milestones, and delay penalties.

Payment Terms

Outlines interim payments, retention money, final payments, and performance securities.

Variation & Change Orders

Describes procedures for managing changes to work scope, design, or unforeseen site conditions.

Dispute Resolution

Provides mechanisms such as arbitration or mediation in case of disagreements.

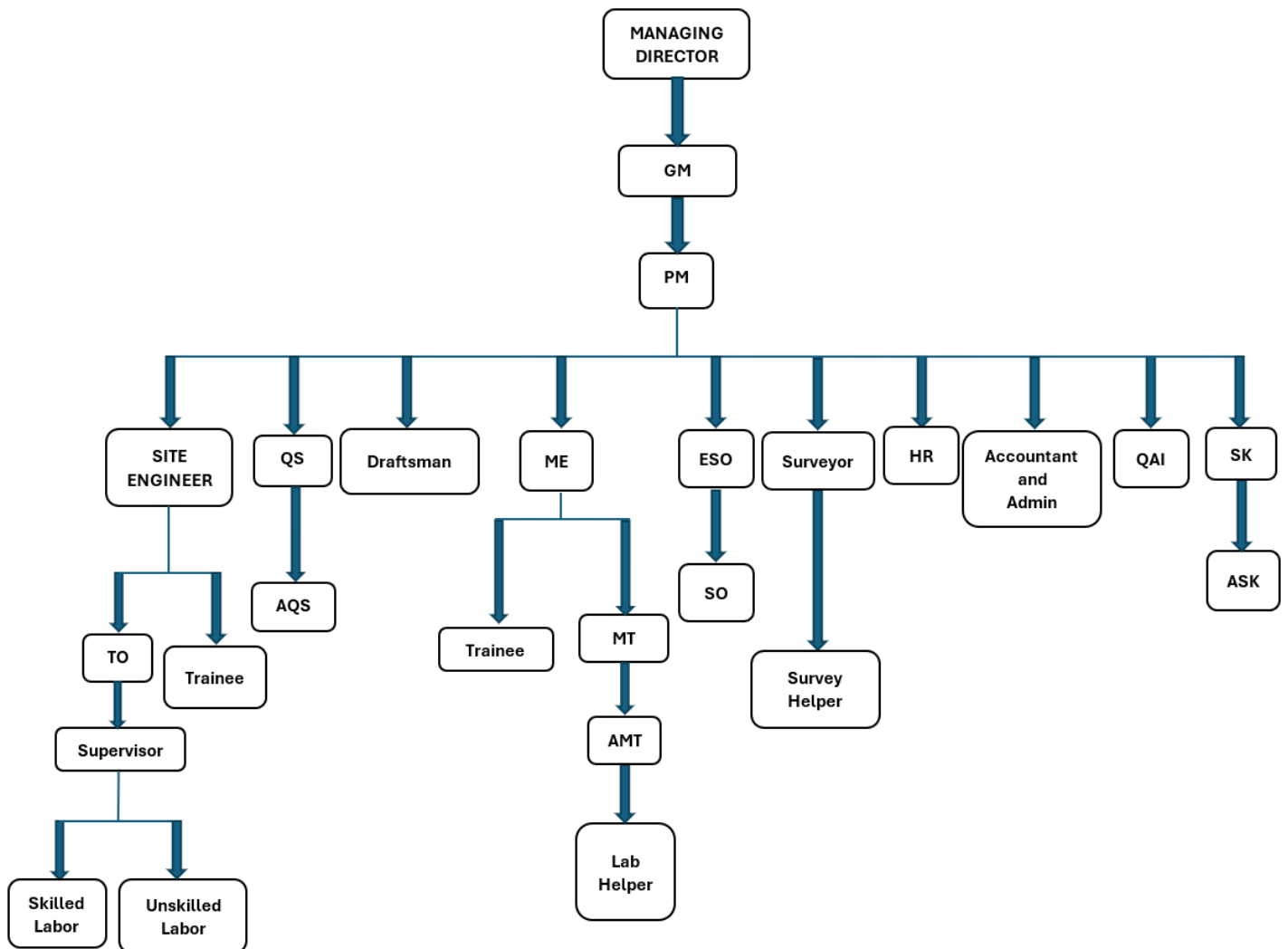
Health, Safety, and Environmental (HSE) Guidelines

Ensures compliance with safety protocols and environmental protection standards.

Donor Compliance

As a World Bank-funded project, the contract must adhere to international procurement, transparency, and reporting standards.

Project Office Staff



1 Managing Director

The highest authority responsible for overall project decision-making and company ownership in our case.

2 General Manager

Supervises all departments, ensuring projects run smoothly and align with company objectives.

3 Project Manager (PM)

Oversees the entire project from planning to completion. Manages resources, budget, and timeline. Coordinates with clients, stakeholders, and regulatory authorities.

4 Site Engineering Team

- **Site Engineer:** Supervises construction activities, ensures technical specifications are met, and manages on-site execution.
 - **TO (Technical Officer):** Assists the site engineer in technical matters and monitors work quality.
 - **Supervisor:** Directly manages site labor and daily work execution.
 - **Skilled Labour:** Performs specialized construction tasks (e.g., masonry, welding, equipment operation).
 - **Unskilled Labour:** Assists with general construction activities such as material handling and cleaning.
 - **Trainee:** Gains practical experience in site engineering.
-

5 Quantity Surveying Team

- **QS (Quantity Surveyor):** Manages project costs, prepares estimates, and ensures financial efficiency. Also prepares the **Interim Payment Application (IPA)** monthly.
 - **AQS (Assistant Quantity Surveyor):** Supports the QS in cost analysis and documentation.
-

6 Drafting & Design Team

- **Draftsman:** Creates technical drawings and site plans; assists engineers with design modifications.
 - **Trainee:** Learns surveying and drafting skills under supervision.
-

7 Material Engineering & Quality Team

- **ME (Material Engineer)** (*also called Quality Assurance Manager – QAM in some projects*): Ensures proper material testing and quality control. Handles mix designs (concrete/asphalt) and ensures compliance with standards.
 - **Trainee:** Receives training in material engineering and assists in report preparation.
 - **MT (Material Technician):** Conducts tests and manages lab equipment under the guidance of the ME.
 - **AMT (Assistant Material Technician):** Supports MT in sample collection, testing, and documentation.
 - **Lab Helper:** Assists with basic lab tasks and sample preparation.
-

8 Environmental & Safety Team

- **ESO (Environmental & Safety Officer):** Oversees environmental management and ensures workplace safety compliance.
 - **SO (Safety Officer):** Monitors on-site safety practices and enforces safety measures.
-

9 Survey Team

- **Surveyor:** Conducts land and roadway surveys to determine elevation, alignment, and boundaries; provides data for the drafting team.
 - **Surveyor Helper:** Assists the surveyor with data collection and documentation.

10 Human Resources

- **HR Officer:** Manages recruitment, payroll, employee welfare, and labor law compliance. Also handles conflict resolution.

11 Finance & Administration

- **Accountant & Admin Officer:** Handles budgeting, financial transactions, record keeping, and general office administration.

12 Quality Assurance

- **QAI (Quality Assurance Inspector):** Ensures all site work complies with relevant standards and limitations. Manages RFI (Request for Inspection) and prepares QA documents alongside IPA.

13 Store Management

- **SK (Storekeeper):** Manages inventory, tools, and materials. Maintains accurate stock records and ensures proper storage practices.
 - **ASK (Assistant Storekeeper):** Supports storekeeping duties and helps with documentation.

ICTAD & Project Spec

Understanding ICTAD Specifications and Their Role in Material Testing

What is ICTAD?

ICTAD (Institute for Construction Training and Development) is the national authority under the Ministry of Construction and Engineering Services in Sri Lanka. It issues mandatory technical specifications, standard bidding documents, and rate schedules used in public infrastructure projects, including road construction.

Relevance of ICTAD in Road Construction Projects

1. Standard Specifications (SCA/5 & others)

These are standard documents issued by ICTAD to guide construction projects in Sri Lanka

- **SCA/5:** This contains the *technical specifications* for building and civil engineering works. It explains the required quality of materials, construction methods, testing procedures, and acceptance criteria.
- **SCA/4:** This document outlines the *conditions of contract*. It includes legal terms, contractor responsibilities, payment conditions, and dispute resolution procedures.
- **SCA/8:** This is the *Schedule of Rates*. It gives the standard unit rates for various construction activities and materials, which helps in cost estimation and billing.

These documents define:

- Material quality requirements (e.g., max flakiness index for aggregates)
- Testing frequencies and acceptance criteria
- Tolerances for compaction, layer thickness, etc.

2. Material Selection

ICTAD specifications influence:

- Whether **River sand, Quarry dust, or M-sand** can be used (based on local availability and test results)

- Approvals for **Type I/II embankment or subbase materials** based on test compliance

3. Frequency & Format of Testing

- ICTAD outlines how **often tests** (like FDT or sieve analysis) must be conducted based on the **volume of work or material batches**
- QA documents and lab reports must follow **ICTAD-prescribed formats**

Project-Specific Specifications vs. ICTAD

While ICTAD sets baseline national standards, **project-specific specifications (from RDA or the consultant)** may tighten these limits depending on:

- **Funding agency requirements (e.g., World Bank)**
- **Local soil conditions**
- **Design pavement life**
- **Environmental conditions**

For example:

- Project specs may require a **higher minimum CBR** than ICTAD.
- Specific tests like **Asphalt Extraction** or **Bitumen Penetration** might have stricter tolerances than the ICTAD default.

Conclusion

ICTAD specifications are the backbone of quality assurance in Sri Lankan infrastructure projects. Understanding and applying them is critical for:

- Accurate test execution
- Material approval
- Compliance in QA documentation

- Preventing rework and disputes

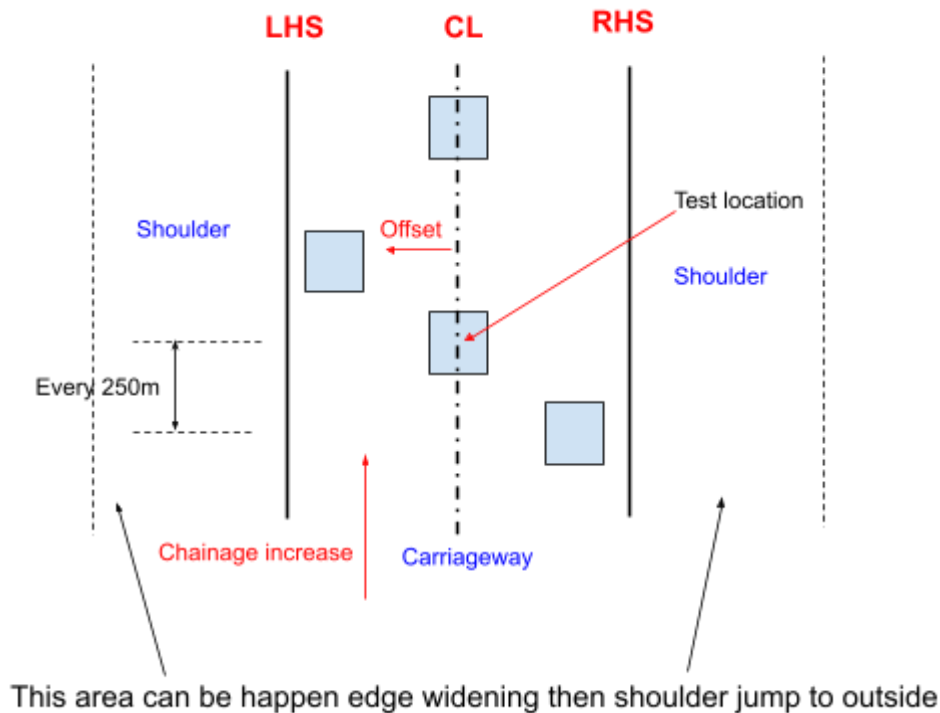
They serve as both **technical benchmarks** and **legal reference points** during construction and auditing phases.

Lab Equipment Calibrate

Initial Site Testing

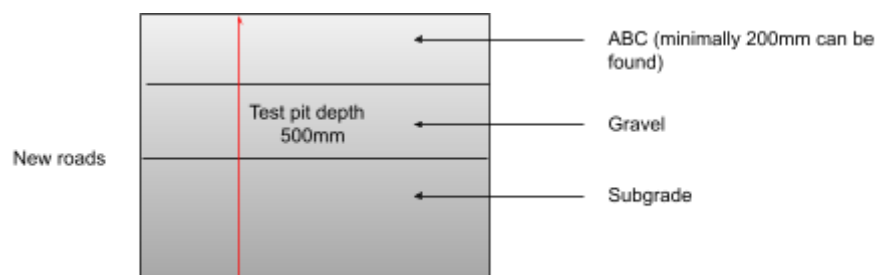
1. Testing Initialization

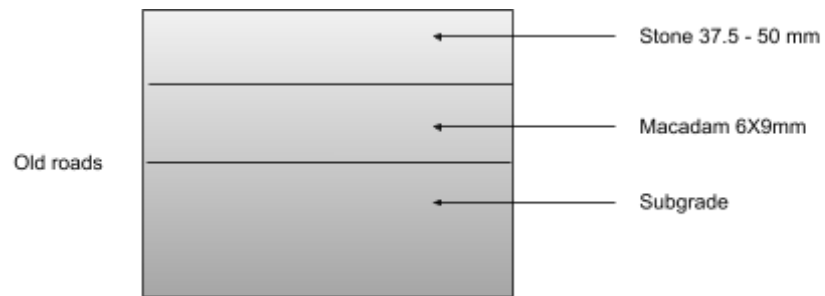
Initially We are performing road investigation (called “**site investigation**”) to assess the **existing road condition** before construction



At Each Test Location:

1. ☒ **DCP Test** (Dynamic Cone Penetration Test):
 - Determines the **in-situ strength** of subgrade and pavement layers.
 - Gives insight into **layer thicknesses** and **compaction quality**.
2. ☒ **Soil Sample Collection** (Called “**Test pit collect**”)
 - We also record the existing layers in the road while collecting samples. This helps to get an idea about the current existing road visually.





From now on i called collected soil sample “Soil Bag”

Lab Testing for Each Soil Bag:

For every Soil bag we do 4 tests in the laboratory.

- **SA** – Sieve Analysis ✓

Grading and particle distribution (SA)

- **PI** – Plasticity Index (via Atterberg Limits) ✓

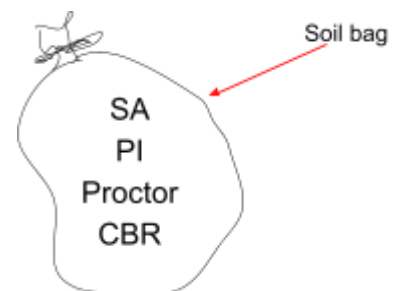
Plasticity and soil behavior (PI)

- **Proctor** – Compaction test (to determine OMC and MDD) ✓

Optimum compaction conditions (Proctor)

- **CBR** – California Bearing Ratio ✓

Bearing capacity for design (CBR)



From the laboratory, they preserve some part of the sample to see visitors or future use.

2. Dynamic Cone Penetration (DCP) Test

The **Dynamic Cone Penetration (DCP) Test** is a widely used in-situ testing method in geotechnical and pavement engineering to assess the strength and stiffness characteristics of subgrade and pavement layers. It is particularly valued for its simplicity, speed, and cost-effectiveness in field conditions.

In this test, a steel cone is driven into the ground by repeated drops of a standard-weight hammer from a fixed height. The penetration per blow, known as the **DCP Index (DPI)**, provides a quantitative measure of soil resistance. This index can be empirically correlated to the **California Bearing Ratio (CBR)**, which is crucial for designing pavement structures and evaluating subgrade quality.

DCP testing helps engineers identify soil layer boundaries, detect under-compacted zones, estimate bearing capacity, and monitor compaction quality. Because it requires no borehole, the DCP test is ideal for rapid assessments across large project sites, particularly for road work, embankments, and shallow foundation investigations.

While highly effective in granular and stiff soils, the test's accuracy can be affected by factors such as rod friction, cone wear, and soil moisture content. Special considerations and modified correlations are needed for cohesive soils, especially clay types with low CBR values.

The DCP Test is standardized under **ASTM D6951**, and its results play a key role in **quality control**, **subgrade evaluation**, and **pavement design optimization**—making it an essential tool in modern infrastructure development.

Read the relevant note to get an idea about the method, interpretation, and specification references!

3. Managerial Problem Identified: Manual DCP Test Report Creation (2024-12-23)

Problem Statement:

The DCP (Dynamic Cone Penetration) test reports are currently created using a manual process, which involves handwritten field logs and manual data entry into spreadsheets or reports. This process is **time-consuming**, **prone to errors**, and **lacks standardization**, affecting efficiency and data reliability in material quality control and decision-making.

4. Test Pit Analysis (Soil Bag Testing)

Introduction

As part of the site investigation process, **soil samples** are collected at each DCP test location through **test pits**. These samples, referred to as “**Soil Bags**,” are carefully labeled and transported to the laboratory for detailed geotechnical analysis.

The primary goal of **test pit analysis** is to determine the **engineering properties** of in-situ materials, evaluate their suitability for use in road construction (e.g., as embankment or sub base), and ensure compliance with **ICTAD specifications**.

Each soil bag represents the actual material encountered in the field, making it a critical link between **field observations** and **laboratory data**. By analyzing these samples, we gain essential insights into:

- Grading characteristics
- Plasticity behavior
- Moisture–density relationships
- Load-bearing capacity

This analysis guides decisions on:

- Material classification (e.g., SB, EMB, SHO)
- Suitability for reuse or replacement
- Stabilization or treatment needs
- Source approval and stockpile categorization

Read the relevant note to get an idea about the **lab tests breakdown** (SA, PI, Proctor, CBR) with method, interpretation, and specification references!

Macadam road



What is a Macadam Road?

A **Macadam road** is a type of road built using **layers of crushed stones**, compacted to form a hard surface. It was invented by **John Loudon McAdam** in the early 1800s.

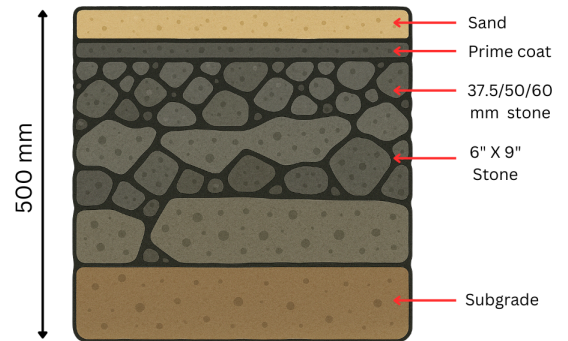


Layers of a Macadam Road



Sand Layer (Optional Wearing Course or Filler)

- May be used as a **blinding layer** or for leveling.
- Sometimes skipped or replaced by bituminous surfacing.



Prime Coat

- Bituminous emulsion sprayed to bind the stone base with the next layer.
- Helps prevent dust and water penetration.



37.5/50/60 mm Stone Layer (Base Course)

- Crushed stones laid and compacted in layers (~100–150mm each).
- Forms the structural base of the road.



6" X 9" Stone Rubble Layer

- Acts as a **foundation** or sub-base.
- Large stones manually packed to provide drainage and load-bearing strength.



Subgrade

- Natural soil at the bottom, compacted to provide support.
- Treated if weak (e.g., with lime or cement stabilization).

Dynamic Cone Penetration

Dynamic Cone Penetration (DCP) Test

Purpose of the DCP Test

The **Dynamic Cone Penetration (DCP) Test** is an economical and effective **in-situ field method** used to evaluate the **strength, stiffness, and bearing capacity** of subgrade and pavement materials. It provides fast data for quality control, layer thickness evaluation, and preliminary pavement and foundation design.



Key Applications

- Measure **in-situ resistance** of soil to penetration.
- Evaluate **compaction quality** of pavement and subgrade layers.
- Estimate **California Bearing Ratio (CBR)** indirectly.
- Determine **soil stiffness, bearing strength, and layer interfaces**.
- Detects **under-compacted fill** in embankments or pavement sections.
- Monitor changes in **granular subgrade condition** over time.
- Aid in **pavement thickness design** and **foundation suitability checks**.
- Support decisions on **subgrade stabilization** or improvement.

Equipment Details (Standard DCP Kit Includes)

Standard Components

- **Steel Cone:** Hardened steel, 60° apex angle, 20 mm diameter.
- **Steel Rods:** Upper and lower rods, typically 5/8" diameter.



- **Hammer:** 8 kg (17.6 lb) dropped from 575 mm (2.26 ft). Dual-mass hammers (4.6 kg + 3.4 kg detachable).
- **Anvil:** Transfers energy from the hammer to cone rod.
- **Graduated Vertical Scale:** With 1 mm resolution or electronic reader.
- **Accessories:** Handle, handguard, 1 m ruler, tommy bars, thread-locking adhesive.



Hammer System

- **Single-Mass:** 8 kg – suitable for general soil types.
- **Dual-Mass:** 4.6 kg (detachable 3.4 kg surcharge). Used for soft soils to avoid over-penetration.

Cone Variants

- **Reusable cones:** Threaded and removable. Replace when worn.
- **Disposable cones:** Useful in expanding or sticky soils. Detached after testing.

Additional Features

- **Electronic logging systems** available for automatic data capture.
- **Extension rods** allow testing depths up to **1200 mm or 5 feet**.
- Apparatus costs range from **\$1,600–\$2,200 USD** for manual kits.

Working Principle

The DCP test is based on **measuring soil resistance** to cone penetration caused by a falling hammer. The **penetration depth per blow** correlates with **soil stiffness** and **bearing strength**, and is used to compute the **DCP Index** and indirectly estimate **CBR** and other parameters.

Energy Calculation:

$$E = mgh$$

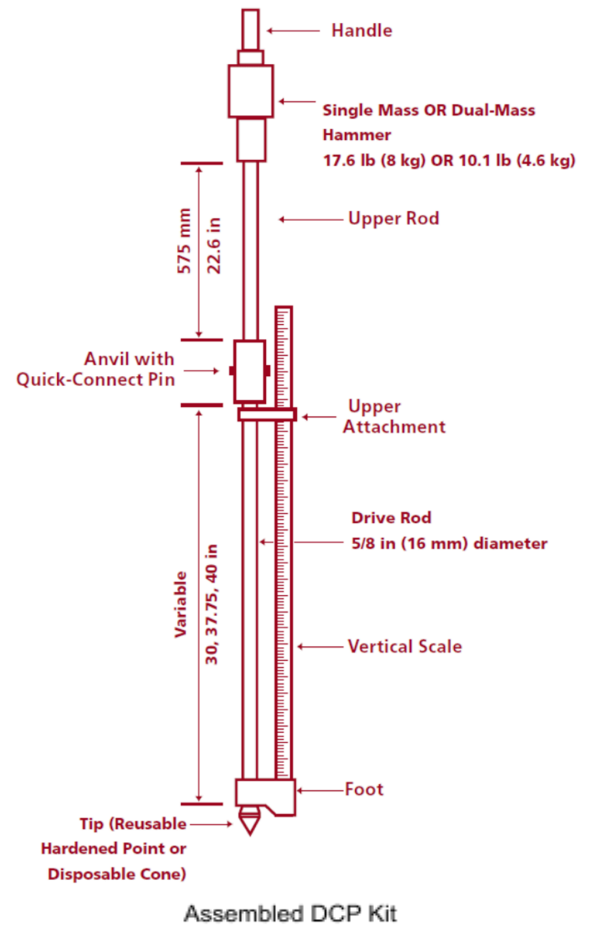
$$E = 8 \times 9.81 \times 0.575 = 45.08 \text{ J (Per Blow)}$$

Where:

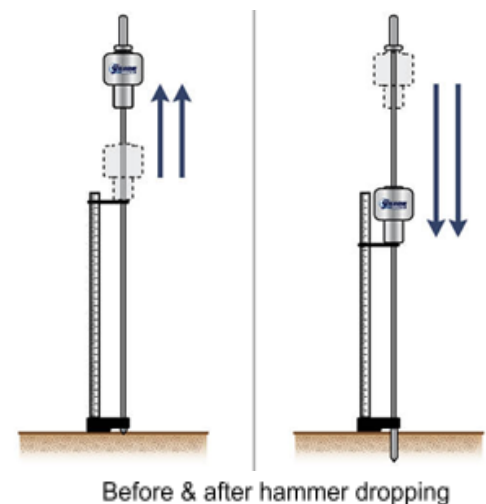
- E = Impact energy (Joules)
- m = Mass of hammer (8 kg)
- g = Gravity (9.81 m/s^2)
- h = Drop height (0.575 m)

Testing Procedure (Step-by-Step)

1.  **Select Location**
 - Choose representative points across the project site.
 - For paved areas, predrill holes to reach the subgrade.
2.  **Prepare Surface**
 - Clear debris. Ensure level contact for cone base.
3.  **Assemble Apparatus**
 - Tighten all joints using tommy bars and apply thread-locking adhesive.
 - Secure the handle, anvil, and cone connections.
4.  **Zero the Device**
 - Place the cone vertically, flush on the ground.
 - Record initial depth reading.
5.  **Begin Testing**



Perform DCP Test



- Raise the hammer to 575 mm and drop freely.
- Avoid pushing or lowering the hammer by hand.
- In cohesive soils, slightly rotate the rod to prevent sticking.

6. 📝 Record Data

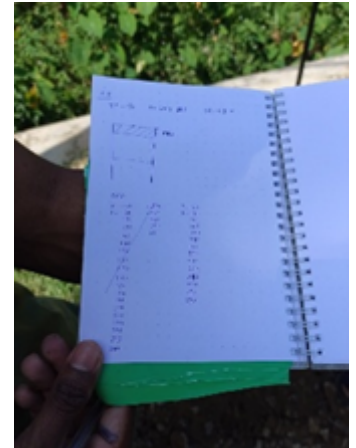
- Record penetration after each blow or after a set number of blows.
- Use tighter intervals for soft soils (every 1–2 blows) and wider for stiff soils (5–10 blows).
- Minimum: 25 mm penetration between data points.

7. 📊 Stop When

- Desired depth is reached (typically 800–1200 mm).
- Cone reaches refusal (penetration <3.175 mm in 10 blows).
- Hard layer or obstruction is encountered.

8. 🔄 Remove Equipment

- Gently tap the hammer upward.
- Avoid aggressive removal to extend instrument life.



Note down DCP values

📊 Data Analysis & Interpretation

📊 DCP Index (mm/blow) - (*DPI*):

$$DPI = \frac{D}{N}$$

Where:

- D = Total penetration depth (mm)
- N = Number of blows

Empirical Correlations to CBR:

Common Approximation (ASTM D6951, Kley et al.):

$$CBR = 292 \times (DPI)^{-1.12}$$

Alternate Logarithmic Formula (for granular soils):

$$\log_{10}(CBR) = 2.48 - 1.057 \times \log_{10}(DPI)$$

 Note: Do not apply these to CL or CH soils. Use ASTM-modified formulas.

CBR-SPT Relationship:

$$\log_{10}(DPI) = -1.05 + 2.03 \times \log_{10}(SPT)$$

Where:

- CBR = California Bearing Ratio
- DPI = DCP Penetration Index
- SPT = Standard Penetration Test (ASTM D 1586-64)

Graphs to Plot:

- **Depth vs. Blow Count** – to detect soil layers.
- **CBR vs. Depth (semi-log)** – to evaluate strength variation with depth.

Interpreting Results

DCP Index (mm/blow)	Soil Strength
Low (1–5)	Very strong soil
Moderate (5–15)	Medium strength soil
High (>15)	Weak or loose soil

- Sharp change in index → Layer interface or material change
- Consistent high index → Soft subgrade / need for stabilization

Equivalent CBR Value

Since multiple CBR values are typically obtained at various depths within a single test location (especially when multiple soil layers are present), a single representative value is often required for analysis or design purposes. In such cases, the **Equivalent CBR Value** is calculated to represent the average strength characteristics of the tested subgrade section.

How to Calculate the Equivalent CBR:

The Equivalent CBR is typically computed using a **weighted average** approach over the required design depth (e.g., top 500 mm or 700 mm), where the contribution of each layer is proportionally considered based on its thickness:

$$CBR_{eq} = \left(\frac{\sum_i L_i CBR_i^{1/3}}{\sum_i L_i} \right)^3$$

Where:

- L_i - Thickness of each layer
- CBR_i - CBR value of each layer

This method ensures the varying strengths of each layer are properly accounted for when determining the representative bearing value at a test point.

This Equivalent CBR Value is crucial for pavement design, subgrade evaluation, and assessing the need for stabilization treatments.

Comparison: SPT vs SCPT vs DCPT

Test	Mode	Sampling	Best For	Notes
SPT (Standard Penetration Test)	Impact-driven	Yes	Loose to dense soils	Requires borehole
SCPT (Static Cone Penetration Test)	Static thrust	No	Very soft soils	Continuous measurement
DCPT (DCP) (Dynamic Cone Penetration Test)	Dynamic drop	No	Granular/stiff soils	Field-portable, no borehole

Factors Affecting Accuracy

- Rod alignment
- Cone tip condition
- Hammer weight and drop height
- Moisture content
- Soil type and compaction level
- Friction along extension rods

- Apex angle (should remain 60°)



Standards & References

- **ASTM D6951** – Standard Method for Shallow Pavement Applications.
- **ICTAD Guidelines** – Sri Lankan road subgrade quality standards.
- **Portland Cement Association (PCA)** – Bearing capacity estimation.
- **USACE Manuals, Powell et al., Heukelom & Klomp** – Soil modulus & CBR relations.



Importance in Road & Foundation Projects

- Verifies **subgrade strength** before construction.
- Guides **design decisions** on stabilization and layer thickness.
- Identifies weak zones and ensures **uniform compaction**.
- Enables rapid, low-cost evaluation across large areas.
- Essential tool for **quality control**, **pavement design**, and **foundation planning**.



Precautions

- Always use **calibrated drop height** (575 mm).
- Avoid testing during **monsoons** due to softening of subgrade.
- Take **repeated readings** to minimize errors.
- Use **thread-lock** to prevent joint loosening during testing.

✓ Key Advantages

- No borehole required – minimal disruption.
- Simple, fast(results within minutes.), and repeatable.
- Low cost, lightweight, Portable and operator-friendly.
- Effective for identifying weak zones and compaction quality.
- Dual-mass systems provide flexibility for different soil types.
- Minimal operator training required.

✗ Limitations

- Not suitable for deep soil investigation (>1.2 m).
- Not ideal for very cohesive or very loose soils.
- Friction in extension rods may affect accuracy at greater depths.

⚠ Special Case: Clay Soils (CL & CH) with Low CBR

In **clay soils** classified as:

- **CL**: Lean clay (low plasticity)
- **CH**: Fat clay (high plasticity)

The **standard DCP–CBR correlation**:

$$CBR = 292 \times (DPI)^{-1.12}$$

does not work reliably when CBR < 10.

Why?

Because clayey soils deform differently — they **don't resist penetration in the same way** as granular soils, and moisture or plasticity affects penetration a lot. That leads to **underestimated or inconsistent CBR values** if you use the standard equation.

✅ ASTM D6951 Solution:

The **ASTM D6951** standard recommends using **alternative regression equations** (based on lab calibration data) for CL and CH soils, or **direct CBR testing** if accurate results are needed.

Some research-backed examples:

- For CL soils:

$$\log_{10}(CBR) = 2.465 - 1.12 \times \log_{10}(DPI)$$

- For CH soils:

$$CBR = 35 \times (DPI)^{-0.6}$$

Note: These values (e.g., 2.465, 1.12, 0.6) **depend on the soil dataset** and may vary project to project. Always calibrate with site-specific CBR tests when possible.

📌 Summary:

- **Standard equation is unreliable for clayey soils with CBR <10.**
- Use **modified equations** from ASTM D6951 or literature.
- Better: **Run lab CBR tests** alongside DCP to calibrate for the site.

📌 Understanding the DCP-SPT Correlation

$$\log_{10}(DPI) = -1.05 + 2.03 \times \log_{10}(SPT)$$

- This empirical relationship allows **comparison between field DCP and borehole-based SPT results**.

- Useful when:
 - No borehole is available, but design requires equivalent **SPT N-values**.
 - **Back-calculating DPI** when **SPT N-values** are available from previous investigations.
- Note: This formula is **site-sensitive** and should be used cautiously—only when **both tests are calibrated for local soils**.

Managerial Problem

1. Managerial Problem Identified: Manual DCP Test Report Creation (2024-12-23)

Problem Statement:

The DCP (Dynamic Cone Penetration) test reports are currently created using a manual process, which involves handwritten field logs and manual data entry into spreadsheets or reports. This process is **time-consuming**, **prone to errors**, and **lacks standardization**, affecting efficiency and data reliability in material quality control and decision-making.

Why This is a Managerial Problem:

1.  **Inefficient Use of Human Resources:**
 - Technicians spend considerable time compiling, plotting, and formatting reports manually.
 - Reduces time available for actual testing and analysis.
2.  **Inconsistency & Human Error:**
 - Manual calculations and transcriptions can introduce errors.
 - Graphs (e.g., depth vs. blow) are often skipped or inaccurately drawn.
3.  **Poor Data Traceability:**
 - Difficult to retrieve past reports or audit the accuracy.
 - No centralized digital database or version control.
4.  **Delayed Decision-Making:**
 - Site engineers and managers receive test results late.
 - Delays in approving subgrade quality or releasing work fronts.
5.  **Lack of Real-Time Insights:**

- No quick visualization or auto-calculated CBR values.
- Misses the chance to flag weak zones immediately on site.

How Can I Present It? (Intern Report or Presentation):

Element	Your Example
Identified Managerial Problem	Manual DCP test report preparation and analysis
Root Cause	Reliance on handwritten logs, Excel files without automation
Impact on Operations	Time delays, data errors, reduced productivity, reporting lag
Affected Stakeholders	Material engineers, QA/QC team, site engineers, consultants
Potential Solution (Suggestion)	Digitize DCP reporting using automated tools, apps, or Excel macros
Benefit If Solved	Faster reporting, reduced error, better resource utilization, instant CBR graphs

Possible Solution Can Propose

-  Design an **automated Excel DCP log sheet** (enter blow data → autograph + CBR).
-  Create a **simple desktop app** in VBA or Python.
-  Even propose a **mobile-based logging tool** (e.g., Google Forms → Sheets → Auto Report).

Justification for Selecting Option 1: Automated Excel DCP Log Sheet

To address the managerial problem of manual and error-prone Dynamic Cone Penetration (DCP) test report creation, I have selected the first solution option: **Designing an automated Excel DCP log sheet**. This approach was chosen based on practical, technical, and organizational considerations observed during my training period.

1. Alignment with Existing Office Practices

- **The engineering staff in our organization already widely uses Excel** for most data logging, calculations, and reporting tasks.
- Staff are **familiar with Excel** interfaces and functions, which reduces the learning curve.
- Integrating automation within Excel leverages existing workflows instead of introducing unfamiliar tools or systems.

2. Simplicity and Accessibility

- The proposed Excel-based automation can be easily deployed **without additional software installations** or changes to current systems.
- **No internet connection or high-performance computers** are required, which suits the current infrastructure where **low-spec office computers are used**.
- The tool remains **lightweight and easy to modify** based on user feedback or future updates.

3. Cost-Effective and Sustainable

- Excel is already licensed and used in the office. No **additional budget** is needed for new software.
- This approach minimizes operational disruption and **maximizes return on existing tools**.
- **No programming expertise** is needed once the macro-enabled sheet is created. Users only input blow data, and outputs (graphs, CBR, depths) are auto-generated.

4. Fast Implementation & Immediate Impact

- The Excel tool can be developed, tested, and deployed **within a short time** frame during the internship period itself.
- **Immediate benefits** can be demonstrated: auto-generated depth vs. blow graphs, estimated CBR, and data consistency.

Conclusion

Given the **technical limitations**, the **need for user-friendliness**, and the **office's reliance on Excel**, the most practical and effective solution is to create a **macro-enabled Excel DCP reporting tool**. This ensures faster, error-free, and standardized reporting without burdening the current system or requiring high-end infrastructure.

Digital Automation of DCP Test Report Creation & Summary

Problem Summary

In our QA/QC workflow, **Dynamic Cone Penetration (DCP)** test data were previously handled using **manual Excel sheets**. This process was:

-  Time-consuming
-  Prone to data entry and plotting errors
-  Lacked consistency and traceability

Each report required:

- Manual entry of blow counts and penetration depths
- Plotting of depth vs. blow charts
- Manual detection of soil layers and estimation of CBR values
- Separate summary sheet preparation across multiple locations

✔ Objective

Develop a fully automated, Excel-based reporting system that:

- 📄 **Standardizes** DCP test report generation
- 📊 **Detects soil layers** automatically
- 📋 **Summarizes reports** into a central summary sheet
- 🔒 **Enforces controlled workflow** (prevents closing until processing is complete)
- 📈 **Visualizes data** and computes CBR seamlessly

🏗️ Project Structure

Workbook Name: `DCP Test Report Generator.xlsm`

Main Sheets:

- 🔒 **Input** – Entry panel & control interface (visible)
- 📄 **DCP_Test_Report_Template** – Hidden; used for generating test sheets
- 📈 **DCP_Test_Summary_Template** – Hidden; used for final summary

📋 Modules Overview

📦 Module	🔧 Purpose
<code>ClearerDCPData</code>	Clears user input fields for new test
<code>DetectorDCPLayers</code>	Performs automatic soil layer detection using RMSD
<code>Generator1TestReport</code>	Creates individual DCP report sheets
<code>Generator2FinishReportCreation</code>	Compiles the final summary sheet

⚙️ Key Features & Logic

🔄 1. Startup Behavior

- **Workbook_Open:**
 - Enables test creation (D24 = TRUE)
 - Clears previous workbook name (D25) and close-lock (D26)
- **Workbook_BeforeClose:**
 - Prevents closing unless test report process is marked finished (D26 = TRUE)

Road ID	COL/65				Only Edit Blue Colour Area	Input Reading Here	
Report No.	RDA/ICDP/LOT3/LAB/65/DCP/001					Blows	Penetration
Date of Test	24/11/2024						
Road Name	Youngma Main Road						
Test Location	Chainage	0	+	120		Select	
Off set (m)	1.3						
LOT number	3						
Total road distar	17.06	Km					
						Create Test Report	
						Finish Test Report Creation	
Contractor	Tested By	W.G.S. Lakmal		MT	Clear Input Data		
	Checked By	Y.S.M Harischandra		ME			
Consultant	Checked By / Witnessed By	R.M.S.K. Bandara		MT			
	Reviewed By	G.D.R Jayasooriya		ME			
New Work Book Needability	TRUE				Default Values TRUE Empty FALSE		
New Work Book Name	#####						
Closeability	#####						
Maximum Allowed MSTD	3						
						When newly start thiswork book check these values	
						Layer detection sensitivity	

Insert zero reading here

2. Input Panel Triggers

Buttons Linked To:

Button	Triggered Sub
Create Test Report	<code>btn_create_test_report_Click</code> → <code>CreateDCPTestReport</code>
Finish Report Creation	<code>btn_finish_test_report_creation_Click</code> → <code>FinishDCPTestReportCreation</code>
Clear Input	<code>btn_clear_input_data_Click</code> → <code>ClearDCPData</code>

3. CreateDCPTestReport Logic

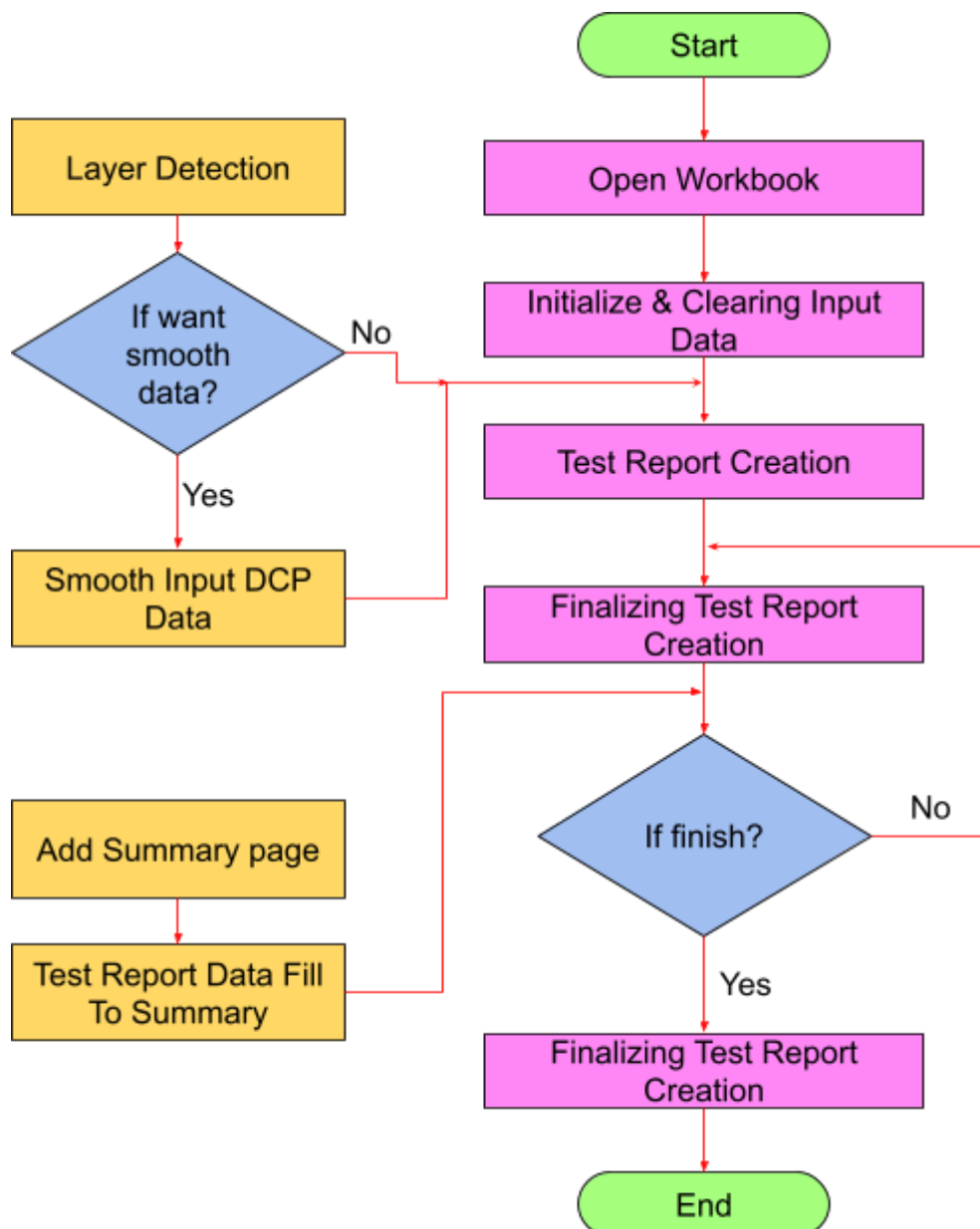
- Reads sanitized Road ID & Report Number
- Creates and names a new sheet: `RD_<RoadID>_DCP_<ReportNo>`
- Calls `DetectDCPLayer()` to identify breakpoints
- Asks user if they want to **smooth the data** (optional interpolation)
- Updates the graph & saves the test report

4. DetectDCPLayer Logic

- Computes **RMSD (Root Mean Square Deviation)** for each segment
- Limits breakpoints (e.g., max 12) to avoid overfitting
- Optionally calls `SmoothDCPData()` to interpolate penetration values for better presentation
- Stores layer boundaries for plotting and summary

5. FinishDCPTestReportCreation Flow

- Opens target workbook listed in [D25](#)
- Deletes default sheet & any previous summary
- Duplicates and fills the summary sheet with:
 - **Shoulder sheets** ($P11 \geq 1.5$)
 - **Carriageway sheets** ($P11 < 1.5$)
- Groups entries by **chainage prefix**, inserts row gaps where prefix changes
- Hides summary template and marks process as complete





Outputs

-  **Individual Report Sheet:** Includes test data, chart, CBR estimates, and layers
-  **Master Summary Sheet:** Compiles key values from all test reports



Benefits Achieved

- ✓ Eliminated manual plotting errors
- ✓ Improved report consistency and readability
- ✓ Ensured all data is tested and summarized before allowing workbook closure
- ✓ Enabled faster report generation → **more tests per day**
- ✓ Facilitated internal approval with organized test logs



Demonstration of Developed Tool: Automated Excel DCP Test Report Generator



Screenshot Description:

The image below showcases the **Input Panel** of the Excel-based automation tool I developed to streamline the DCP Test Report process. This panel serves as the central user interface for all interactions and is designed with simplicity and clarity in mind to suit on-site and office use by engineers and technicians.



Workflow Demonstration (via Screenshot):

1. **Enter basic test info** → Road ID, Report No., Chainage, offset, etc.
2. **Click "Create Test Report"** → Auto-generates report sheet with:
 - Depth vs. Blow graph
 - Detected soil layers
 - CBR calculation
 - Optionally smooth input dcp data according to detected layers
3. **Repeat for other locations**

4. Click "Finish Report Creation" → Summary sheet attach and fill

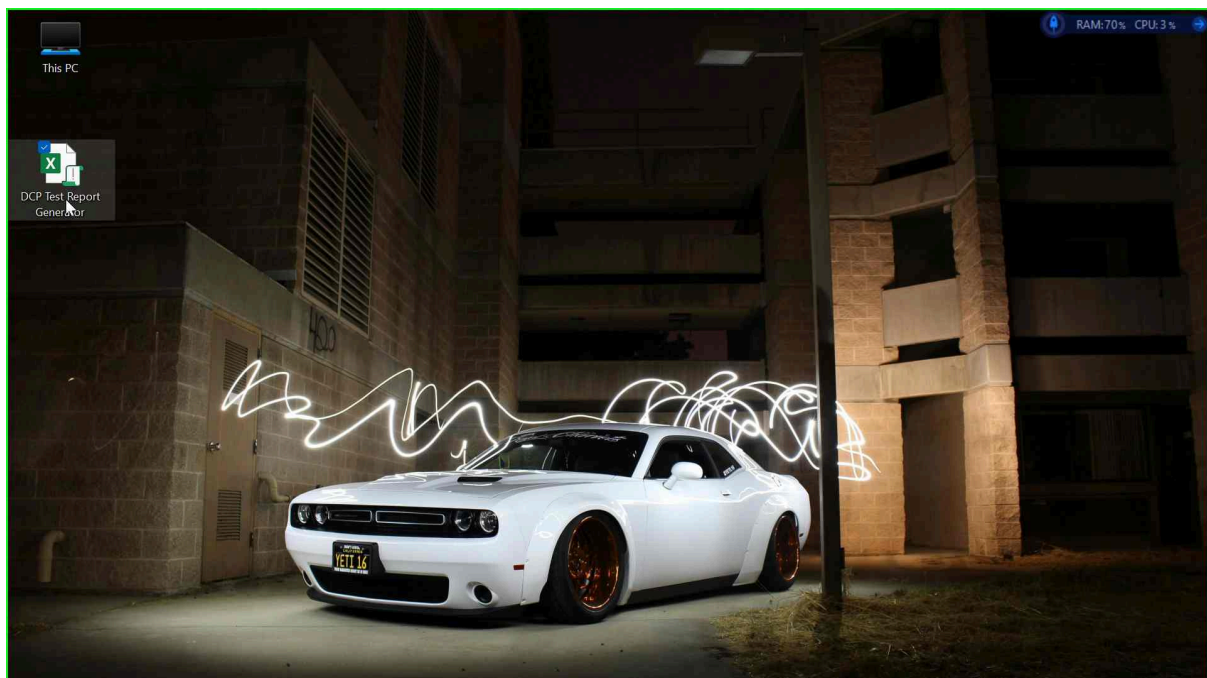
💡 Why This Interface Matters:

- **User-Friendly:** Minimal input required; all processing is automated.
- **Error-Free:** Eliminates manual plotting and summary issues.
- **Standardized Output:** All reports follow a consistent format, aiding both field review and managerial approval.
- **Locked Flow Control:** Ensures proper testing, reporting, and finalization before workbook closure.

🔗 This screenshot demonstrates the actual implementation and functionality developed during my internship period. It exemplifies how automation can address real-world managerial inefficiencies in material quality control workflows.

1. Open the Automated Workbook

Launch the macro-enabled Excel file titled “**DCP Test Report Generator.xlsm.**”



2. Enter Basic Test Information

Fill in the Road ID, Report Number, and DCP input data.

👉 (Optional) Click the **“Clear Input”** button to reset previous test data before starting a new entry.

The screenshot shows the DCP Test Report Generator Excel spreadsheet. The input fields are as follows:

Field	Value
Report No.	RDA/ICDP/LOT3/LAB7/-/DCP/002
Date of Test	4/18/2025
Road Name	Uduwapitiya Road
Test Location	0 + 430 LHS
Off set (m)	1.4
LOT number	3
Total road distant	17.06 Km
Tested By	W.G.S. Lakmal
Checked By	Y.S.M Harischandra
Checked By / Witnessed By	R.M.S.K. Bandara

The buttons are:

- Create Test Report (Green)
- Finish Test Report Creation (Yellow)
- Clear Input Data (Pink)

The data table is:

Blows	Penetration
0	160
1	210
1	232
1	255
1	278
1	300
1	308
1	316
1	325
1	333
1	341
1	349
1	357
1	365
1	374
1	382

An arrow points to the first row of the data table with the text "Insert zero reading here".

The screenshot shows the DCP Test Report Generator Excel spreadsheet with a success message dialog box displayed. The dialog box text is:

Microsoft Excel

DCP data cleared successfully.

OK

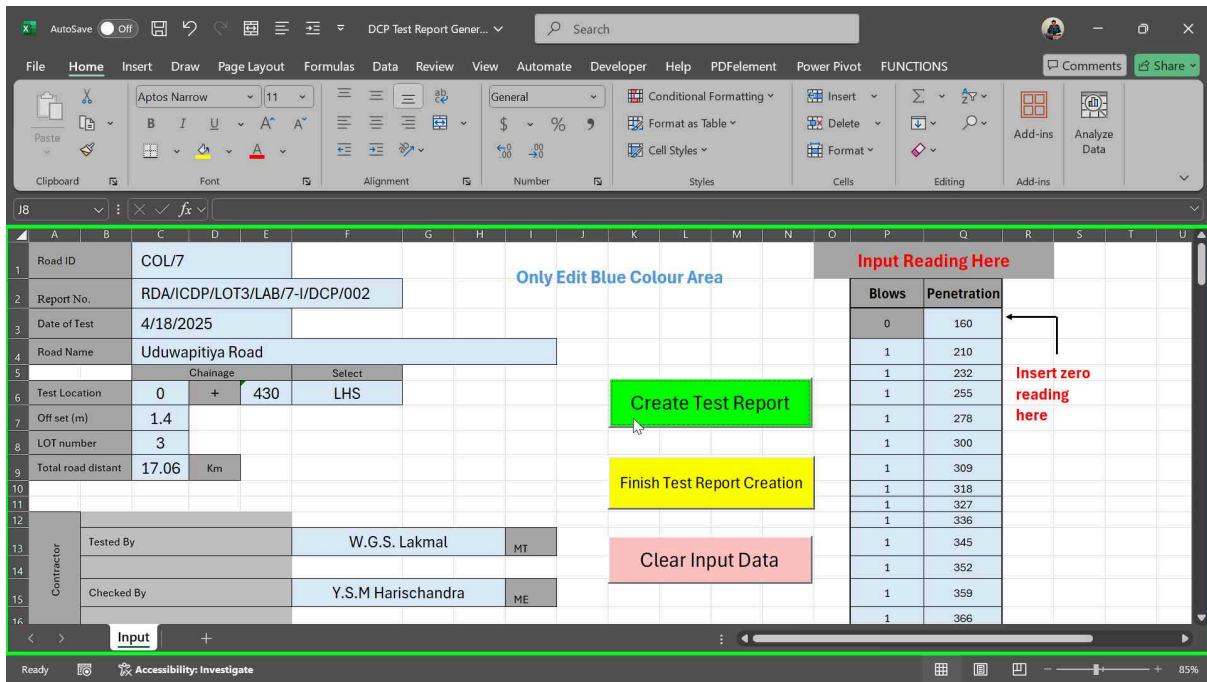
The buttons are:

- Create Test Report (Green)
- Finish Test Report Creation (Yellow)
- Clear Input Data (Pink)

The data table is empty.

3. Create a Test Report

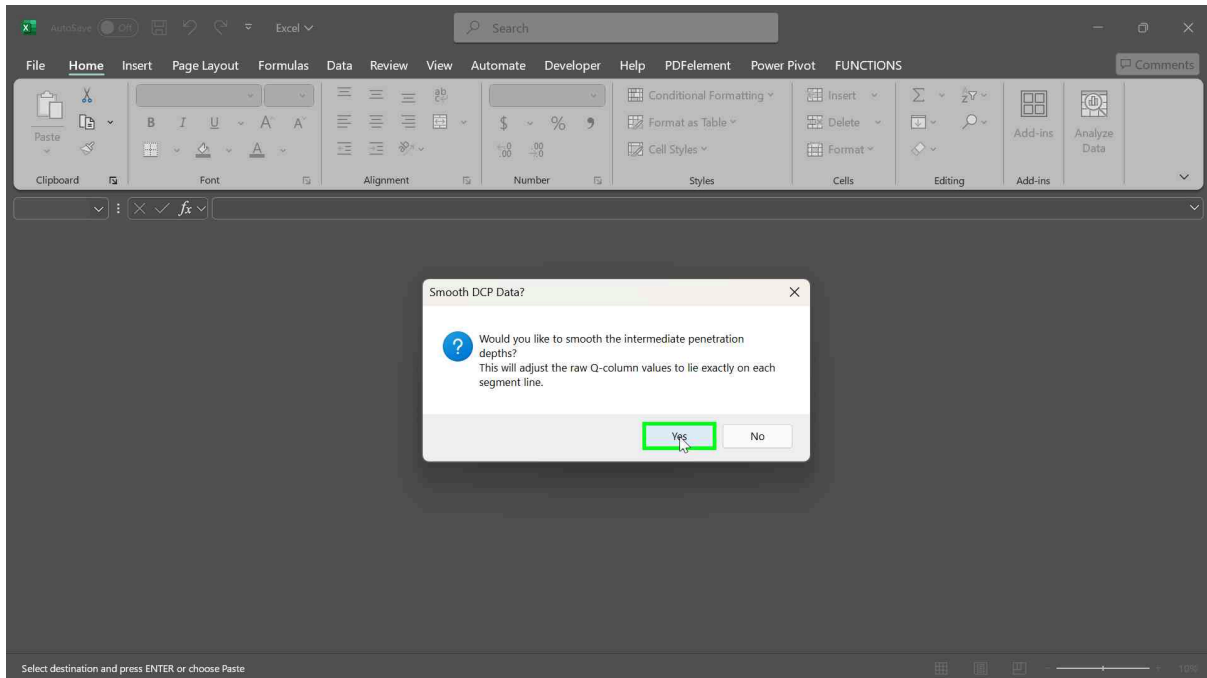
Click the **“Create Test Report”** button to generate a new report sheet. This will trigger automatic layer detection and graph generation.



4. Optional Data Smoothing Prompt

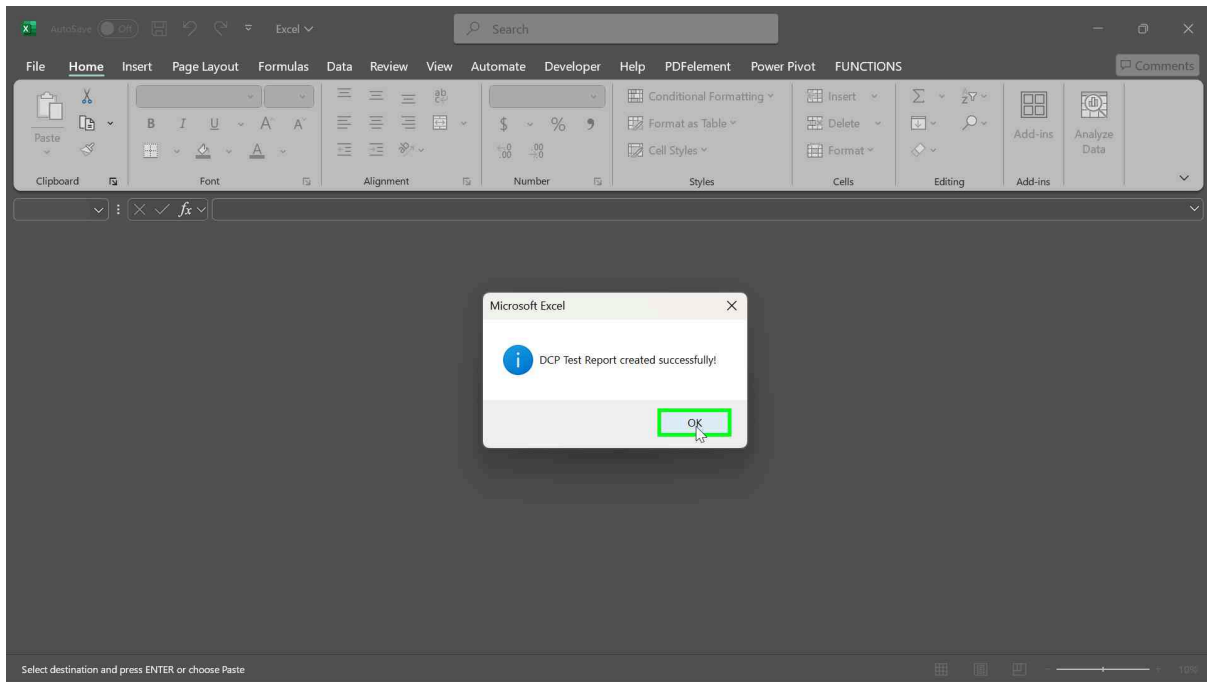
The system will ask if the user wants to **smooth the data** for better presentation.

✚ If confirmed, interpolation is applied to the penetration values without altering the blow counts.



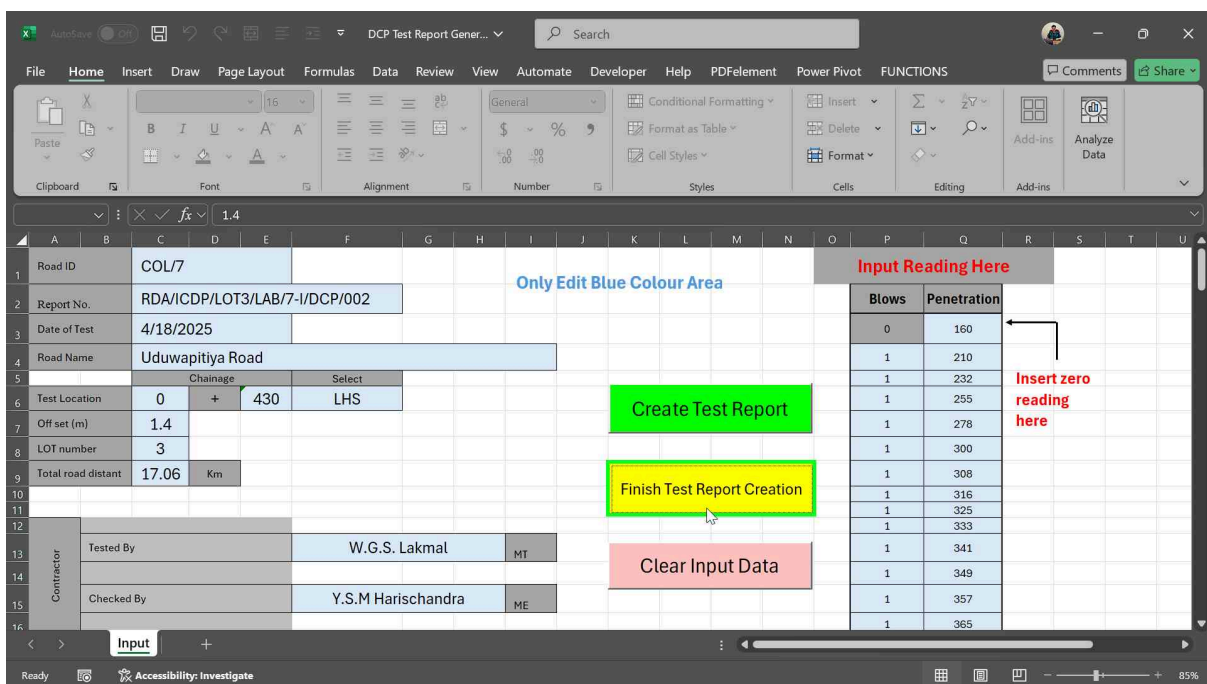
5. Report Creation Confirmation

A success message will appear, confirming that the individual DCP report was generated.



6. Generate Final Summary

After creating reports for all required chainages, click **“Finish Report Creation”** to compile the results into a centralized summary sheet.



Only Edit Blue Colour Area

Input Reading Here

DCP Test Report Summary Sheet successfully created!

Finish Test Report Creation

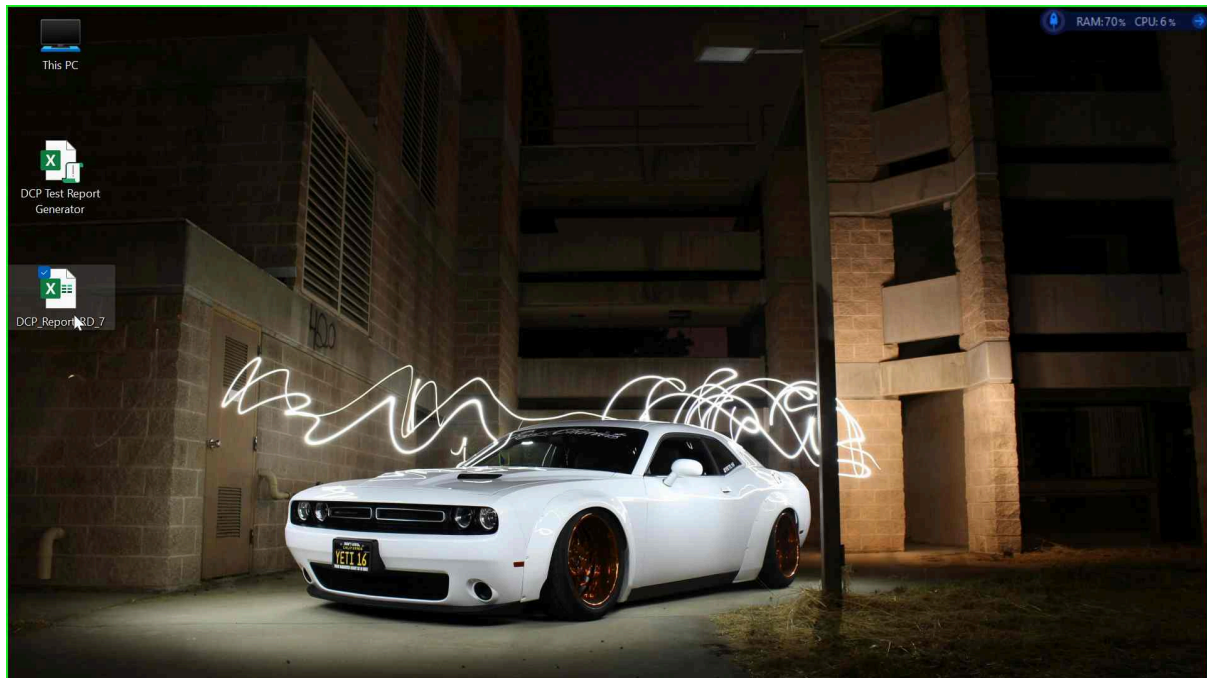
Clear Input Data

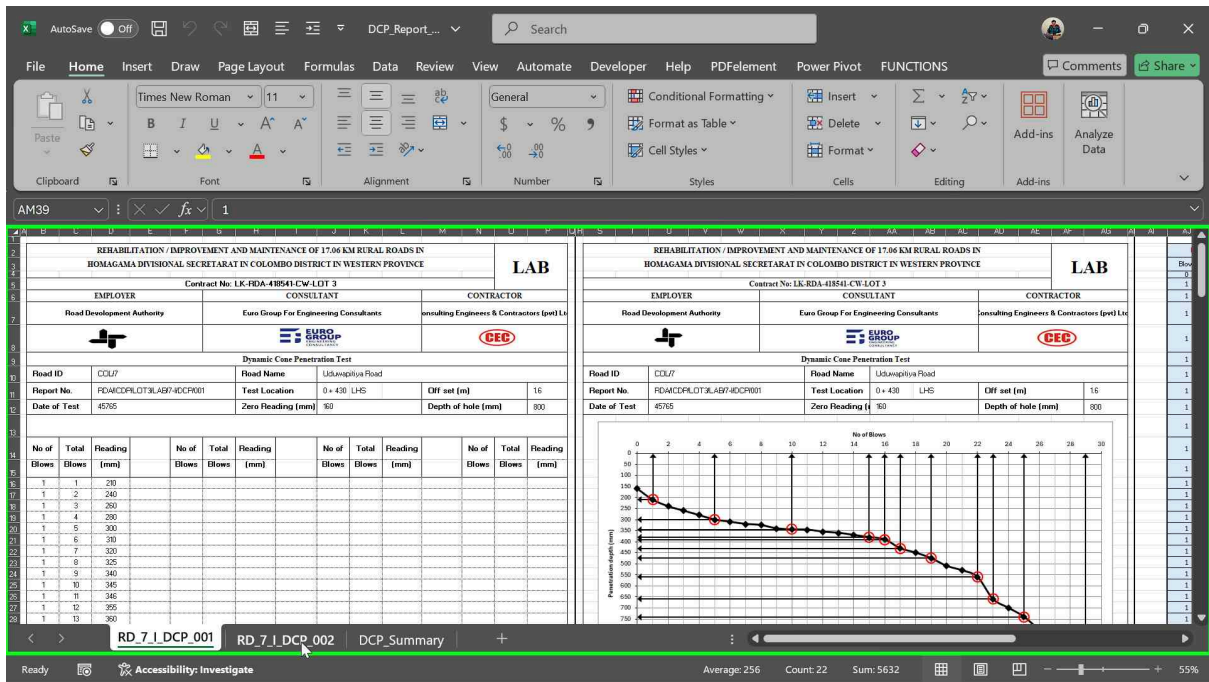
Blows	Penetration
0	160
1	210
1	232
1	255
1	278
1	300
1	308
1	316
1	325
1	333
1	341
1	349
1	357
1	365

Insert zero reading here

7. Access the Final Output

The completed and saved file will be available in the **same folder** where the original automated workbook is stored.





AutoSave Off

DCP_Report...

Search

FileHomeInsertDrawPage LayoutFormulasDataReviewViewAutomateDeveloperHelpPDFelementPower PivotFUNCTIONS

Clipboard

Font

Alignment

Number

Styles

Cells

Editing

Add-ins

Analyze Data

General

Conditional Formatting

Format as Table

Cell Styles

Insert

Delete

Format

Sum

Find & Replace

Comments

Share

B17

=IF(AA17<>"",INDIRECT("\$AA17 & "!\$D\$11"), "")

REHABILITATION / IMPROVEMENT AND MAINTENANCE OF 17.06 KM RURAL ROADS IN HOMAGAMA DIVISIONAL SECRETARAT IN COLOMBO DISTRICT IN WESTERN PROVINCE																					
Contract No: LK-RDA-418541-CW-LOT 3																					
EMPLOYER					CONSULTANT					CONTRACTOR											
Road Development Authority					Euro Group For Engineering Consultants					Consulting Engineering & Contractor (pvt) Ltd.											
																					
ROAD INVESTIGATION DCP TEST RESULT SUMMARY - SHOULDER																					
Road ID		COL7			Road Name		Udumapattana Road														
DCP Results																					
Test Report No	Date of Test	Test Location			CBR																
		Change	Side	Offset (m)	Layer 01 Layer Tk (mm) CBR	Layer 02 Layer Tk (mm) CBR	Layer 03 Layer Tk (mm) CBR	Layer 04 Layer Tk (mm) CBR	Layer 05 Layer Tk (mm) CBR	Layer 06 Layer Tk (mm) CBR	Layer 07 Layer Tk (mm) CBR	Layer 08 Layer Tk (mm) CBR									
RD4/CDPLOT3LAB74DCP001	45765	0+430	LHS	1.6	90.0	11.2	45.0	29.8	35.0	38.8	10.0	26.3	40.0	6.1	45.0	11.2	85.0	8.8	100.0	2.3	8

Test Pit Analysis

Laboratory Tests on Soil Bags (Test Pit Analysis)

Each collected **Soil Bag** is subjected to a standard series of **four laboratory tests**, as required by ICTAD and other road construction guidelines. These tests determine the fundamental properties of the soil, helping engineers classify the material and assess its suitability for use in embankment, sub base, or shoulder layers.

1 Sieve Analysis (SA) – Grading Test

Purpose:

To determine the **particle size distribution** of the soil.

Procedure:

- Dry the sample.
- Pass it through a set of **standard sieves** (e.g., 37.5 mm, 19 mm, 4.75 mm, 425 μm , 75 μm).
- Weigh material retained on each sieve.
- Plot a **grading curve** (percentage passing vs sieve size).

What It Tells Us:

- Classifies the material as **well-graded, poorly graded, or gap-graded**.
- Helps determine if the soil is **suitable for sub base or embankment**.
- Fine content (passing 75 μm) is important for drainage and compaction.

Specification Reference:

- ICTAD specifies grading limits for SB, EMB, and SHO in terms of % passing each sieve.

2 Plasticity Index (PI) – Atterberg Limits Test

Purpose:

To determine the **plastic behavior** of fine-grained soil.

Procedure:

- Conduct **Liquid Limit (LL)** and **Plastic Limit (PL)** tests.
- Compute: $PI = LL - PL$

What It Tells Us:

- **Low PI** → Soil is non-plastic or slightly plastic (good for compaction).
- **High PI** → Soil is clayey and may expand/shrink (problematic for subgrade).

PI Range	Plasticity Description
0–6	Non-plastic to slightly plastic (good)
7–17	Medium plasticity (acceptable for some fills)
>17	Highly plastic (may require stabilization)

Specification Reference:

- ICTAD often requires $PI < 12$ for embankment and < 6 for sub base materials.

3 Proctor Compaction Test – OMC & MDD

Purpose:

To determine the **Optimum Moisture Content (OMC)** and **Maximum Dry Density (MDD)** for compaction.

Procedure:

- Compact soil in layers inside a mould using a standard hammer.
- Vary moisture content in each trial.
- Plot **dry density vs. moisture content** curve.
- Identify peak point:
 - **OMC** = Moisture at peak
 - **MDD** = Peak dry density



What It Tells Us:

- Guides field compaction:
 - Site compaction is usually done to **95–100% of MDD**.
- Shows how sensitive the soil is to moisture during compaction.



Specification Reference:

- ICTAD requires meeting a **minimum % of MDD** (usually $\geq 95\%$) for approval.

California Bearing Ratio (CBR)



Purpose:

To assess the **bearing capacity** of soil, especially subgrade and pavement layers.



Procedure:

- Soak and compact soil sample in a mould.
- Penetrate with a plunger at standard rate.
- Record load vs. penetration.
- Calculate:
$$\text{CBR (\%)} = \frac{\text{Load on specimen}}{\text{Standard Load}} \times 100$$
$$\text{CBR (\%)} = \frac{\text{Load on specimen}}{\text{Standard Load}} \times 100$$

$$\text{CBR (\%)} = \frac{\text{Standard Load on specimen}}{\text{Load on specimen}} \times 100$$



What It Tells Us:

- Indicates how well the soil can **support wheel loads**.
- Used directly in **pavement thickness design**.

CBR (%)	Strength Classification
>30	Very strong (excellent sub base)
10–30	Medium strength (common for SB/EMB)
<10	Weak soil (may require stabilization)



Specification Reference:

- ICTAD generally recommends:
 - **CBR > 8%** for sub base
 - **CBR > 5%** for embankment fills

Summary Table

Test	Main Property Measured	Key Use	ICTAD Focus
Sieve Analysis	Particle size distribution	Grading & classification	% Passing limits
Plasticity Index	Soil plastic behavior	Material suitability & swelling	PI limits
Proctor	OMC & MDD	Compaction control	Achieving required %MDD
CBR	Load-bearing strength	Pavement design	CBR $\geq$ 5%–8% depending on layer

CBR Test

Comprehensive Guide to California Bearing Ratio (CBR) Test

Introduction

The **California Bearing Ratio (CBR) Test** is a widely used penetration test for evaluating the strength of subgrade soil and base materials for road and pavement design. It provides a measure of the load-bearing capacity of soils, guiding engineers in determining the thickness of pavement layers.

Purpose of the CBR Test

- Determine the bearing capacity of soils used in road construction.
 - Classify soils based on strength for subgrade and sub-base layers.
 - Provide data for pavement thickness design as per standards (e.g., RDA, AASHTO, IRC).
-

Types of CBR Test

1. **Laboratory CBR Test** – Conducted on prepared soil samples (remoulded or soaked).
 2. **Field CBR Test** – Performed in-situ directly on compacted subgrade/base.
-

Equipment Required (Laboratory Test)

- CBR mould (150 mm dia. × 175 mm height)
- Spacer disc, collar, and filter paper
- Loading machine (capacity: 50 kN or more)
- Penetration plunger (50 mm dia.)
- Weights for surcharge (if needed)

- Proving ring or load cell
 - Dial gauges for penetration and load
 - Soaking tank (for soaked test)
 - Compaction tools (Proctor rammer, balance)
-

Test Procedure (Laboratory CBR)

A. Specimen Preparation

1. Mix soil at desired moisture content (OMC from Proctor test).
2. Compact the soil in CBR mould in three layers using Proctor hammer.
3. Place filter paper and surcharge weights.
4. For soaked CBR: Soak specimen in water for **4 days**.

B. Testing

1. Place mould under the loading frame with plunger on soil surface.
2. Apply load at 1.25 mm/min penetration rate.
3. Record loads at every **0.5 mm** of penetration up to **12.5 mm**.
4. Plot load vs. penetration curve.
5. Calculate CBR at **2.5 mm** and **5.0 mm** penetration:

$$\text{CBR} = \frac{\text{Test load}}{\text{Standard load}} \times 100\%$$

- Standard load at 2.5 mm = 1370 kg
 - Standard load at 5.0 mm = 2055 kg
 - Take higher value between 2.5 mm and 5.0 mm CBR
-

Interpretation of Results

CBR (%)	Soil Strength	Usage in Road Construction
0–5	Very poor	Not suitable without stabilization
6–10	Poor	Subgrade with improved layer
10–20	Fair	Suitable for sub-base
20–50	Good	Suitable for base course
>50	Excellent	Rare for natural soil

Reporting Format

A standard CBR report should include:

- Sample identification and location
 - Moisture content and dry density
 - Soaking condition (if applicable)
 - Load vs. penetration graph
 - CBR at 2.5 mm and 5.0 mm
 - Final reported CBR value
 - Remarks on suitability for pavement design
-

Precautions

- Maintain proper compaction and moisture level.
 - Ensure uniform surcharge and seating load.
 - Use proper calibration for load and penetration gauges.
 - Avoid disturbances during soaking and testing.
-

✓ Conclusion

The CBR test is a critical tool in highway engineering to assess the structural capacity of soil. It helps determine the appropriate pavement thickness and identify needs for soil improvement. A well-executed CBR test leads to safer and longer-lasting road infrastructure.

This guide serves engineers, lab technicians, and project teams involved in road design and material evaluation.

Sandun

CBR Test in Test Pit Analysis – Complete Guide

What is the CBR Test?

The **California Bearing Ratio (CBR)** test measures the **bearing capacity of soil** and is a critical parameter used in the **design of pavement thickness**.

It evaluates how well the subgrade soil can **resist penetration** under load, mimicking vehicle wheel loads.

Where is CBR Used in Test Pit Analysis?

In **road construction**, test pits are excavated to visually and physically examine the existing subgrade. From these pits:

- **Soil samples** are collected and
- **CBR tests** are performed (either in the field or the lab)

Why CBR is Important in Subgrade Evaluation:

Purpose	Reason
 Pavement Design	CBR is directly used in determining thickness of sub-base and base courses .

 Subgrade Strength	Indicates whether the natural soil can support the pavement layers or needs replacement/improvement.
 Moisture Sensitivity	Helps assess how soil behaves under soaked (wet) and unsoaked conditions.
 Construction Decisions	Low CBR may lead to decisions like soil stabilization , replacement , or thicker pavement layers .

Types of CBR Tests:

1. Laboratory CBR Test

- Performed on **soil samples collected from test pits**
- Samples are **compacted into molds** (usually 3 layers, standard or modified effort)
- Tested under **soaked or unsoaked** conditions

2. Field (In-situ) CBR Test

- Done directly in the test pit bottom
 - Penetration piston pushed into the **natural soil**
 - Reflects **actual field condition**, but less controlled than lab test
-



CBR Test Procedure (Lab – Simplified):

1. Sample Preparation:

- Soil from test pit is **air dried** and **sieved**.
- Compacted in a mold in **3 layers** at **optimum moisture content**.

2. Soaking (if required):

- Mold is soaked for **4 days** (standard for soaked CBR).

3. Penetration Test:

- Load applied via a piston at **1.25 mm/min rate**.
- Load is recorded at **2.5 mm** and **5.0 mm** penetration.

4. CBR Calculation:

$$\text{CBR}(\%) = \frac{\text{Test Load}}{\text{Standard Load}} \times 100$$

$$\text{CBR}(\%) = \frac{\text{Test Load}}{\text{Standard Load}} \times 100$$

- Standard loads:

- 2.5 mm: 1370 kg
- 5.0 mm: 2055 kg



Interpreting CBR Values:

CBR Value (%)	Subgrade Quality	Recommendation
> 20%	Excellent	Suitable for high-traffic roads
10–20%	Good	Suitable with standard pavement
5–10%	Fair	May require sub-base improvement
< 5%	Poor	Soil replacement or stabilization needed



Tip in Test Pit Analysis:

- **Use multiple CBR values** from different test pits per chainage.
- Compute **Equivalent CBR** if multiple layers or values are present.

Example:

If three values for a location:

$\text{CBR}_1 = 6\%$, $\text{CBR}_2 = 9\%$, $\text{CBR}_3 = 12\%$

Then average or weighted CBR may be calculated (based on layer thickness if applicable).



CBR vs DCP:

- DCP test results can be **correlated** with CBR (empirical formulas).
- Faster and cheaper than lab CBR, often used **as screening** or in **dense grid**.

Second time generation

Week 7 – California Bearing Ratio (CBR) Test Summary

This week, I gained hands-on experience with the **California Bearing Ratio (CBR) Test**, one of the most important tests used in road construction to evaluate the **strength of subgrade, subbase, and base course materials**. The CBR value is a critical design input for determining pavement layer thickness.



Purpose of the CBR Test

- To determine the **bearing capacity** of soils and aggregates.
 - To evaluate the **support strength of subgrade** and other pavement layers.
 - To assist in the **design of flexible pavements** using CBR-based methods.
 - Required in both **lab testing** and **field compaction control**.
-



Test Standards Used

- **Laboratory CBR Test:**
 - ASTM D1883
 - BS 1377 Part 4
 - ICTAD Specification Section 1602

- **Field CBR (if conducted in-situ):**
 - Test pit prepared and compacted to field MDD
 - Penetration testing with surcharge weights
-

Laboratory CBR Test – Step-by-Step Procedure

1. Sample Preparation

- Soil sample is air-dried and passed through a 20 mm sieve.
- Compacted in a CBR mould in **3 layers** using **Proctor compaction energy**.
- The **Optimum Moisture Content (OMC)** and **Maximum Dry Density (MDD)** are determined using the Proctor Test.

2. Soaking (If Required)

- The specimen is soaked for **4 days** in water to simulate worst-case conditions (standard for subgrade design).

3. Penetration Test

- A piston is forced into the compacted soil at a rate of **1.25 mm/min**.
 - Loads are recorded at **2.5 mm** and **5.0 mm** penetrations.
 - Load vs. penetration curve is plotted.
-

CBR Calculation

$$\text{CBR (\%)} = \left(\frac{\text{Load on sample at 2.5 or 5.0 mm (N)}}{\text{Standard load (N)}} \right) \times 100$$

Penetration Depth	Standard Load (N)
2.5 mm	1370 N
5.0 mm	2055 N

The higher of the two penetration values is taken as the CBR value, unless 5.0 mm load is much greater.

 CBR Classification (Typical Subgrade Strength)

CBR Value (%)	Subgrade Class	Description
< 2	S1	Very Weak
3–4	S2	Weak
5–7	S3	Medium Strength
8–14	S4	Strong
15–29	S5	Very Strong
≥ 30	S6	Exceptionally Strong

 Site Observations and Relevance

- **Test Pit samples** were collected from various roads and borrow areas.
 - Samples were **prepared and soaked as per RDA requirements**.
 - Subgrade CBR values from the Homagama project typically ranged from **6% to 11%**, indicating **medium to strong** support (S3–S4 class).
 - These values were used in pavement design charts to determine **required pavement thickness**.
-

 Use of CBR in Pavement Design

- CBR value is used along with **traffic class (ESALs)** to determine:
 - Thickness of sub-base and base layers
 - Need for subgrade improvement or stabilization
- If CBR is **<5%**, stabilization or replacement is usually recommended.

Conclusion

The CBR Test is fundamental in pavement engineering. This week helped me understand its full testing cycle, from **sample preparation to penetration testing**, and its **real-world application in flexible pavement design**. The results directly impact layer thickness decisions, making this test a cornerstone of road material evaluation under **ICTAD and RDA** standards.

PI Test

Comprehensive Guide to Plasticity Index (PI) Test in Road Construction

Introduction

The **Plasticity Index (PI)** is a vital parameter in geotechnical and highway engineering. It measures the plasticity range of fine-grained soils and indicates the soil's ability to undergo deformation without cracking or crumbling. This guide outlines the complete procedure, significance, and interpretation of the Plasticity Index Test in road construction.

What is Plasticity Index?

The Plasticity Index (PI) is the numerical difference between the **Liquid Limit (LL)** and **Plastic Limit (PL)** of soil:

$$PI = LL - PL$$

- **Liquid Limit (LL)**: The water content at which soil changes from plastic to liquid state.
 - **Plastic Limit (PL)**: The water content at which soil changes from semi-solid to plastic state.
 - **Plasticity Index (PI)**: Indicates the moisture range within which the soil remains plastic.
-

Importance in Road Projects

- Helps classify soil using the **Unified Soil Classification System (USCS)** or **AASHTO system**.
 - Indicates **expansive behavior**, which is critical for subgrade stability.
 - High PI soils tend to **swell/shrink**, leading to pavement distress.
 - Assists in determining **soil suitability** for embankments, subgrade, and backfill.
-

Equipment Required

- Casagrande liquid limit device
 - Grooving tool
 - Glass plate for rolling threads
 - Spatula and mixing tools
 - Oven (105–110°C)
 - Moisture content containers
 - Balance (accuracy 0.01 g)
-

Test Procedure

The Plasticity Index is determined by measuring both the **Liquid Limit (LL)** and **Plastic Limit (PL)** of the soil. These are found through separate tests described below.

A. Liquid Limit (LL) – Identification Process

Purpose: To find the water content at which the soil changes from a plastic to a liquid state.

Steps:

1. **Prepare Soil Paste:** Take a portion of soil (passed through 425 μm sieve), add distilled water gradually, and mix into a smooth paste.
2. **Fill Casagrande Cup:** Place the soil paste evenly into the brass cup of the Casagrande apparatus.
3. **Form Groove:** Using a standard grooving tool, cut a straight groove through the center of the soil.
4. **Operate Device:** Turn the handle at a rate of 2 revolutions per second so that the cup repeatedly lifts and drops.
5. **Count Blows:** Note the number of blows required for the two halves of the soil to come together for a length of about 13 mm.
6. **Repeat:** Perform at least 4 trials at different water contents (adjusting soil consistency).
7. **Plot Flow Curve:** Plot water content (y-axis) vs. number of blows (log scale, x-axis).

8. **Determine LL:** Read the water content corresponding to **25 blows** from the graph. That is the **Liquid Limit**.

B. Plastic Limit (PL) – Identification Process

Purpose: To find the water content at which the soil transitions from a semi-solid to a plastic state.

Steps:

1. **Prepare Soil Sample:** Take a small quantity of soil (passed through 425 μm sieve) and moisten it.
2. **Roll Threads:** Roll the moist soil on a clean glass plate or tile using your fingers into threads about **3 mm** in diameter.
3. **Observe Behavior:** If the soil starts crumbling at 3 mm diameter, stop rolling.
4. **Collect Sample:** Take this crumbled soil and determine its water content by oven drying.
5. **Repeat:** Conduct the test with at least **three trials** and average the moisture content to get the **Plastic Limit**.

These two water content values allow you to calculate the **Plasticity Index (PI = LL - PL)**, which is crucial for classifying and assessing soil for road construction.

Calculation

$$PI = LL - PL$$

Example:

- Liquid Limit (LL) = 42%
- Plastic Limit (PL) = 22%
- Plasticity Index (PI) = $42 - 22 = 20\%$

Interpretation of PI Values

PI Value	Plasticity Description	Soil Behavior
0–5	Non-plastic	Good for subgrade
6–15	Low plastic	Suitable with good drainage
16–30	Medium plastic	May need stabilization
>30	High plastic	Poor – not recommended

Reporting Format

A standard PI test report should include:

- Sample ID and location
 - Liquid Limit (%), Plastic Limit (%)
 - Calculated Plasticity Index (PI)
 - Soil description (color, texture)
 - Remarks on usability for subgrade
-

Precautions

- Soil should be well mixed and sieved through 425 µm sieve.
 - Avoid rapid drying or excess water during preparation.
 - Maintain consistent groove dimensions in LL test.
-

Conclusion

The Plasticity Index Test is a key indicator of a soil's engineering behavior. In road construction, it aids in assessing **subgrade suitability** and need for **soil stabilization**. Proper execution and interpretation of the test contribute to the long-term durability of pavement structures.

This guide is essential for laboratory technicians, field engineers, and quality controllers involved in highway and pavement construction.

Plasticity Index (PI) Test – Complete Guide

1. What is Plasticity?

Plasticity is the ability of soil, especially fine-grained (clayey) soil, to undergo **permanent deformation without cracking** when water is present.

It is caused by:

- Presence of **clay particles** (which are negatively charged)
 - **Adsorption of water** by clay (water sticks to the clay particles due to polarity)
-

2. What is Plasticity Index (PI)?

The **Plasticity Index (PI)** is the **numerical difference between Liquid Limit (LL) and Plastic Limit (PL)**:

$$PI = LL - PL$$

It represents the **range of water content** over which the soil remains **plastic**.

3. Why is PI Important?

Application	Importance
Soil classification	Helps classify soil in AASHTO/USCS systems
Subgrade design	Indicates strength and volume change potential
Construction behavior	High PI = high shrink/swell risk, poor workability
Drainage	Soils with high PI tend to retain water

◆ 4. Atterberg Limits Overview

Limit	Definition
Liquid Limit (LL)	Water content at which soil just begins to flow
Plastic Limit (PL)	Water content at which soil starts to crumble when rolled into threads
Shrinkage Limit (SL)	Max water content where further drying doesn't reduce soil volume

◆ 5. Test Procedures

📌 A. Liquid Limit (LL) Test (Casagrande Method)

Equipment:

- Casagrande apparatus
- Grooving tool
- Moisture cans
- Balance

Procedure Summary:

1. Air-dry and sieve soil ($< 425 \mu\text{m}$).
2. Mix with water to form a paste.
3. Fill Casagrande cup to ~1 cm depth.
4. Cut groove using tool.
5. Drop cup at ~2 rev/sec until groove closes (10 mm length).
6. Record number of blows and take sample for moisture content.
7. Repeat for 4–5 trials with varying moisture content (blows: 15–35).

8. Plot **flow curve** (moisture vs. log of blows).
 9. From graph, read **LL = water content at 25 blows**.
-

B. Plastic Limit (PL) Test

Equipment:

- Glass plate
- Moisture cans
- Balance

Procedure Summary:

1. Take ~8 g of soil.
 2. Roll on glass plate into thread.
 3. If thread < 3 mm without cracking, knead to reduce water and repeat.
 4. Stop when thread crumbles at 3 mm.
 5. Collect and measure moisture content.
 6. Repeat 3 times. Average = **Plastic Limit (PL)**.
-

6. Sample Calculation

Example:

- Liquid Limit (LL): **35.2%**
- Plastic Limit (PL): **21.1%**
- **PI = 35.2 - 21.1 = 14.1%**

7. Interpreting Plasticity Index (PI)

PI Range	Description	Engineering Behavior
$PI < 7$	Low plasticity	Good subgrade, minimal shrink/swell
$7 \leq PI \leq 17$	Medium plasticity	Moderate strength, some volume change
$PI > 17$	High plasticity	Poor workability, risk of swelling/shrinking

8. Liquid Limit Classification

Liquid Limit (LL)	Description	Notation
$< 35\%$	Low plasticity	L
35–50%	Intermediate	I
50–70%	High	H
70–90%	Very high	V
$> 90\%$	Extremely high	E

Application in Road Construction

- **Subgrade material** with **low to medium PI** is preferred for better strength and durability.
- **High PI soils** may need stabilization (lime, cement) or replacement with better material.

Proctor Compaction Test

Proctor Compaction Test – Determining OMC & MDD

Purpose

The **Proctor Compaction Test** is a laboratory method used to determine the **Optimum Moisture Content (OMC)** at which a soil achieves its **Maximum Dry Density (MDD)**. In road construction and geotechnical engineering, achieving the correct compaction is essential for:

-  Ensuring **stability and bearing capacity**
-  Reducing **permeability and water infiltration**
-  Minimizing **settlement and erosion**

The test provides a **reference compaction curve** that guides field engineers to achieve specified compaction levels (typically $\geq 95\%$ of MDD) during site work.

Compactive Energy Calculation

The **compactive effort (E)** is the energy per unit volume applied during the test:

$$E = \frac{W \times H \times N \times L}{V}$$

Where:

- **W** = Hammer weight (kg)
- **H** = Drop height (m)
- **N** = Number of blows per layer
- **L** = Number of layers
- **V** = Mold volume (m^3)

Example Values:

- Standard Proctor:
 $W = 2.5 \text{ kg}, H = 0.305 \text{ m}, N = 25, L = 3, V = 0.000943 \text{ m}^3$

- Modified Proctor:
Higher weight, height, and layers
⇒ $E \approx 2700 \text{ kJ/m}^3$

Test Standards

The Proctor test is standardized in two main versions: the Standard Proctor (ASTM D698, AASHTO T99) and the Modified Proctor (ASTM D1557, AASHTO T180). Both follow the same fundamental procedure but with different compaction energies. The Modified Proctor was introduced in 1958 to cater to higher compaction needs (e.g. for heavy aircraft runways and highways), because soils that met Standard Proctor densities were found to still deform under heavier loads. The table below summarizes the key differences between the standard and modified Proctor tests:

Test Type	Standard Proctor	Modified Proctor
Standard	ASTM D698 / AASHTO T99	ASTM D1557 / AASHTO T180
Hammer Weight	2.5 kg	4.5 kg
Drop Height	305 mm (12 in)	457 mm (18 in)
No. of Layers	3	5
Blows per Layer	25	25
Compactive Energy	$\sim 593 \text{ kJ/m}^3$	$\sim 2700 \text{ kJ/m}^3$
Use Case	Light-duty compaction (e.g., residential)	Heavy-duty compaction (e.g., highways, airports)

The Modified Proctor uses a hammer roughly twice as heavy dropped from a higher height, and compacts the soil in more layers. As a result, it delivers about $4\frac{1}{2}$ times the energy of the Standard Proctor. The increased compactive effort produces a higher soil density and typically a lower optimum moisture for a given soil. In practical terms, the Modified Proctor better represents the compaction achievable by modern heavy compaction equipment (and is often specified for airport pavements, highways, and other heavily loaded structures), whereas the Standard Proctor is used for lighter compaction requirements (such as residential or minor works) or as a baseline for general fills. Engineers choose which Proctor standard to use based on project specifications and the level of compaction needed.

Equipment Used

1. Compaction mold with base
2. Manual rammer (2.5 kg or 4.5 kg)
3. Scale (mass measurements)
4. Extruder (to remove compacted soil)
5. Mixing pan
6. Trowel/Scoop
7. Drying oven (~105–110°C)
8. No. 4 sieve
9. Moisture cans
10. Graduated cylinder
11. Straight edge

Test Procedure (Step-by-Step)

1. **Sample Preparation**
 - Air-dry soil, pass through No. 4 sieve
 - Divide into 4–6 subsamples for varying moisture contents
2. **Water Addition**
 - Add specific amounts of water to each subsample
 - Mix thoroughly to ensure uniform moisture
3. **Compaction**
 - Place soil in mold in specified layers (3 or 5)
 - Compact each layer with standard blows using the hammer

- Trim excess soil flush with mold top

4. Weighing & Moisture Sampling

- Weigh mold with compacted soil (record moist weight)
- Extract soil and take top & bottom samples for moisture content
- Dry in oven at 105–110°C for 24 hours

5. Repeat

- Increase moisture by ~2% each trial
- Repeat compaction until dry density decreases → covers full compaction curve



Data Analysis

1. Calculate Moisture Content

$$w = \frac{W_{\text{wet}} - W_{\text{dry}}}{W_{\text{dry}}} \times 100\% \quad w = \frac{W_{\text{wet}} - W_{\text{dry}}}{W_{\text{dry}}} \times 100\%$$

2. Calculate Wet & Dry Densities

$$\rho = \frac{W_{\text{moist}}}{V}; \rho_d = \frac{W_{\text{dry}}}{V} \quad \rho = \frac{W_{\text{moist}}}{V}; \rho_d = \frac{W_{\text{dry}}}{V}$$

- ρ = Wet density (g/cm³)
- ρ_d = Dry density (g/cm³)
- w = Moisture content (decimal form)

3. Plot the Compaction Curve

- **x-axis:** Moisture Content (%)
- **y-axis:** Dry Density (g/cm³)
- Identify:
 - **OMC** → Peak moisture content

- **MDD** → Maximum dry density

Zero Air Voids Curve

Also plot the **Zero Air Voids (ZAV) Curve**, which represents the theoretical max density if all air voids are removed.

$$w_{sat} = \frac{(\rho_w \rho_d - 1) G_s}{\rho_w (G_s - 1)} \times 100$$

$$w_{sat} = \left(\frac{\rho_w}{\rho_d} - \frac{1}{G_s} \right) \times 100$$

Where:

- ρ_d = dry density
- G_s = specific gravity of soil solids (usually ≈ 2.65 – 2.70)
- ρ_w = density of water ($\approx 1.0 \text{ g/cm}^3$)

The actual curve should always fall **below** the ZAV line.

Key Outputs

- **OMC (%)** – Optimum moisture for best compaction
- **MDD (g/cm³)** – Maximum achievable dry density
- **Compaction Curve** – Dry density vs. moisture content
- **ZAV Curve** – Theoretical saturation reference

Why It Matters

- Field compaction specs (e.g., 95% MDD) are based on this lab result
- Guides site team whether to **add or remove moisture**
- Directly influences **pavement strength, durability, and drainage**
- Applicable to:
 - Subgrade

- Sub Base (SB)
- Embankment (EMB)
- Shoulder (SHO)

SA

What is Effective Size (d_{10})?

- The **effective size** is the **particle diameter** at which **10% of the soil** (by weight) is **finer** (passed) in the grain size distribution curve.
 - It's a **critical indicator** of **drainage**, **permeability**, and **soil behavior**.
-

Why d_{10} Is Important in Road Subgrade Analysis:

1. Permeability Indicator

- d_{10} is directly related to **permeability (k)** of granular soils (especially sandy soils).
- **Higher d_{10} → higher permeability**, which is good for **drainage** and avoiding **water retention** in the subgrade.

2. Foundation Suitability

- A well-graded subgrade (with suitable d_{10} , C_{1u} , and C_c) provides better **load distribution** and **reduced settlement**.
- Helps assess whether the subgrade can support the pavement layers without excessive deformation.

3. Used in Soil Classification

- Part of the criteria for **USCS (Unified Soil Classification System)**:
 - Determines if soil is **well-graded** or **poorly graded**.
 - Helps classify as GW, SW, etc.

4. Empirical Design Parameters

- Some **empirical correlations** (e.g., Hazen's formula for permeability) use d_{10} :

$$k = C \times d_{10}^2$$

(where k is permeability and C is a constant depending on units)

5. Filter Design & Drainage Layers

- In pavement design, **filter layers** and **drainage layers** often use d_{10} criteria to prevent **migration of fine particles**.

✓ Summary

Aspect	Importance of d_{10}
Drainage	Higher $d_{10} \rightarrow$ better permeability
Soil Strength	Linked to compaction and shear behavior
Classification	Helps define grading and soil group
Pavement Performance	Ensures stability and avoids pumping
Filter Design	Prevents soil clogging and piping

◆ Uniformity Coefficient (C_u)

✓ Formula:

$$C_u = \frac{d_{60}}{d_{10}}$$

📌 Meaning:

The **Uniformity Coefficient (C_u)** indicates the **range of particle sizes** in a soil sample. It tells us whether the soil is:

- **Well-graded** (particles of many sizes), or
- **Poorly graded** (particles are mostly similar in size).

⚠️ Why It's Important:

Value of C_u	Soil Type	Interpretation
$C_u < 4$ (for gravel) or < 6 (for sand)	Poorly graded	Uniform size particles – bad for compaction and strength

$C_u \geq 4$ (gravel), ≥ 6 (sand)

Well-graded Good mix of particle sizes – **better compaction, density, and strength**

In **subgrade evaluation**, a higher C_u generally indicates **better load-bearing capacity** and **improved stability**.

Coefficient of Gradation (C_k or C_c)

Formula:

$$C_k = \frac{d_{30}^2}{d_{60} \times d_{10}}$$

Meaning:

The **Coefficient of Gradation (C_k)** (also known as **Coefficient of Curvature**) evaluates the **smoothness and shape** of the gradation curve between d_{10} and d_{60} .

Why It's Important:

Value of C_k	Interpretation
$1 \leq C_k \leq 3$	Well-graded soil
Outside this range	Poorly graded soil (gap-graded or skewed distribution)

Use in Road Subgrade:

- When both **C_u** and **C_k** are within desirable limits, the subgrade soil is considered **well-graded**, which is ideal for road construction.
 - Well-graded soils:
 - Achieve **higher density** on compaction
 - Offer **better load distribution**
 - **Minimize settlement** over time
-

✓ Summary Table:

Parameter	Formula	Indicates	Good Range
Cu	$C_u = \frac{d_{60}}{d_{10}}$	Soil grading spread	$Cu \geq 4$ (gravel), $Cu \geq 6$ (sand)
Ck (Cc)	$C_k = \frac{d_{30}^2}{d_{60} \times d_{10}}$	Curve shape & gradation	$1 \leq Ck \leq 3$
d₁₀	Effective size	Permeability, fines control	Higher d ₁₀ → Better drainage

Material Sourcing

Borrow Pit Material

When sourcing Sub Base (SB), Embankment (EMB), and Shoulder (SHO) materials from a Borrow Pit, the process is as follows:

Step 1: Finding the Borrow Area

- Identify suitable **Borrow Area** with required licenses.
 - The license must belong to the project company or authorized supplier.
-

Step 2: Borrow Pit Investigation (Initial Testing)

- **Lab helpers** collect **three soil samples ("Soil Bags")** from three different locations within the borrow area. ICTAD has which area should be covered.
 - Conduct essential lab tests for each soil bag:
 -  **SA** (Sieve Analysis)
 -  **PI** (Plasticity Index)
 -  **Proctor Test** (Optimum Moisture Content & Maximum Dry Density)
 -  **CBR Test** (California Bearing Ratio)
 - If all tests pass according to **ICTAD specifications**,
 -  A **Material Approval** (called Borrow pit approval) is sent to **RDA**
 -  If **RDA approves**, the material  **can be transported to the stockpile**
-

Step 3: Material Transportation

- Materials are transported using a **3 cubic meter (m³) lorry** to the company's stockpile site.

- Ensure clear separation of different material types in stockpiles:
 - 🪨 Upper Sub Base (SB Type 1)
 - 🪨 Lower Sub Base (SB Type 2)
 - 🌱 Embankment Material (EMB Type 1)
 - 🌱 Embankment Material (EMB Type 2)
 - 🛣️ Shoulder Material (SHO)
 - 🚧 Aggregate Base Course (ABC) (this is not borrow pit)
- Store the materials separately and neatly as per **ICTAD guidelines**.

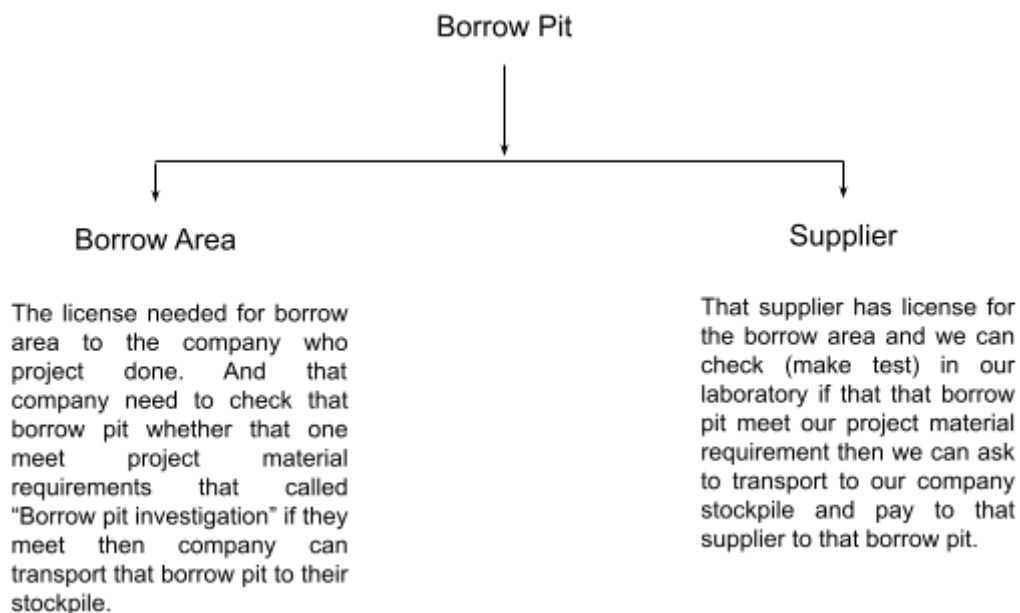
Step 4: Stockpile Testing (Confirmation Testing)

- Before using the material at the site, conduct **confirmation testing** again from stockpile samples as per **ICTAD Sec.1602**.
- ICTAD clearly mentions the testing **frequency and specification limits** that must be followed.
- If these **bulk tests** pass again,

📦 **A Material Approval** (called stockpile approval) is sent to the **RDA**

✅ Upon **RDA Approval**, the material is cleared for **site usage**

Important Points on Supplier vs. Own Borrow Area



❶ Own Company Borrow Area:

- License held directly by the company.
- Company directly responsible for borrow pit investigation.
- Once tests pass, directly transport material to its own stockpile.

❷ Supplier's Borrow Area:

- Supplier holds the borrow pit license.
- Supplier provides their own test pass reports.
- **BUT:** Company must independently verify tests in their own lab before approval.
- If verified tests pass, the supplier is instructed to transport the material to the company's stockpile.
- Payment is made to the supplier only after verifying materials that meet ICTAD requirements.
- Once material is delivered to the stockpile, the rest of the procedure (confirmation testing, storing separately, ICTAD compliance) remains the same.

Stockpile Material Approval Procedure

Before any construction material (e.g., SB, EMB, SHO, DGAB, etc.) is used, it must undergo a formal approval process from RDA. This process ensures compliance with ICTAD Specifications and project-specific requirements. So we **cannot use it directly** at the site until it goes through the following important steps:



1. Material Sampling and Laboratory Testing (Bulk Material Testing)

- Once the material is delivered or prepared (e.g., in a stockpile or direct from the source(borrow pit area)), representative samples are collected for testing.
- **Laboratory** tests are carried out according to **ICTAD Specifications** (Section 1602).

- Typical tests:
 -  **Grading (Sieve Analysis)**
 -  **Plasticity Index (PI)**
 -  **Proctor Compaction**
 -  **CBR** (California Bearing Ratio)
 -  **Moisture Content & Specific Gravity** (sometimes) depending on material type.
-

2. Compare with ICTAD Standards

- The lab results are compared with **project material specifications**.
 - ICTAD gives specific **limits and tolerances** for each material (e.g., allowable % passing in sieve sizes, PI limits, etc.).
-

3. Source Approval Request

If test results meet all relevant standards:

- A formal approval request is submitted to the Engineer or relevant authority (e.g., RDA).
 - This is a **formal step** to **document that the material is acceptable** for use on the project.
 - The source or stockpile may be inspected as part of this process.
-

4. Approval Granted → Ready for Site Use

- Once the material is **approved**, it's added to the **"Approved Source List."**
- It becomes eligible for use in site works such as embankment filling, sub-base laying, etc.

In Case of Failure

If the material fails to meet specifications:

- It **cannot be used** at the site.
- The material must be either removed, reprocessed, or retested until it passes the necessary criteria.

Stockpile

ICTAD (sec 105, 202, 505 A, 505B, 506, 507, 801)

When storing road construction materials like **Upper Sub-Base (USB Type 1)**, **Lower Sub-Base (LSB Type 2)**, **Embankment Material (EMB Type 1 & 2)**, **Shoulder Material (SHO)**, and **Aggregate Base Course (ABC)** in **stockpiles**, there are **standard procedures** — especially under **ICTAD**, **RDA**, or **general highway engineering practice** — to ensure quality and avoid contamination.

General Procedure for Stockpiling Road Construction Materials:

1. Site Selection

- Choose a **dry, well-drained area**.
 - Preferably near the site, but **away from water bodies** and **flood-prone zones**.
 - Proper access for loading/unloading.
-

2. Separation & Marking

- **Each material must be stored separately.**
 - **Provide clear signboards** with:
 - Material Type (e.g., USB Type 1)
 - Source (quarry name)
 - Date of delivery
 - Test status (e.g., “Approved” / “Rejected”)
-

3. Stockpile Shape & Height

- Follow **conical** or **trapezoidal** shape (not elongated piles).
- Maximum height:

- Usually not more than **3 meters**.
 - Avoid steep sides — helps prevent **segregation**.
-

4. Base Preparation

- Stockpiles should be built on a **hard, clean surface**:
 - Compacted earth
 - Concrete pad
 - Geo-membrane / tarpaulin (if on soft ground)
 - This prevents **mixing with soil or mud**.
-

5. Drainage & Protection

- Provide **side drains** around the stockyard.
 - Use **tarpaulin or plastic sheets** in rainy weather.
 - Avoid material washing away or becoming soaked.
-

6. Sampling & Testing

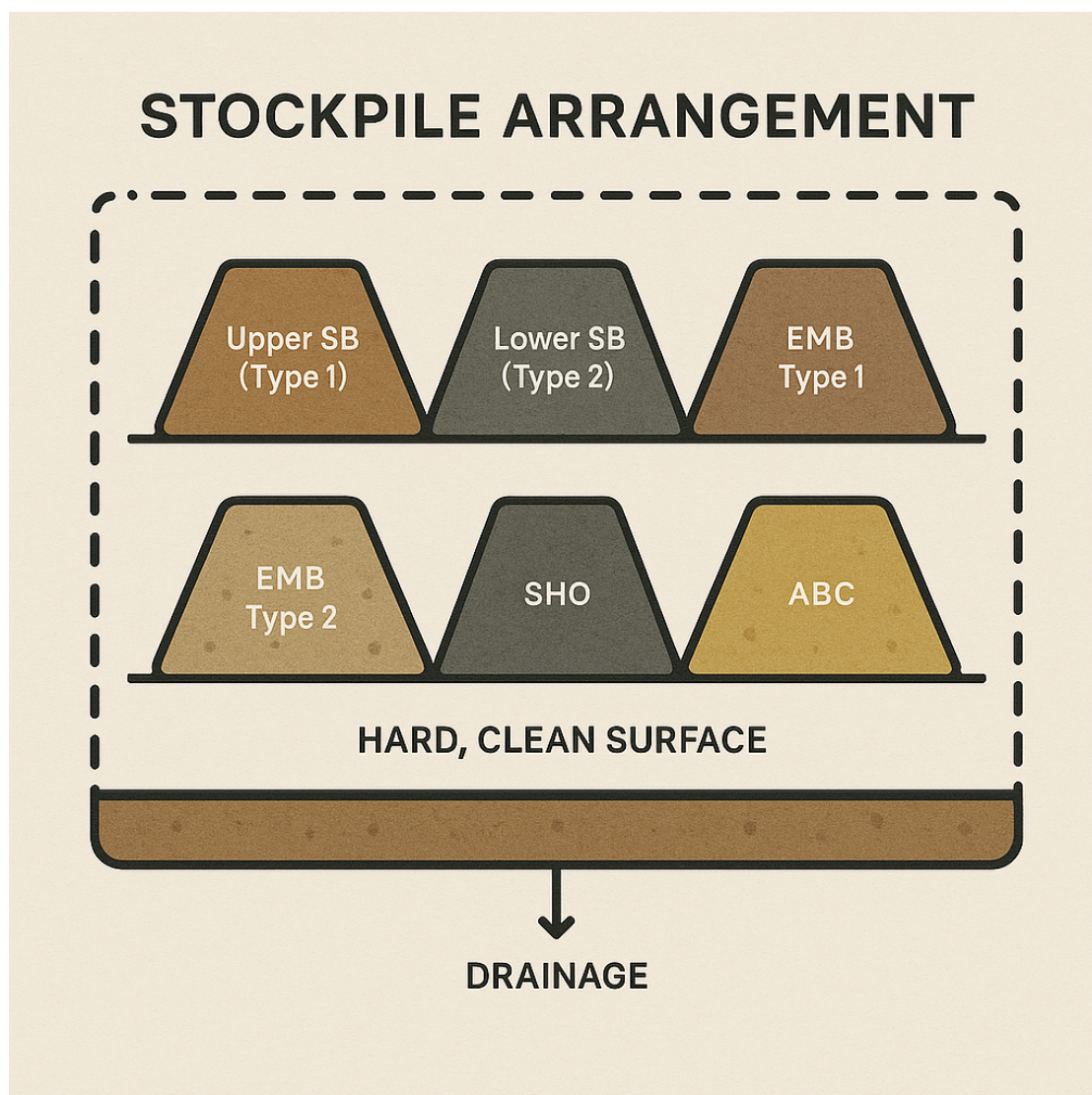
- Before use, take samples for:
 - **Gradation**
 - **Plasticity Index**
 - **Moisture Content**
 - **CBR/Proctor**, etc.
- Store tested and approved material separately if needed.

7. Avoid Mixing

- Maintain a **minimum 1–1.5 m gap** between different materials.
 - If using machines to move materials, clean the buckets/blades before switching material types.
-

Bonus Tip: Sequence of Use

Use the “**First In, First Out**” (FIFO) method so older materials are used first, especially for **moisture-sensitive** layers.



Those are **commonly stored materials** in roadway construction stockpiles.

✓ **ABC – Aggregate Base Course**

- Crushed stone or gravel.
- Used as a **structural layer** below the asphalt.
- Provides **strength and drainage**.
- Typically well-graded and compacted.

✓ **SB – Sub Base**

- Lower quality material than ABC.
- Placed **below ABC**.
- Provides additional **load distribution** and **drainage**.
- Can be natural gravel, sand, or other approved material.

✓ **EMB – Embankment Material**

- Used to **build up the road** to the desired elevation.
- Usually made of **soil, rock, or borrowed material**.
- Needs to be **well-compacted**.
- Can also include **cut-and-fill operations**.

✓ **Shoulder Material**

- Placed along the **sides of the pavement**.
- Can be the same as ABC, or a **less expensive material** like gravel or crushed rock.
- Provides **lateral support** to the pavement.
- Also helps in **drainage and safety**.



Method Statement for Stockpiling Road Construction Materials

Title: Stockpiling of Construction Materials (USB, LSB, EMB, ABC, SHO)

Location: [Project Name / Site]

Prepared by: [Site Engineer / Contractor]

Date: [DD/MM/YYYY]

1. Scope

This method statement outlines the procedure for proper receiving, storing, and maintaining road construction materials to prevent contamination, ensure quality, and support efficient site operations.

2. Materials Covered

- Upper Sub-Base (USB) Type 1
- Lower Sub-Base (LSB) Type 2
- Embankment Material (EMB) Type 1 & 2
- Shoulder Material (SHO)
- Aggregate Base Course (ABC)

3. Responsibilities

Role	Responsibility
Site Engineer	Supervise location, separation, and compliance.
Storekeeper	Record deliveries and quantities.
QA/QC Engineer	Approve material quality and testing.
Machine Operator	Handle material loading/unloading safely.

4. Equipment Used

- Dump trucks
 - Front-end loader or backhoe
 - Surveying tools
 - Signboards
 - Tarpaulins (during rain)
-

5. Procedure

a. Site Preparation

- Clear and level ground.
- Provide compacted base or tarpaulin layer.
- Construct side drains for rainwater diversion.

b. Stockpile Arrangement

- Assign specific areas for each material with a minimum 1m gap.
- Display signboards showing material type, source, and test status.
- Pile height should not exceed 3m to avoid segregation.

c. Material Unloading

- Unload approved materials only.
- Visually check for contamination or mixing.

d. Moisture Management

- During rain, cover piles with tarpaulin.

- Avoid water stagnation around piles.

e. Testing and Approval

- Collect random samples as per ITP.
 - Do not use materials until test results are approved.
-

6. Safety Measures

- Use PPE (helmets, boots, gloves) during loading/unloading.
 - Maintain clear paths for machinery.
 - No unauthorized persons in stockyard.
-



Stockpile Inspection Checklist

Item	Yes/No	Remarks
Approved location used?		
Signboards in place?		
Materials clearly separated?		
Stockpile height < 3m?		
Tarpaulin used during rain?		
Drainage around stockpile okay?		
No contamination observed?		
Samples tested and approved?		
FIFO maintained?		

ABC Material

  **ABC Material is NOT from Borrow Pit**  

 **What is ABC?**

ABC – Aggregate Base Course

- A **crushed stone or gravel material** used as a **strong foundation layer** under asphalt.
 - Provides **load-bearing strength** and good **drainage**.
-

 **Where Does ABC Come From?**

-  **NOT** collected from natural borrow pits like SB, EMB, or SHO.
 -  Usually sourced from:
 -  **Quarries**
 -  **Crusher plants**
 -  **Licensed aggregate suppliers**
-

 **How is it Supplied?**

- Delivered in bulk by **truckloads** from **approved quarries** or **crusher plants**.
- Must meet the **grading, strength, and durability specs** in **ICTAD**.
-  **Lab Testing** required for:
 - Sieve Analysis (Grading) - SA 
 - Proctor Test – for OMC and MDD 
 - Plasticity Index (PI) – should be non-plastic or within limits 
 - California Bearing Ratio (CBR) – check bearing capacity 

- Flakiness Index (FI) – control flaky particles 📏
- Aggregate Impact Value (AIV) – toughness to impact 💣
- Los Angeles Abrasion Value (LAAB) – resistance to abrasion 🧱
- Ten Percent Fines Value (TFV) – crushing resistance 💣
- Aggregate Crushing Value (ACV) – strength check 🪨
- Specific Gravity & Water Absorption – density and durability check 💧

🏗️ DGAB – Dense Graded Aggregate Base ✅

📖 Full Form:

DGAB = Dense Graded Aggregate Base

📌 Key Details:

- DGAB is a type of **ABC (Aggregate Base Course)** but more specific.
 - Called “*dense graded*” because:
 - It contains a **wide range of aggregate sizes**, from fine dust to coarse stones.
 - These mix together tightly to form a **strong, well-compacted base**.
-

🧱 DGAB Properties:

Property	Description
Material Type	Crushed stone / gravel
Grading	Wide range (dust to ~37.5 mm)
Strength	High load-bearing
Compaction	Very good due to dense grading

Usage	Base layer for roads, before asphalt
Source	Quarry / Crusher plant
Lab Tests	Grading (SA), LAA, ACV, SG, etc.

✅ Why It's Better Than Regular ABC

- DGAB offers better **compaction**, **stability**, and **drainage** than loose or poorly graded ABC.
 - It's the **preferred base material** in many national standards and is mentioned in **ICTAD specs**.
-

💡 Fun Fact:

Some places still use "ABC" as a general term, but **DGAB** is the **more accurate engineering term** for base layers that meet proper gradation and compaction standards.

Stone-Related Material Terms

Stone-Related Material Terms

1 Aggregate (Agg) – The General Term

- **Definition:** A general name for **granular materials** used in construction.
 - **Includes:** Crushed stone, gravel, sand, recycled concrete, etc.
 -  **Used in:** Concrete, asphalt, road base, drainage, etc.
-

2 Fine Aggregate (Fine Agg)

- **Definition:** Aggregates that **pass through a 4.75 mm sieve**.
 - **Examples:** Natural sand, crusher dust, manufactured sand.
 - **Use:** Fills voids between coarse aggregates, used in concrete and asphalt.
-

3 Coarse Aggregate (Coarse Agg)

- **Definition:** Aggregates that **stay on a 4.75 mm sieve**.
 - **Examples:** Crushed stone (10 mm, 20 mm, 40 mm), gravel.
 - **Use:** Major component of concrete and base layers.
-

4 Gravel

- **Definition:** Natural rounded stone, usually from riverbeds.
- **Shape:** Round or sub-rounded (not angular).
- **Use:** Drainage layers, sub base (if permitted), sometimes in concrete.

-  May be **less preferred** than crushed stone in road bases due to round shape (less interlock).

5 Crushed Stone (Manufactured Aggregate)

- **Definition:** Rock mechanically crushed in quarries.
- **Shape:** Angular with sharp edges.
- **Use:** Stronger interlocking → ideal for **DGAB / ABC**, concrete, asphalt.

6 ABC (Aggregate Base Course) / DGAB

- **Definition:** A specific mixture of **coarse and fine crushed stone** to create a **dense base layer**.
- **Also called:** DGAB (Dense Graded Aggregate Base).
- **Use:** Placed under asphalt, pavers, or concrete to carry load.

Quick Comparison Table

Term	Size Range	Shape	Natural/Crushed	Common Use
Aggregate	Any size	Varies	Natural/Crushed	General term for stone materials
Fine Aggregate	< 4.75 mm	Fine sand-like	Both	Concrete, asphalt, filler
Coarse Agg	> 4.75 mm	Gravel/stone	Both	Concrete, road layers
Gravel	Usually 5–50 mm	Rounded	Natural	Sub base, drainage

Crushed Stone	Varies	Angular	Crushed	Strong base layers, concrete, asphalt
ABC / DGAB	0–37.5 mm (mixed)	Angular & fines	Crushed	Base layer below asphalt

Summary

-  **Aggregate (Agg)** = General term for stone/sand mix
-  **Fine Agg** = Small particles like sand
-  **Coarse Agg** = Bigger pieces like gravel/stones
-  **Gravel** = Natural, rounded stone
-  **Crushed Stone** = Angular, strong material from quarries
-  **ABC / DGAB** = Specific mix of fine + coarse crushed stone used as base layer

Gravel and soil

The difference between **gravel** and **soil** mainly lies in **particle size**, **composition**, and **engineering behavior**:

♦ 1. Particle Size

- **Gravel:**
 - Coarse particles > 2 mm in diameter
 - Includes **pebbles, cobbles, and boulders**
- **Soil:**
 - Mixture of **smaller particles** like:
 - **Sand** (0.075–2 mm)
 - **Silt** (0.002–0.075 mm)
 - **Clay** (< 0.002 mm)

♦ 2. Composition

- **Gravel:**
 - Mostly **rock fragments** (quartz, granite, limestone, etc.)
 - Very low in organic matter
- **Soil:**
 - Contains **minerals, organic material, water, and air**
 - Can support plant growth due to nutrients and organics

♦ 3. Water Behavior

- **Gravel:**
 - **Drains quickly**, low water retention
 - Doesn't compact easily
- **Soil:**
 - **Holds water**, especially if clay is present
 - May compact and swell depending on moisture content

◆ 4. Engineering Properties

- **Gravel:**
 - High **bearing capacity**, good for **road bases and foundations**
 - Low plasticity (doesn't deform easily)
- **Soil:**
 - Strength varies with type (e.g., clay vs. sand)
 - Often needs stabilization in construction

◆ 5. Classification (Geotechnical Engineering)

According to the **Unified Soil Classification System (USCS)**:

- **Gravel** = Coarse-grained soil with >50% retained on No. 4 sieve
- **Soil** = May refer broadly to **all unconsolidated materials**, but often implies **finer particles** like sand, silt, and clay

Tab 30

During this week, I was actively involved in understanding how road construction materials are sourced, tested, and managed before they are used on-site. This includes material investigation from borrow pits, testing compliance with ICTAD standards, and maintaining a well-organized stockpile system.

Borrow Pit Investigation

To source materials like Sub Base (SB), Embankment (EMB), and Shoulder (SHO), the company can either:

1. **Use Own Borrow Area** – Must hold a valid license.
2. **Use a Supplier's Borrow Pit** – Supplier must provide license and sample access.

For either case, soil samples are collected from different locations within the pit and tested for:

- **Sieve Analysis (SA)**
- **Plasticity Index (PI)**
- **Proctor Compaction Test**
- **California Bearing Ratio (CBR)**

Only if results meet ICTAD requirements, **Borrow Pit Approval** is submitted to the RDA for clearance.

Site Selection & Supplier Testing

If using a **supplier's source**, we independently verify the material through our lab testing before accepting any delivery. Upon passing results, the supplier is authorized to deliver material to our site stockpile. Payment is made **only after approval**.

For **own-source borrow pits**, the company performs all testing internally and directly manages transport and stockpile buildup.

Stockpile Management

Once materials are transported to our project site, they are stored in separate, clearly identified stockpiles to prevent cross-contamination. Before being used in road layers, each material is subjected to **confirmation (bulk) testing** from the stockpile. These tests again include:

- Grading
- PI
- CBR
- Proctor Compaction

Approval is then sought from the RDA to include the stockpile material in the “**Approved Material List.**”

Material Types & Specifications

All materials must follow project-specific and ICTAD (Section 1602) standards.

Material Type	Sub-Types	Notes
Sub Base (SB)	Type I & Type II	Used below base layer. Type based on CBR & grading requirements.
Embankment (EMB)	Type I & Type II	Selected based on strength and plasticity criteria.
Shoulder (SHO)	–	Must ensure stability and drainage along pavement edge.
ABC	–	Not from borrow pit; sourced from quarry. Requires strict grading.
Concrete Aggregate	Coarse & Fine	Used in concrete works; must meet specific gravity & shape tests.
Sand (Fine)	River Sand, Quarry Dust, M-Sand	Depends on site availability and spec; tested for cleanliness & gradation.

Conclusion

Week 9 gave me deep exposure to the material approval workflow—starting from borrow pit selection or supplier testing, to ICTAD-compliant lab evaluations, and proper stockpile management. Understanding this process is vital for quality control and smooth construction progress.

Industrial Problem

Project Summary – Asphalt Thickness Reduction Study

Our company, Consulting Engineering and Contractors (Pvt) Ltd., is currently undertaking a road construction project in Homagama, Sri Lanka. As part of this project, the Road Development Authority (RDA) has requested a study to assess the feasibility of reducing the asphalt layer thickness as a cost-saving measure.

The project consists of four lots, and we are responsible for Lot 2, which includes the road segment labeled COL/2, spanning approximately 4.7 kilometers. For this study, we have selected two separate 450-meter sections from different locations along the COL/2 road to carry out the research.

The RDA's main objective is to reduce construction costs, with one proposed method being the reduction of asphalt layer thickness. However, before proceeding, we aim to evaluate other cost-reduction options and examine their practicality within the Sri Lankan context. The goal is to identify whether asphalt thickness reduction is the most effective and feasible solution. My idea is that asphalt is the highest cost in road construction. my plan is make a asphalt mix design reducing highest valuable materials

Currently, a 40 mm thick asphalt layer is being applied. The RDA has proposed trialing two reduced thicknesses: 25 mm and 30 mm. For this study, one section will be constructed using a 25 mm layer, and the other with a 30 mm layer. I have selected the 25 mm section for further evaluation.

I am a trainee material engineer at the company, and this is the project I have been assigned for my internship.

General Knowledge

Road Project General Information:

This page contains general information about the road project.

Earth Drain, U-Drain, and Leaderway

Difference Between Earth Drain, U-Drain, and Leaderway (Leederway)

1. Earth Drain

A simple, unlined open channel dug into the earth.

- Key Features:
- Made by excavating soil.
- Cheapest option.
- Common in rural areas or temporary setups.
- Prone to erosion, siltation, and weed growth.
- Requires regular maintenance.
- Use:
- Used where the budget is tight and stormwater flow is minimal or temporary.

2. U-Drain

A concrete channel shaped like a "U", used to carry water.

- Key Features:
- Precast or cast-in-situ concrete.
- Smooth surface allows faster and more controlled flow.
- Durable and long-lasting.
- Can be covered or left open.
- Less maintenance than earth drains.
- Use:
- Ideal in urban areas, roads, or industrial zones where a clean, controlled drainage system is needed.

3. Leaderway (Leederway)

A lined side drain (stone/brick/concrete) that runs alongside roads to guide surface water to a drain or natural outlet.

- Key Features:
- Usually lined with concrete, stone, or bricks.
- Positioned to collect water from the pavement/road shoulder.
- Designed with proper gradient and alignment.
- Sometimes includes U-drains or other types of channels.
- Use:
- Acts as a transition path for stormwater, collecting water from surfaces and directing it to the main drain or culvert.

Comparison Table

Feature	Earth Drain	U-Drain	Leaderway (Leederway)
Material	Soil only	Concrete	Brick, stone, or concrete lining
Cost	Low	High	Medium to high
Durability	Low	High	Medium to high
Maintenance	High	Low	Moderate
Usage	Rural/Temporary	Urban/Permanent	Roadside drainage system
Water Flow Control	Poor	Excellent	Good

Bonus Tip

Sometimes a U-drain is used as a leaderway in modern designs—especially in cities—because it's more efficient and neater.

Retaining wall

The retaining wall project was an integral part of the site's design, **to stabilize the slope and manage water flow effectively.**

75 mm THK. FILTER LAYER

MINIMUM 600mm THICK 50-200 DRY STONE LINING

Ø 110mm (TYPE 600) PVC PIPE WEEP HOLES AT 1500 CRS. STAGGERED (300x600) CLAY PUDDLE

CONC. GR. 15(19)

CONC. GR. 25(20)

75mm THK. SCREED CONCRETE

600 600

1200 1200 1250 1850

G.L.

150 350 650 300 75

VARIES

1100

Y10-01-200 Y10-02-300

SCALE 1:50

- **Material Used:** 75 mm thick filter material (granular type).
- The filter layer is used to prevent soil from clogging the drainage system while allowing water to pass through.

- **Material Used:** Stones of size 50-200 mm, minimum 600 mm thick.
- Dry stone lining provides a sturdy yet permeable barrier for water drainage while supporting the structure.

3. PVC Pipe Weep Holes

- **Material Used:** $\varnothing$ 110 mm PVC pipes (Type 600) spaced at 1500 mm centers.
- These pipes help drain water collected behind the wall, reducing hydrostatic pressure.

4. Clay Puddle

- **Material Used:** Compacted clay, dimensions 300 mm $\times$ 600 mm.
- Clay is impervious to water and prevents seepage through the wall.

5. Concrete Layer

- **Material Used:**

Concrete Grade 15 (19 mm aggregate) for the base layer.

Concrete Grade 25 (20 mm aggregate) for structural stability.

- The different grades ensure cost-effective use of materials while maintaining strength and stability.

6. Reinforcement Steel

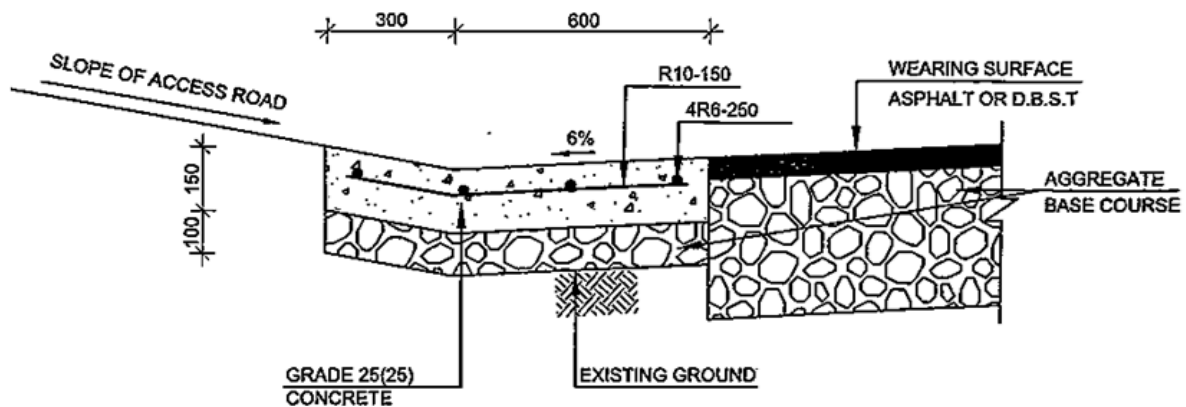
- **Material Used:** Reinforcement bars Y10-01-200 and Y10-02-300.
- Reinforcement improves tensile strength and resists cracking.

7. Screed Concrete

- **Material Used:** 75 mm thick screed concrete.
- Provides a smooth and level surface for further construction or finishing.

Tab 18

Precast Dish Drain



The pre-cast dish drain project was designed to manage surface runoff effectively, ensuring proper drainage along access roads and other critical areas. The system integrates pre-cast concrete components, reinforcement, and a base preparation to guarantee longevity and functionality.

Let's road length as L.

1. Base Preparation

- **Material Used:** Compacted subgrade and aggregate base.
- Provides a stable and leveled foundation for the installation of pre-cast elements.
- Quantity Calculations not need here.

2. Pre-Cast Dish Drain Units

- **Material Used:** Pre-cast concrete dish drain sections.
- Pre-cast units reduce construction time and ensure uniformity in size and quality.
- Quantity Calculation:

§ Number of units: .

§ Volume per unit: Provided by manufacturer's specifications.

3. Reinforcement Steel

- **Material Used:** Steel reinforcement bars (e.g., R10-150 or equivalent).
- Reinforcement enhances the tensile strength of the pre-cast units.
- Quantity Calculation:
 - § Based on the manufacturer's design and spacing of the bars.

4. Wearing Surface

- **Material Used:** Asphalt or D.B.S.T.
- Protects the drain edges and ensures a smooth transition between the drain and the road.
- Quantity Calculation not needed here

5. Jointing and Sealing

- **Material Used:** Concrete grout or sealant.
- Ensures watertight joints between pre-cast sections.
- Quantity Calculation not needed here.

6. Slope and Alignment

- **Design Specifications:** 6% slope (as per the drawing).

- Ensures efficient water flow and prevents stagnation.

FDT

Comprehensive Guide to Field Density Test (FDT) for Road Layers

Introduction

The **Field Density Test (FDT)** is a critical quality control procedure used in road construction to ensure that each compacted layer (such as subgrade, sub-base, and base) meets specified density requirements. Adequate compaction improves the load-bearing capacity, stability, and longevity of the road structure. This guide outlines the complete process of performing and evaluating FDT for road construction projects.

1. Purpose of Field Density Testing

- Verify if the compacted layer meets the **specified dry density**.
 - Prevent structural failures due to under-compacted materials.
 - Ensure compliance with project specifications and standards.
-

2. Common Test Methods

There are two widely used methods for FDT in road projects:

a. Sand Replacement Method (Standard Method)

Used where the soil is granular or fine-grained. Suitable for embankment, subgrade, and base layers.

b. Core Cutter Method (For cohesive soils only)

Used primarily in laboratory or for fine cohesive soils in the field. Not ideal for granular or coarse materials.

3. Equipment Required (Sand Replacement Method)

- Sand pouring cylinder with calibrated cone
- Metal tray with a central hole (approx. 10 cm diameter)

- Clean, dry standard sand (with known density)
 - Weighing balance
 - Digging tools (trowel, brush)
 - Moisture content containers
 - Oven or moisture meter
-

✓ 4. Test Procedure (Step-by-Step)

a. Preparation

- Level the surface of the compacted layer.
- Place the tray on the surface and cut a clean hole (approx. 10 cm diameter, 10–15 cm depth).
- Collect the excavated soil and place in a container for moisture and weight measurement.

b. Weighing Operations

- Weigh the sand pouring cylinder filled with standard sand (W1).
- Place it on the hole and allow sand to fill the hole completely.
- Weigh the cylinder again after pouring (W2).
- Also, perform a calibration by filling a standard calibration cone to find the weight of sand in cone alone.

c. Measurements

- Weight of excavated soil (wet) = W3
- Moisture content = $(W_{\text{wet}} - W_{\text{dry}}) / W_{\text{dry}} \times 100$
- Volume of hole = $(W1 - W2 - W_{\text{cone}}) / \text{Sand density}$
- Dry Density = Weight of dry soil / Volume of hole

5. Calculations

Let:

- W_s = Weight of dry soil (g)
- V = Volume of the hole (cm^3)

Dry Density (g/cm^3):

$$\rho_d = \frac{W_s}{V}$$

Degree of Compaction (%):

$$\text{Degree of Compaction} = \frac{\rho_d}{\rho_{d(\max)}} \times 100$$

Where:

- $\rho_{d(\max)}$ = Maximum dry density from Proctor Test

6. Acceptance Criteria

Each road layer has a specific compaction requirement, usually defined as a percentage of the **Maximum Dry Density (MDD)** obtained from a Modified or Standard Proctor Test. Proper compaction ensures the strength and durability of the pavement structure.

Road Layer	Minimum Compaction Requirement
Subgrade	90–95% of MDD
Edge Widening Subgrade	95% of MDD
Material Filling	
└─ Embankment	90–95% of MDD
└─ Sub-base	95% of MDD

└─ Shoulder	95% of MDD
Aggregate Base Course (ABC Layer)	98% of MDD

7. Documentation

A typical FDT report includes:

- Location and layer details
 - Test method used
 - Moisture content
 - Bulk and dry density
 - Degree of compaction
 - Test result (pass/fail)
-

Common Issues and Precautions

- Ensure the hole is not too shallow or irregular.
 - Use standard, uniformly graded dry sand.
 - Protect the test spot from water, vibration, and contamination.
 - Check calibration of sand cone regularly.
-

Conclusion

FDT is essential for ensuring the structural reliability of road layers. By following the standard procedure, engineers can confirm whether the compaction meets the design intent, thereby preventing costly failures and rework.

This guide enables field technicians, quality inspectors, and engineers to confidently execute and interpret FDT results for road construction layers.



Concrete Works

Comprehensive Guide to Concrete Testing in Road Projects

Introduction

What is Concrete?

Concrete is a composite construction material made by mixing cement, water, fine aggregates (sand), and coarse aggregates (gravel or crushed stone). It hardens over time due to a chemical reaction known as hydration, resulting in a strong and durable material.

Ready Mix vs Nominal Mix

- **Nominal Mix:** A traditional method using approximate volume-based ratios (e.g., 1:2:4 for M15), commonly used in small-scale works and where high precision is not critical. In many road projects, especially in minor structures, the **Highway Schedule of Rates (HSR)** guideline is followed. HSR provides standard mix ratios and specifications for commonly used grades of concrete to ensure consistency, ease of estimation, and suitability for public infrastructure. While it simplifies field batching, it may lack the rigorous control of a formal design mix.
- **Ready Mix Concrete (RMC):** A factory-batched and quality-controlled product delivered to site. The mix is proportioned by weight using a design mix method (e.g., M25, M30) and is ideal for road projects due to its uniformity, consistency, and controlled strength.

Why is Concrete Used in Road Construction?

Concrete is extensively used in road construction due to its high compressive strength, durability, versatility, and relatively low maintenance. It is ideal for:

- Rigid pavements
- Kerbs and side drains
- Box culverts and retaining walls
- Shoulder slabs and medians

Concrete can withstand heavy loads and adverse weather conditions, making it suitable for critical structural components in highways and road infrastructure.

Improving Concrete Properties with Chemical Admixtures

To enhance workability, strength, durability, and setting characteristics, chemical admixtures are added to concrete. Common types include:

- **Plasticizers / Superplasticizers:** Improve workability without increasing water content
- **Retarders:** Delay setting time, useful in hot climates
- **Accelerators:** Speed up setting and early strength gain
- **Air-entraining agents:** Introduce microscopic air bubbles to improve freeze-thaw resistance
- **Waterproofing agents:** Reduce permeability and enhance durability

These admixtures allow engineers to tailor the concrete mix to suit environmental conditions, structural requirements, and construction timelines.

Impact of Water on Concrete Strength Water plays a dual role: it is essential for hydration but excess water weakens concrete. A **low water-cement ratio (W/C)** increases strength, while high W/C results in reduced strength, segregation, and increased permeability. A typical W/C ratio ranges from 0.4 to 0.5 for road concrete.

Effect of Admixtures on Workability Admixtures like plasticizers and superplasticizers improve workability by dispersing cement particles. This enables easier placement without increasing water content, thus maintaining strength and durability.

Setting Time of Concrete

- **Initial setting time:** Approx. 30–90 minutes after mixing
- **Final setting time:** Approx. 6–10 hours Setting time can be adjusted with admixtures depending on ambient conditions and transportation delays in ready mix applications.

Testing of Concrete Materials (Before Mixing) To ensure concrete quality, materials must be tested:

- **Cement:** Fineness, setting time, soundness, compressive strength
- **Fine aggregates (sand):** Sieve analysis, silt content, specific gravity
- **Coarse aggregates:** Sieve analysis, aggregate crushing value, water absorption
- **Water:** pH level, total solids, organic matter (must be clean and potable)

Concrete Mix Design Concrete mix design is the process of determining the optimal proportion of cement, water, fine and coarse aggregates to achieve the required strength and workability.

Steps for Mix Design

1. **Define Target Strength:** Add a margin (typically 5 MPa) above characteristic strength.
2. **Choose W/C Ratio:** Based on strength and exposure conditions.
3. **Select Water Content:** From IS chart based on slump requirement.

4. **Calculate Cement Content:** Using W/C ratio and selected water.
5. **Choose Aggregate Proportions:** Balance based on density, gradation, and workability.
6. **Adjust for Admixtures:** Deduct water or cement if using plasticizers.
7. **Trial Mix:** Prepare sample batches and test slump and strength.
8. **Finalize Mix:** Based on achieving desired slump and 28-day compressive strength.

Proper mix design ensures durable, economical, and high-performance concrete suitable for the demands of road construction.

1. Key Properties of Concrete to be Tested

- **Workability**
 - **Compressive strength**
 - **Consistency**
 - **Air content** (if air-entrained)
 - **Density and water absorption**
 - **Curing and setting time**
-

2. Concrete Tests in Road Projects

a. Slump Test – Workability Assessment

The Slump Test is a simple field test used to measure the workability or consistency of fresh concrete before it sets. It helps ensure that the concrete has the right water-cement ratio and is suitable for placement, compaction, and finishing on site. The test is widely used in road construction, building works, bridges, and precast applications as part of routine quality control.

Why use a concrete slump test?

The main reason to perform the test is to ensure batches of the same concrete are of constant quality and strength. The more water is added to a mix, the weaker the concrete gets. If customers require a higher workability/more flowing concrete (a higher slump result) we need to increase the cement content to ensure the concrete still reaches its target strength.

This test is just one of the ways we ensure the quality of our concrete. Water is the enemy when it comes to concrete strengths.

How do you carry out a slump test?

This easy test is carried out in just seven stages, using a slump cone, a steel rod, and a measuring stick. Here's how you carry out the test:

- Place the cone on a flat, smooth, horizontal surface and stand on the footholds either side to ensure the cone is planted firmly on the ground
- Fill the cone in three layers, using the steel tamping rod to compact - or tamp - the concrete after each layer in an even, uniform manner (this should be done 25 times per layer).
- Once the cone is filled, remove any overflowing concrete from the top, making sure the concrete fills the cone exactly to the top level. Remove any spilled concrete from the base of the cone, too.
- Lift the cone vertically, using a slow and steady motion until the cone is clear of the concrete. Place the cone upside down on the surface next to the concrete.
- The concrete will subside or slump. To measure the slump, place the steel rod across the top of the upturned cone so it overhangs the concrete.
- Now, measure from the rod down to the top of the slump. The level of slump is measured to the nearest 5mm.

Results of the slump test

Depending on the water-cement ratio of the mix, the concrete slump will fall under one of four categories

- **True slump** - the concrete largely retains its cone shape, demonstrating that the mix is cohesive and its workability isn't too high.
- **Zero slump** - the concrete retains its shape completely. This shows that the mix is very dry (this kind of concrete is best used in road construction).
- **Shear slump** - the top half of the concrete subsides dramatically, leaning to one side, meaning the mix has workability but low cohesion. The mix may have too much water content and can be retested after being amended
- **Collapse** - the mix doesn't retain its shape at all and completely collapses. This means the water-cement ratio is too high and needs to be fully amended.

b. Concrete Cube Test – Compressive Strength Verification

This test is the standard method for determining the compressive strength of concrete used in road works.

- **Sample Preparation (Cube Casting):**
 - Cast three 150 mm × 150 mm × 150 mm cubes for each sample.
 - Fill the mould in three layers, tamping each layer with 25 strokes.
 - Surface is leveled and moulds are stored in a controlled environment for 24 hours.
 - After 24 hours, demould and place in curing tank until testing.
- **Testing Ages:** 7 days, 14 days (optional), and 28 days (standard).
- **Testing Procedure:**
 - Use a compression testing machine (CTM).
 - Apply load gradually until failure.
 - Record maximum load and calculate compressive strength using:

$$f_c = \frac{P}{A}$$

where:

 - f_c = compressive strength (MPa)
 - P = maximum load (N)
 - A = area of specimen face (mm²)
- **Acceptance Criteria:**
 - Minimum 28-day average strength ≥ specified strength (e.g., 25 MPa, 30 MPa).
 - No individual cube should fall below 85% of target strength.



3. Acceptance Criteria for Slump and Cube Tests

For road construction, concrete quality is assessed using key parameters based on testing outcomes. The criteria below reflect common acceptance standards used in the industry:



Slump Test (Workability Check)

- The acceptable **slump range** depends on the concrete placement method and structural element.
- For general road structures (e.g., kerbs, drains, and shoulders), the slump range is typically:
 - **25–75 mm** for medium workability (manually placed sections)
 - **75–100 mm** for higher workability (sections requiring pumping or deep placement)
- If slump is outside the specified range:
 - Reject the batch
 - Adjust water content or admixture and retest

Cube Compressive Strength Test

- Minimum strength must meet or exceed the **design compressive strength**.
- Standard target strengths: **25 MPa, 30 MPa**, or higher depending on structural use
- **Evaluation at 28 days:**
 - **Average of 3 cubes** $\geq$ specified strength
 - **No individual cube** $<$ 85% of specified strength
- If cubes fail:
 - Conduct testing on retained samples
 - Use rebound hammer/core cutting for in-situ verification

These criteria ensure that concrete placed in roadworks can endure structural loads and environmental effects as intended.

4. Field Records and Reporting

- Cube register/logbook (date, location, mix ID)
- Slump test records

- Cube curing tank record
- Test certificates from laboratory
- Site instruction records (if rejection or rework needed)

Common Issues and Prevention

Issue	Cause	Prevention
Low cube strength	Poor mix, curing issues	Follow correct proportions & curing
High slump / segregation	Excess water	Maintain W/C ratio, retest before pour
Incomplete compaction	Lack of vibration	Ensure proper vibration during casting
Cracks in cubes	Improper demoulding or curing	Handle carefully and cure consistently

Conclusion

Concrete quality testing is essential to ensure structural integrity and durability in road-related works. Adhering to proper testing procedures—from slump to cube strength—helps prevent failure, delays, and costly repairs. This guide empowers field engineers, supervisors, and QA teams to carry out systematic and compliant concrete testing at all stages of a road project.

Sandun

Here is your professionally written **Week 18 Internship Diary Entry**, focused on **Concrete Cube Casting and Compressive Strength Testing**, which is a vital quality control process in road and structural concrete works under RDA/ICTAD supervision.

Week 18 – Concrete Cube Casting and Compressive Strength Testing

This week, I was involved in the **casting and testing of concrete cubes**, which is the standard method for verifying the **compressive strength** of concrete used in structural works such as **culvert headwalls, drain bases, and side walls**.

Purpose of Cube Testing

Concrete cube tests are conducted to:

- Verify if concrete achieves its **design strength** (e.g., 25 MPa for M25)
 - Ensure compliance with **project specifications and ICTAD standards**
 - Monitor **quality of Ready Mix Concrete (RMC)** and site-mixed concrete
 - Validate the performance of curing and placement procedures
-

Cube Casting Procedure (On Site)

1. Sampling

- Concrete is collected from the middle portion of the delivery or pour.
- The sample is representative of the batch.

2. Mould Preparation

- Moulds (typically 150 mm x 150 mm x 150 mm) are cleaned, oiled, and assembled.
- Placed on a firm and level base.

3. Filling the Mould

- Filled in **three layers**, each layer **compacted 25 times** using a standard tamping rod.
- Surface is leveled and smoothed.

4. Initial Curing

- Moulds are kept in a shaded area for 24 hours and covered with damp cloths or polythene.

5. Demoulding & Curing

- After 24 hours, cubes are demoulded and placed in a **curing tank with clean water** at 27°C ±2°C until the testing date.

Testing Procedure – Compressive Strength Test

- **Standard:** BS EN 12390 / ASTM C39
- **Testing ages:** 7 days and 28 days (sometimes 3 days for early strength check)
- The cube is placed in a **compression testing machine** and loaded until failure.
- The maximum load is recorded and used to calculate **compressive strength**:

$$\text{Compressive Strength (MPa)} = \frac{\text{Max Load (N)}}{\text{Area of Cube Face (mm}^2\text{)}}$$

Example:

- Max load = 450,000 N
- Area = 150 mm × 150 mm = 22,500 mm²
- Strength = 450,000 ÷ 22,500 = **20 MPa**

Acceptance Criteria (per ICTAD)

Grade of Concrete	7-Day Strength	28-Day Target Strength
M20	≥ 15 MPa	≥ 20 MPa
M25	≥ 17 MPa	≥ 25 MPa

M30

≥ 20 MPa

≥ 30 MPa

Concrete is accepted if average strength of 3 cubes meets the target and no individual cube falls below **85%** of target.

Site Observations

- Cubes were cast from each concrete delivery (RMC and site mix).
 - Each cube was properly labeled (date, road no, structure, mix grade).
 - The test results were documented and compared with design values.
 - In one case, a 28-day cube failed below 85%, and the consultant was notified for further investigation.
-

Conclusion

The cube casting and testing process is essential for ensuring that concrete delivered and placed meets required structural standards. This week enhanced my understanding of quality assurance in concrete works, including **sampling, curing, documentation, and strength verification** in accordance with ICTAD and RDA requirements.

Let me know if you'd like all 18 weeks combined into a formatted PDF internship report or ready-to-print document.

Pavement Design

Comprehensive Guide to Pavement Design

Introduction

Pavement design is a critical component of road construction projects. It ensures that the road can withstand expected traffic loads and environmental conditions over its intended lifespan. This guide explains the complete step-by-step process of designing a **flexible pavement**, based on common methodologies and engineering standards such as those outlined in the **Road Development Authority (RDA) of Sri Lanka** pavement manuals.

1. Project Understanding and Road Classification

Before starting the design:

- Identify the **location** and **classification** of the road.
- Assess **terrain type** (plain, rolling, mountainous).
- Note **climatic conditions** and **soil characteristics**.
- Determine the **functional classification** of the road (e.g., arterial, collector, local road).

This background is essential to choose suitable materials and design parameters.

2. Traffic Data Collection & Analysis

A pavement must be designed to carry the expected **traffic load** throughout its life. Here's how to estimate the traffic:

a. Conduct Traffic Surveys

- Use **manual or automatic counters** to count the number and type of vehicles.
- Focus on **Commercial Vehicles per Day (CVPD)**.

b. Convert to Design Traffic in ESALs

Design traffic is expressed in **Equivalent Standard Axle Loads (ESALs)** using the formula:

$$ESALs = \frac{365 \times A \times ((1 + r)^n - 1)}{r} \times F$$

Where:

- **A** = Initial daily commercial vehicle count
- **r** = Annual traffic growth rate (typically 5-7%)
- **n** = Design life (in years, typically 15-20)
- **F** = Vehicle Damage Factor (VDF), based on axle load surveys or RDA standards

This gives the total expected traffic loading the pavement must support.

✓ c. Identify the Traffic Class

Based on the cumulative number of ESALs calculated, traffic is categorized into classes. This helps to select the appropriate pavement thickness from design charts. Common classifications are:

Traffic Class	ESAL Range (Millions)	Description
T1	< 0.1	Very low traffic volume
T2	0.1 – 0.3	Low traffic volume
T3	0.3 – 1.0	Medium traffic volume
T4	1.0 – 3.0	Medium-high traffic volume
T5	3.0 – 10.0	High traffic volume
T6	> 10.0	Very high traffic volume

The appropriate traffic class determines the minimum thickness and type of materials required for each pavement layer.

3. Subgrade Soil Evaluation

The **subgrade** is the natural soil foundation for the pavement and plays a critical role in the pavement's structural performance.

a. Soil Sampling and Testing

- Collect soil samples from the site at representative intervals.
- Conduct **CBR (California Bearing Ratio)** tests following **ASTM D1883**.
- Perform laboratory or field CBR tests after proper moisture conditioning and compaction.
- Use the **lowest CBR value** observed along the alignment for conservative design.

b. Subgrade Class Identification

Subgrade soils are classified into strength classes based on their CBR values. These classes are referenced in pavement design charts to determine appropriate pavement layer thickness. The classification system below follows the standard used by many national road authorities:

Subgrade Class	CBR Range (%)	Description
S1	$\leq 2\%$	Very weak subgrade
S2	3 – 4%	Weak subgrade
S3	5 – 7%	Medium strength
S4	8 – 14%	Strong subgrade
S5(a)	15 – 19%	Very strong subgrade
S5(b)	20 – 29%	Very strong subgrade
S6	$\geq 30\%$	Exceptionally strong

This classification ensures that the subgrade strength is accurately factored into the structural design. Lower classes (S1, S2) typically require thicker pavement structures or soil improvement methods to meet design performance criteria.

c. Soil Improvement (if required)

If the subgrade is classified as **S1 or S2**, soil stabilization or improvement may be required:

- **Mechanical stabilization:** Blending with gravel or crushed stone.
- **Chemical stabilization:** Lime or cement treatment.
- **Replacement:** Removing weak soil and replacing with better material.

Proper subgrade preparation is essential to the long-term performance of the pavement.

The **subgrade** is the natural soil foundation for the pavement.

4. Pavement Type Selection

Choose between:

- **Flexible Pavement** (bitumen-based, layered system): Common in many countries.
- **Rigid Pavement** (concrete slab): Used for heavy-duty or long-lasting roads.

Flexible pavement is generally more cost-effective and easier to maintain, which makes it suitable for most public road projects.

5. Pavement Layer Thickness Design

One of the critical steps in flexible pavement design is determining the appropriate thickness of each pavement layer to safely distribute traffic loads to the subgrade. This is where the concept of **Structure Number (SN)** becomes important.

a. Understanding Structure Number (SN)

The **Structure Number (SN)** is a numerical index that represents the overall structural strength of a pavement. It is calculated using the thickness and strength (layer coefficients) of each pavement layer:

$$SN = a_1 \cdot D_1 + a_2 \cdot D_2 + a_3 \cdot D_3 + \dots$$

Where:

- **a_1, a_2, a_3** = Structural layer coefficients (strength per unit thickness)

- D_1, D_2, D_3 = Thickness of each layer (in inches)

The required SN depends on traffic loading (ESALs) and subgrade support conditions. It ensures that the designed pavement is neither over- nor under-built.

✓ b. Structural Layer Coefficients (Standard Values)

Below is a table of standard structural coefficients derived from pavement design guidelines. These coefficients represent the relative structural contribution of each material layer per unit thickness (typically per inch):

Layer Description	Structural Coefficient (a)
Single surface dressing	0.10
Sand seal, hot bitumen application	0.05 – 0.10
Double surface dressing	0.20
Asphalt concrete (high stability)	0.40
Hot rolled asphalt (surfacing)	0.35 – 0.40
Flexible bituminous surfacing	0.35 – 0.40
Bitumen macadam roadbase	0.30
Bitumen bound base / asphalt treated base	0.28
Penetration macadam	0.20
Dense graded crushed stone base (CBR 110)	0.14
Water bound / dry bound macadam	0.14
Single sized aggregate base	0.125
Granular base (CBR 80)	0.12
Cement/lime stabilized base (3.0–6.0 MPa)	0.18 – 0.27
Cement/lime stabilized base (1.5–3.0 MPa)	0.13 – 0.18
Cement/lime stabilized base (CBR 80)	0.12
Cement/lime stabilized subbase (0.75–1.5 MPa)	0.11
Granular subbase (CBR 30)	0.11
Granular subbase (CBR 20)	0.095

Capping layer (CBR 15)	0.09
Granular subbase / capping layer (CBR 8)	0.07

These coefficients should be used when computing the Structure Number (SN) for flexible pavement design. Selection depends on material quality, CBR, and stabilization method. Lower coefficients are used for weaker materials and higher for stiffer, better-performing materials.

c. Using SN in Design

Design manuals (such as AASHTO) provide SN values required for different combinations of ESALs and subgrade CBR or Resilient Modulus (MR). The designer selects appropriate layer materials and thicknesses such that their combined SN meets or exceeds the required SN.

This method provides flexibility and optimization in pavement layer selection while ensuring structural adequacy.

Pavement Design Alternatives (Charts)

Based on the RDA Pavement Design Manual, there are multiple standard pavement configurations available depending on traffic class, subgrade condition, material availability, and intended performance. These are typically represented in the form of charts:

Chart 1: Single Sized Aggregate Base / Bituminous Applications

Used in lighter traffic situations where high-quality, single-sized aggregates and bituminous layers are available.

Chart 2: Granular Base / Double Bituminous Surface Treatment and Sand Seal (Surface Dressing – SD)

This configuration is commonly used for **low to moderate traffic roads**, especially in rural or secondary networks. It includes:

- A well-compacted **granular base** (e.g., ABC material)
- A **double surface dressing** (two coats of bitumen and chips)
- A final **sand seal** to enhance binding and surface quality

It is **cost-effective**, quick to construct, and suitable for roads with traffic classes ranging from T1 to T4 depending on subgrade condition.

Chart 3: Granular Base / Asphalt Concrete Surfacing

Ideal for roads with **medium to high traffic volumes**, combining a durable granular base with an asphalt surfacing layer. Offers longer service life and better resistance to environmental wear.

 Chart 4: Bitumen Bound Base / Double Bituminous Surface Treatment and Sand Seal

Used in high-strength base applications with relatively thin bituminous surfacing. The base layer is stabilized or bound with bitumen or asphalt, giving it a higher structural contribution.

👉 In the current pavement design scenario, **Chart 2** has been selected. This approach balances durability with cost-effectiveness and is appropriate for the expected traffic and subgrade conditions. It is widely used for rural roads and rehabilitation projects where resources are limited, but reliable service performance is required.

 Example: Design Section and Existing Pavement Consideration

When designing a new pavement using Chart 2 (Granular Road Base / Double Bituminous Surface Treatment and Sand Seal), it is common to calculate the **Structure Number (SN)** of both the proposed new design and the existing road structure.

 Design Section (Chart 2 Configuration)

Layer Thickness	Layer Type	Structural Coefficient
20 mm	Surface Dressing	0.20
175 mm	Granular Road Base	0.12
125 mm	Granular Sub-base	0.095

Calculated Structure Number:

$$SN = (20 \times 0.20) + (175 \times 0.12) + (125 \times 0.095) = 36.875$$

 Existing Section (Damaged or Deteriorated)

Layer Thickness	Layer Type	Structural Coefficient
50 mm	Macadam	0.14
100 mm	DGAB (CBR > 80)	0.12

Existing SN:

$$SN = (50 \times 0.14) + (100 \times 0.12) = 19 \quad SN = (50 \times 0.14) + (100 \times 0.12) = 19$$

Effective Contribution of Existing Structure

Due to deterioration and reduced integrity of old pavement, only **25% of its structural value** is considered:

$$\text{Effective SN of Existing Section} = 19 \times 0.25 = 4.75 \quad \text{Effective SN of Existing Section} = 19 \times 0.25 = 4.75$$

Required Structure Number (after considering existing structure):

$$\text{Required SN} = 36.875 - 4.75 = 32.125 \quad \text{Required SN} = 36.875 - 4.75 = 32.125$$

This approach allows designers to account for remaining strength in existing pavements while still meeting the structural requirements of the design traffic and subgrade condition.

Widening Area Design Consideration

In projects involving widening of existing macadam roads, it is important to ensure that the proposed widening section meets or exceeds the structural strength of the existing carriageway.

Design Section for Widening (Chart 2 Based)

Based on RDA Chart 2 (Granular Road Base / Double Bituminous Surface Treatment and Sand Seal):

Layer Thickness	Layer Type	Structural Coefficient
20 mm	Surface Dressing	0.20
175 mm	Granular Road Base	0.12
125 mm	Granular Sub-base	0.095

Structure Number of Design Section:

$$SN = (20 \times 0.20) + (175 \times 0.12) + (125 \times 0.095) = 36.875 \quad SN = (20 \times 0.20) + (175 \times 0.12) + (125 \times 0.095) = 36.875$$

Matching Existing Pavement Strength

In field inspections, the existing pavement was found to have an **average 100 mm thick ABC layer**. To ensure the edge widening has equivalent strength, the minimum structural number of **12** (100×0.12) must be achieved.

Thus, for widening compatibility, a **100 mm thick DGAB layer** is proposed to replicate or exceed the existing layer's performance.

Proposed Section for Widening

Considering unevenness and field constraints, the following structure is recommended for the widening section:

Layer Thickness	Layer Type	Structural Coefficient
40 mm	Asphalt Concrete	0.40
150 mm	DGAB (CBR > 80)	0.12
100 mm	DGAB (CBR > 80)	0.12

Proposed Section Structure Number:

$$SN = (40 \times 0.40) + (150 \times 0.12) + (100 \times 0.12) = 46$$

Since the required structure number of 36.875 is exceeded, this section provides adequate strength and can be accepted as suitable for edge widening construction.

Flexible pavement includes the following layers:

1. **Wearing Course** (top asphalt layer)
2. **Binder Course** (intermediate asphalt)
3. **Base Course** (crushed aggregate)
4. **Sub-base** (granular material)
5. **Subgrade** (compacted natural soil)

a. Reference Design Charts

- Use appropriate pavement design manuals (e.g., RDA Manual, AASHTO Guide, IRC:37).
- Match **design ESALs** and **Subgrade Strength Class** to get layer thickness.

Example:

For **T1(b)** and **S4**, charts recommend:

- Sub-base = 125 mm
 - Granular Road Base = 175 mm
 - Surface Dressing = 20mm
-

 6. Drainage Consideration

Water is the enemy of pavement. Drainage design includes:

- **Cross slope** (camber) of 2-3% for water runoff.
- **Side drains** or culverts to prevent water accumulation.
- **Subsurface drainage** if soil is poorly draining or water table is high.

Poor drainage can cause pavement failure regardless of good design.

☐ **7. Final Documentation**

Prepare detailed design documentation:

- Cross-sectional **drawings** showing layer composition and thickness.
 - **Material specifications** for each layer.
 - **Quantity estimates** for budgeting.
 - **Construction notes** on compaction, testing, and quality control.
-

 Sample Pavement Design Summary

Layer	Thickness (mm)	Material Description
Wearing Course	40	Asphalt Concrete (AC)
Binder Course	60	Dense Bitumen Macadam (DBM)

Base Course	200	Crushed Aggregate Base
Sub-base	150	Granular Sub-base (ABC material)
Subgrade	-	Compacted Natural Soil (CBR = 5%)

Tools & Standards Used

- **RDA Pavement Design Manual – Sri Lanka (2004 Edition)**
 - **ASTM D1883** – CBR Test Procedure
 - **IRC:37** (Indian Roads Congress)
 - **AASHTO Guide for Design of Pavement Structures**
-

Conclusion

Pavement design is a structured engineering task involving traffic estimation, soil strength analysis, and material layer configuration. By following this guide and using pavement design standards, one can develop a cost-effective and durable flexible pavement suitable for diverse road conditions.

This document ensures that any engineer, trainee, or stakeholder can understand the full process and independently prepare a pavement design for practical projects.

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Week 13 – Pavement Design Summary

Introduction

Pavement design ensures a road structure can withstand traffic loads and environmental effects over its design life. This process follows **RDA**, **ICTAD**, and **AASHTO** guidelines, with focus on material quality, traffic volume, and subgrade condition.

1. Project Understanding

- Assess **location, terrain, climate**, and **functional road class**.
 - Classify the road (arterial, collector, rural, etc.) to guide material selection and layer thickness.
-

2. Traffic Data Analysis

- Conduct traffic surveys to determine **Commercial Vehicles per Day (CVPD)**.
 - Convert traffic into **Equivalent Standard Axle Loads (ESALs)**.
 - Categorize traffic into classes (T1–T6) to determine required pavement strength.
-

3. Subgrade Evaluation

- Perform **CBR tests** on site soils.
 - Classify subgrade strength (S1–S6) based on CBR:
 - **S1 ≤ 2%** – Very weak
 - **S6 ≥ 30%** – Very strong
 - For weak subgrades, apply **mechanical/chemical stabilization** or replacement.
-

4. Pavement Type

- **Flexible Pavement** (bitumen-based) – Cost-effective, commonly used.
 - **Rigid Pavement** – For heavy-duty or long-life roads.
-

5. Pavement Layer Design

Structure Number (SN) Method:

$$SN = a_1 \cdot D_1 + a_2 \cdot D_2 + a_3 \cdot D_3 + \dots$$

- a = layer coefficient (strength/unit thickness)
- D = thickness (in inches)

Coefficient Examples:

Layer Type	Coefficient (a)
Asphalt Concrete	0.40
Granular Base (CBR 80–100)	0.12–0.14
Granular Sub-base (CBR 30)	0.11
Surface Dressing	0.20

Design SN depends on ESALs and subgrade class.

6. Design Charts (RDA Pavement Design Manual)

Common options:

- **Chart 2:** Granular base + Double Surface Dressing + Sand Seal
→ Used for T1–T4 traffic class in rural roads
- **Chart 3:** Granular base + Asphalt surfacing → Medium to high traffic
- **Chart 4:** Bitumen-bound base + surface dressing → High strength

Design Example

Selected Structure (Chart 2):

- Surface Dressing: 20 mm ($a = 0.20$)
- Granular Base: 175 mm ($a = 0.12$)
- Sub-base: 125 mm ($a = 0.095$)
→ **SN = 36.875**

Existing Pavement (Macadam 50 mm + DGAB 100 mm):

→ $SN = 19 \times 0.25$ (degraded) = 4.75
→ **Required SN = 36.875 – 4.75 = 32.125**

Widening Design

To match existing pavement strength (ABC 100 mm → SN = 12), new layers proposed:

- Asphalt Concrete: 40 mm ($a = 0.40$)
 - DGAB: 250 mm total ($a = 0.12$)
→ **SN = 46 ≥ 36.875** → Acceptable
-

7. Drainage Design

- Ensure 2–3% camber for runoff.
 - Use side drains, culverts, and subdrains as needed.
 - Drainage is critical to pavement life.
-

8. Final Documentation

- Cross-sections and design drawings.

- Material specifications.
 - Quantity estimates.
 - Construction and QC notes.
-

Standards & References Used

- RDA Pavement Design Manual (SL)
 - AASHTO Guide (1993)
 - IRC:37 – Indian Roads Congress
 - ICTAD SCA/5
 - ASTM D1883 (CBR Test)
 - Overseas Road Notes 31 & 40 (TRRL UK)
-

Conclusion

This pavement design process ensures structurally sound, cost-effective, and durable flexible pavements tailored to Sri Lankan road conditions. The method integrates field data, engineering calculations, and national standards for successful implementation.

FI

Comprehensive Guide to Flakiness Index Test

Introduction

The **Flakiness Index Test** is a standard procedure used in both road and concrete construction to evaluate the shape characteristics of coarse aggregates. In concrete applications, aggregate shape significantly influences workability, compaction, strength, and long-term durability. This test identifies the percentage of flaky particles—those whose least dimension is less than 0.6 times the mean size. A high percentage of flaky particles can reduce interlocking, impair cement bonding, and weaken the concrete structure.

Purpose of the Test

- Evaluate the shape quality of coarse aggregates used in concrete and road construction.
 - Ensure aggregates meet flakiness limits for strength, workability, and durability.
 - Prevent poor packing and segregation in fresh concrete due to excess flaky particles.
 - Maintain reliable bonding between cement paste and aggregate.
-

Equipment Required

- Flakiness Index Gauge (Thickness Gauge)
 - Weighing Balance (accuracy 0.1 g)
 - Standard IS Sieves: 63 mm, 50 mm, 40 mm, 31.5 mm, 25 mm, 20 mm, 16 mm, 12.5 mm, 10 mm, 6.3 mm
 - Sample of coarse aggregate (approximately 5 kg)
 - Tray and brush
-

Test Procedure

1. Sample Preparation:

- Sieve the aggregate sample into standard size fractions (e.g., 63–50 mm, 50–40 mm, etc.).
- Discard particles retained on 63 mm sieve and passing through 6.3 mm sieve.

2. Weighing Size Fractions:

- Weigh at least **200 particles** from each fraction (or a minimum of 50 pieces if availability is low).
- Record the total weight of particles in each fraction (W1).

3. Flakiness Measurement:

- Pass each particle from the fraction through the corresponding slot on the flakiness gauge.
- Particles passing through the gauge are considered **flaky**.
- Collect and weigh all flaky particles (W2).

4. Repeat the process for all sieve fractions.

Calculation

$$\text{Flakiness Index (FI)} = \frac{\sum W_2}{\sum W_1} \times 100$$

Where:

- $\sum W_2$ = Total weight of flaky particles
- $\sum W_1$ = Total weight of all particles used in the test

Acceptance Criteria

Application Area	Max. Flakiness Index (%)
Plain Cement Concrete (PCC)	25
Reinforced Cement Concrete (RCC)	15–20
Concrete Pavement Layers	15–20
Granular Sub-base / WMM / GSB Layers	30–35

Exceeding these values may reduce concrete strength, compaction efficiency, and overall mix performance.

Reporting Format

A Flakiness Index test report should include:

- Project name, location, and sample details
 - Sieve fraction data
 - Weight of each size fraction (W1)
 - Weight of flaky particles in each fraction (W2)
 - Individual and total Flakiness Index values
 - Remarks on compliance with specification
-

Precautions

- Use dry and clean aggregates.
 - Ensure proper calibration of flakiness gauge.
 - Avoid double counting or reusing broken particles.
-

Conclusion

The Flakiness Index Test is essential for evaluating the shape of coarse aggregates used in concrete and pavement construction. Lower flakiness ensures better compaction, packing

density, bonding with cement, and ultimately stronger and more durable concrete. The test plays a key role in quality control, especially for structural and load-bearing elements.

This guide is intended to assist QA/QC engineers, lab technicians, and site supervisors in correctly performing and evaluating flakiness index tests for aggregate compliance and concrete performance.

AIV

Comprehensive Guide to Aggregate Impact Value (AIV) Test

Introduction

The **Aggregate Impact Value (AIV) Test** is a mechanical test used to determine the impact strength or toughness of coarse aggregates. It measures the resistance of aggregates to sudden shock or impact, which is crucial for aggregates used in concrete, pavement base, and sub-base layers. Lower AIV indicates tougher and more impact-resistant aggregates, desirable in heavy traffic and structural concrete applications.

Purpose of the Test

- Assess the aggregate's resistance to impact or sudden load.
 - Determine the suitability of aggregates for concrete and road construction.
 - Ensure aggregates meet specifications for high-impact areas like highways and airfields.
-

Equipment Required

- Aggregate Impact Testing Machine (with cylindrical cup and hammer)
 - IS Sieves: 12.5 mm and 10 mm
 - Weighing Balance (accuracy 0.1 g)
 - Tamping rod (10 mm diameter, 230 mm length)
 - Oven (100°C to 110°C)
 - Measuring cylinder
 - Tray and brush
-

Test Procedure

1. Sample Preparation

- Sieve the aggregate sample to collect particles passing through 12.5 mm and retained on 10 mm sieve.
- Oven dry the sample at 105°C–110°C for 4 hours and cool to room temperature.
- Weigh 350–400 g of the prepared aggregate.

2. Filling the Cup

- Fill the cylindrical cup in three layers, tamping each layer 25 times using the tamping rod.
- Level the surface and fix the cup in position under the hammer.

3. Impact Application

- Raise the hammer to 380 mm and allow it to fall freely onto the aggregates.
- Apply **15 blows** at regular intervals.

4. Sieve Analysis After Impact

- Remove and sieve the impacted sample through a **2.36 mm IS sieve**.
- Weigh the material passing (W_2) and retained (W_1) separately.

Calculation

$$AIV (\%) = \frac{W_2}{W_1 + W_2} \times 100$$

Where:

- W_2 = Weight of aggregate passing 2.36 mm sieve
- $W_1 + W_2$ = Total weight of aggregate tested



Acceptance Criteria

Application Area	Max. AIV (%)
Wearing Course (bituminous)	30
Base Course (granular)	40
Concrete Works (non-structural)	45
Concrete (high-strength)	25–30

Lower AIV values are preferred for concrete and surface courses subjected to dynamic loads.



Reporting Format

An AIV test report should include:

- Project and sample details
 - Date of test, location, sample source
 - Weight of oven-dried sample
 - Weight retained and passing 2.36 mm sieve
 - Aggregate Impact Value (%)
 - Compliance with relevant standard (e.g., IS 2386 Part IV)
-



Precautions

- Ensure hammer drop height is consistent.
- Oven-dry aggregates uniformly.
- Avoid repeated use of impacted samples.

- Clean and reset equipment between tests.

Conclusion

The Aggregate Impact Value Test helps assess aggregate durability against impact loading. It is a vital quality control measure in construction projects using concrete or granular layers. Aggregates with lower AIV are more suitable for concrete and surface layers exposed to traffic loads.

This guide supports QA/QC engineers, lab technicians, and site supervisors in correctly performing and interpreting AIV test results.

LAAY

The LAAV test, or Los Angeles Abrasion Value Test, is a standard method used to assess the wear resistance of aggregate materials like those used in road construction. It determines the percentage of aggregate that breaks down into smaller sizes due to abrasion and impact from steel balls within a rotating drum. Lower LAAV values indicate higher resistance to abrasion and a tougher aggregate.

TFV

Comprehensive Guide to Ten Percent Fines Value (TFV) Test

Introduction

The **Ten Percent Fines Value (TFV) Test** is a strength test that measures the resistance of coarse aggregates to crushing under gradually applied compressive load. Unlike the Aggregate Crushing Value (ACV) test which calculates the percentage of crushed material, the TFV test determines the load at which 10% fines (material passing a 2.36 mm sieve) are generated. It is particularly useful for assessing aggregates used in concrete, asphalt, and base/sub-base layers.

Purpose of the Test

- Assess the crushing strength of coarse aggregates.
 - Identify aggregates suitable for heavy-duty concrete or pavement layers.
 - Compare aggregate strength across different sources.
-

Equipment Required

- Compression testing machine (200 kN or higher capacity)
 - Steel cylinder (150 mm dia., 130 mm height)
 - Plunger and base plate
 - Weighing balance (accuracy 0.1 g)
 - 2.36 mm IS sieve
 - Tamping rod
 - Oven (100°C to 110°C)
 - Aggregate sample (typically 2.8 – 3.2 kg)
-

Test Procedure

1. Sample Preparation

- Obtain aggregate passing 14 mm and retained on 10 mm sieve.
- Oven-dry the sample at 105°C–110°C and cool to room temperature.
- Weigh exactly 2.5 kg of the sample.

2. First Trial (To Estimate Load for 10% Fines)

- Fill the steel cylinder with the dried sample in three layers, tamping each layer 25 times.
- Place the plunger on top and apply load gradually over 10 minutes until visible crushing is achieved.
- Remove and sieve the crushed material using a 2.36 mm sieve.
- Weigh fines passing through the sieve.
- Calculate percentage fines to estimate the load required for exactly 10% fines.

3. Final Test (Main Determination)

- Repeat the crushing procedure with three different loads, adjusting to produce close to 10% fines.
- For each load, weigh and record the mass of fines generated.
- Use interpolation to determine load corresponding to exactly 10% fines.

Calculation

Ten Percent Fines Value (TFV)=Load at which 10% of sample passes 2.36 mm sieve

If test results do not directly yield 10%, interpolation is applied using:

$$TFV = P + \left(\frac{10 - Y}{X - Y} \right) (Q - P)$$

Where:

- P = Load corresponding to % fines just below 10
- Q = Load corresponding to % fines just above 10
- X = % fines from Q
- Y = % fines from P

 Acceptance Criteria (Typical Guidelines)

Application Area	Minimum TFV (kN)
Concrete (Heavy Structures)	≥ 150
Bituminous Layers	≥ 100
Base/Sub-base (Granular)	≥ 50

Note: Specifications may vary by country or project requirements.

 Reporting Format

A TFV test report should include:

- Project name, date, sample ID, test method
- Aggregate size range (e.g., 10–14 mm)
- Oven-dried weight of sample
- Applied loads and % fines for each trial
- Final interpolated load for 10% fines (TFV)
- Compliance with relevant specifications

 Precautions

- Ensure uniform sample compaction in the cylinder.
 - Apply load gradually to prevent sudden failure.
 - Use properly calibrated compression testing equipment.
 - Clean and dry all sieves and apparatus before use.
-

Conclusion

The Ten Percent Fines Value (TFV) Test is a critical indicator of the crushing strength of coarse aggregates. It ensures that only durable, mechanically sound materials are used in high-stress construction layers like base courses and structural concrete. Properly conducted TFV testing supports long-term performance of infrastructure.

This guide is essential for lab engineers, QA/QC teams, and field technicians evaluating aggregate strength for concrete and pavement applications.

ACV

Flakiness Index (FI)

1 Flakiness Index (FI)

- **Standard:** BS 812 Part 105
- **Purpose:** Measures the percentage of flaky particles in coarse aggregates, which affect compaction and interlocking strength.
- **ICTAD Requirement:**
 - **FI ≤ 30%** (for ABC and concrete aggregate)

✓ Aggregates with high flakiness can cause weak interlock and reduce pavement durability. Aggregates must be angular but not flat.

2 Aggregate Impact Value (AIV)

- **Standard:** BS 812 Part 112
- **Purpose:** Evaluates the toughness of aggregate — its ability to resist sudden shock or impact loads.
- **ICTAD Requirement:**
 - **AIV ≤ 30%** for base course aggregates
 - For surface layers, a lower AIV may be required depending on design

✓ A lower AIV indicates stronger, more impact-resistant aggregate — ideal for high-traffic roads.

3 Los Angeles Abrasion Value (LAAV)

- **Standard:** ASTM C131 / BS 812 Part 113
- **Purpose:** Measures aggregate resistance to abrasion and wear under traffic loads using a rotating drum and steel balls.
- **ICTAD Requirement:**

- **LAAV $\leq$ 40%** (general requirement for ABC aggregates)

✓ Lower LAAV indicates better resistance to wear, important for maintaining road surface strength and reducing rutting over time.

Inspection Workflow

Here is a professional and concise **Week 14 Internship Diary Entry** covering **Road Construction Activities**, including the **Daily Progress Report (DPR)** and **Request for Inspection (RFI) Process**, with a mention of your contribution to **Excel-based automation** for RFI and QA documentation.

Week 14 – Road Construction Activities and Inspection Workflow

This week, I was involved in observing and assisting with site documentation processes related to road construction progress and quality assurance. This included preparation and submission of the **Daily Progress Report (DPR)** and handling of the **Request for Inspection (RFI)** workflow in compliance with RDA procedures.

Daily Progress Report (DPR)

The **Daily Progress Report** is an essential document that summarizes all major construction activities performed at the site each day.

- The **Site Engineer** prepares and finalizes the DPR based on inputs from supervisors and site records.
 - The DPR must be **submitted on the day prior** to the planned inspection or activity for review and scheduling.
 - It includes:
 - Road section and chainage
 - Type of work (e.g., ABC laying, compaction)
 - Equipment used
 - Manpower
 - Weather conditions
 - Material details
-

Request for Inspection (RFI) Process

The RFI ensures that the **Consulting Engineer or Resident Engineer (RE)** verifies and approves each stage of construction before proceeding.

Step-by-Step RFI Workflow:

1. **DPR + RFI Form Submission:**
After preparing the DPR, the Site Engineer submits the RFI to the Resident Engineer's office along with the DPR.
2. **Site Inspection:**
The Consulting Site Engineer (RE's representative) visits the site to inspect the reported work.
3. **Inspection Outcome:**
 - If approved, the RFI is **signed and returned** to the contractor.
 - If not approved, reasons are noted, and corrections must be made.
4. **Final Filing:**
The signed RFI is added to the **RFI Register** and retained as official proof of work inspection and approval.

Automation of RFI & QA Documentation

To improve accuracy and efficiency in managing RFI records and monthly QA reports, I contributed to developing an **Excel-based automation tool** that:

- Automatically logs RFI details by road number, date, and activity.
- Generates summary reports for **monthly QA submission**.
- Reduces manual errors and helps maintain compliance with **RDA and consultant tracking systems**.

Conclusion

This week gave me hands-on experience in the **administrative and quality assurance aspects of road construction**, particularly the coordination between contractor and

consultant teams. I also applied my technical skills to streamline the process through Excel-based documentation tools, enhancing project efficiency and traceability.

Let me know if you'd like to start drafting **Week 15 – Compaction Testing & ABC Laying**, or if you'd like help formatting this report.

Sand Cone Method (FDT)

Week 15 – Compaction Testing Using Sand Cone Method (FDT)

This week, I learned about the **Field Density Test (FDT)** using the **Sand Cone Method**, which is a standard in-situ testing procedure to verify soil compaction. Compaction is a critical aspect of road construction, ensuring the stability and long-term performance of subgrade and base layers.

Field Density Test (FDT) – Sand Cone Method

Purpose:

The Sand Cone Test is used to determine the **in-place dry density** of compacted soil or aggregate layers. It verifies whether field compaction meets the **specified percentage of the Maximum Dry Density (MDD)** obtained from the **Proctor Test**.

Test Standards:

- ASTM D1556
- BS 1377: Part 9
- ICTAD Specification Section 1602

Equipment Used:

- Sand cone apparatus
 - Clean, dry calibrated sand (uniform grading)
 - Metal tray with a central hole
 - Balance (0.01 g sensitivity)
 - Moisture container
-

Applications Observed in Site Work

Subgrade Preparation

- Conducted after trimming and compacting the natural ground.
- Target compaction: $\geq 95\%$ of MDD for mainline, $\geq 93\%$ for confined/trench areas.
- Field FDT ensures uniform compaction before placing sub-base materials.

Edge Widening Subgrade

- Since it's narrow and adjacent to existing roads, achieving compaction is challenging.
- FDT is critical to ensure edge stability and to match the existing pavement strength.

Filling Over Prepared Subgrade

- Observed filling works using **Type I EMB**, **Type II Sub-base**, and **Shoulder Material**.
- FDT ensures each layer achieves target compaction before placing the next.

ABC Laying

- After laying and rolling the Aggregate Base Course, FDT is done at random locations.
- Compaction target for ABC: $\geq 98\%$ of MDD
- Proper compaction prevents future rutting and pavement failure.

Compaction Thickness & Limitations

According to **ICTAD Specifications**:

- **Layer Thickness:**
 - Should not exceed 150 mm before compaction (for EMB and SB)
 - May be limited to 100 mm in case of hand compaction or confined areas
- Each layer must be **tested and approved** before placing the next
- Over-compaction or wet material can cause density loss or instability

✓ Conclusion

The Sand Cone Method is a reliable and essential field test for confirming soil and aggregate compaction. This week provided practical exposure to compaction quality control across various layers—from subgrade to ABC. Understanding and applying these tests ensures structural integrity and long-term pavement performance.

Let me know when you're ready to move on to **Week 16 – Concrete Works (Slump & Properties)** or if you'd like this turned into a printable report.

4o

You said:

procedure and sample calculation

ChatGPT said:

Here is the **Field Density Test (Sand Cone Method)** explained with a **clear step-by-step procedure** and a **sample calculation**, ideal for your internship diary or QA documentation.

Sand Cone Method – Field Density Test (FDT)

Purpose

To determine the in-situ dry density of compacted soil or granular layers and compare it with the laboratory **Maximum Dry Density (MDD)** obtained from the Proctor Test. This helps verify whether compaction meets the specified percentage (e.g., $\geq 95\%$).

Test Procedure

✓ Preparation

1. Calibrate the Sand Cone Apparatus

- Weigh the apparatus filled with calibrated dry sand.
- Determine the weight of sand required to fill the cone and base tray hole.

- Record this as the **Cone Correction**.

2. Select and Prepare the Test Spot

- Level a small area of the compacted surface.
- Place a metal tray with a circular hole over the test point.

3. Excavate the Test Hole

- Dig a small hole through the tray using a chisel and hammer.
- Collect and store the excavated soil in a moisture container (for moisture content test).

4. Determine Hole Volume Using Sand

- Place the inverted sand cone over the hole.
- Open the valve to allow sand to fill the hole.
- After flow stops, close the valve and weigh the remaining apparatus.



Measurements Required

Parameter	Symbol	Unit
Initial weight of sand + cone	W_1	grams
Final weight after test	W_2	grams
Cone correction weight	W_c	grams
Weight of excavated soil	W_s	grams
Moisture content (from lab)	w	% (decimal)
Sand density (from calibration)	ρ_s	g/cm^3



Sample Calculation

Example Input Data:

Measurement	Value
W_1 (initial sand + cone)	6500 g
W_2 (final sand + cone)	5200 g
W_c (cone correction)	200 g
W_s (excavated soil)	1600 g
Moisture Content (w)	8% (0.08)
Calibrated Sand Density (ρ_s)	1.52 g/cm ³

Step-by-Step Calculation

1. **Sand Used in Test Hole (W_f):**

$$W_f = (W_1 - W_2) - W_c$$

$$W_f = (6500 - 5200) - 200 = \mathbf{1100 \text{ g}}$$

2. **Volume of Test Hole (V):**

$$V = W_f / \rho_s = 1100 / 1.52 = \mathbf{723.68 \text{ cm}^3}$$

3. **Wet Density (ρ_{wet}):**

$$\rho_{wet} = W_s / V = 1600 / 723.68 = \mathbf{2.211 \text{ g/cm}^3}$$

4. **Dry Density (ρ_{dry}):**

$$\rho_{dry} = \rho_{wet} / (1 + w)$$

$$\rho_{dry} = 2.211 / (1 + 0.08) = 2.211 / 1.08 = \mathbf{2.048 \text{ g/cm}^3}$$

5. **Compaction Percentage:**

$$\% \text{ Compaction} = (\rho_{dry} / \text{MDD}) \times 100$$

If MDD from Proctor Test = 2.20 g/cm³:

$$\% \text{ Compaction} = (2.048 / 2.20) \times 100 = \mathbf{93.09\%}$$

Result Interpretation

- If project spec requires $\geq 95\%$ compaction, this sample **fails**.
- The contractor must recompact and retest the area.

Let me know if you'd like this formatted for a QA sheet or added to your report with diagrams and forms.

40

You said:

Should not exceed 150 mm before compaction

why?

ChatGPT said:

Great question!

The **reason why the layer thickness should not exceed 150 mm before compaction** (as per **ICTAD specifications** and many international standards like AASHTO and ASTM) is to ensure **uniform compaction and structural integrity** of the layer.

Why 150 mm is the Limit for Uncompacted Layer Thickness?

1. Achieve Uniform Compaction Throughout the Layer

- **Compaction equipment (like rollers or plate compactors)** can only transmit energy effectively to a certain depth.
- If the layer is too thick (e.g., >150 mm), the **lower part of the layer may not get compacted properly**, resulting in soft zones.
- This causes **uneven settlement** and reduces the pavement's load-bearing capacity.

2. Prevent Shear Failure and Rutting

- Under-compacted lower layers may **shear** or compress under traffic loads.
- This can cause **pavement deformation**, such as **rutting, cracking**, or potholes.

3. Ensure Moisture Distribution

- For proper compaction, the material should be at or near **Optimum Moisture Content (OMC)**.
 - In thick layers, **moisture may not distribute evenly**, leading to dry zones and poor compaction.
-

✓ 4. Compliance with Lab-Determined Properties

- Proctor Test (used to determine MDD and OMC) is done on **layers around 100–150 mm thick**.
 - Field compaction should match lab conditions to ensure consistent and reliable density results.
-

ICTAD / Industry Practice

Layer Type	Max Loose Thickness (Before Compaction)
Subgrade Fill	150 mm
Subbase / Base Layer	150 mm
ABC Layer	100–150 mm (based on project spec)
Manual Compaction	≤ 100 mm (due to limited energy)

If thicker layers are needed (e.g., 300 mm), they should be **placed and compacted in two or more layers** (each ≤150 mm) to achieve the desired density.

! Consequences of Ignoring This

- FDT will show **low dry density**, requiring rework.
 - May lead to **RFI rejection** by the Consultant or RE.
 - Delays in project and increased costs.
-

Summary

"The 150 mm rule ensures that each layer gets fully compacted, matches lab values, and contributes to a strong, durable pavement."

Concrete Works and Slump Test

Week 16 – Concrete Works and Slump Test

This week, I was engaged in understanding the properties of fresh concrete and how they are controlled during construction. I also observed and assisted with the **Slump Test**, which is a standard on-site method used to measure the **workability of concrete**.

Concrete – Key Properties and Parameters

Concrete is a composite material whose performance depends on several **interrelated parameters**, especially at the fresh (plastic) and hardened stages.

Important Properties of Fresh Concrete:

Property	Description	How to Control It
Workability	Ease with which concrete can be mixed, placed, and compacted.	Adjust water content, use admixtures
Slump	A measure of consistency/workability (via Slump Test).	Controlled by water-cement ratio
Setting Time	Time taken for concrete to start and finish setting.	Affected by cement type, temperature, admixtures
Segregation	Separation of aggregates from the mix.	Controlled by proper mix design
Bleeding	Water rising to the surface after placing.	Reduced by using proper grading & admixtures

Key Properties of Hardened Concrete:

Property	Controlled By
Compressive Strength	Water-cement ratio, curing, mix proportion
Durability	Cement type, exposure conditions
Shrinkage/Cracking	Curing quality, water content

Slump Test – On-Site Workability Check

The **Slump Test** is a simple field test to measure the **consistency** or **workability** of fresh concrete. It is widely used in site quality control for **every concrete batch**.

Test Method:

- **Standard:** ASTM C143 / BS EN 12350-2
- **Equipment:**
 - Slump cone (300 mm high, 200 mm base, 100 mm top)
 - Tamping rod (16 mm diameter)
 - Base plate
 - Scale for measurement

Procedure:

1. Moisten the cone and base plate.
2. Fill the cone in 3 layers, each tamped 25 times.
3. Level the top surface and remove the cone vertically.
4. Measure the vertical slump (subsidence) of the concrete.

Slump Value Interpretation:

Slump (mm)	Workability	Use Case
0–25	Very Low	Mass concrete, road base layers
25–50	Low	Pavement and footings
50–100	Medium	General construction
100–150	High	Complex formwork, congested steel

Field Observations

- During site visits, slump tests were conducted before pouring concrete into structural elements (culverts, headwalls).
 - If the slump was outside the specified range (e.g., 75–100 mm), adjustments were made on-site:
 - Too low → water/admixture added (within limits)
 - Too high → batch rejected or re-mixed
-

Conclusion

This week gave me insight into **how fresh concrete properties affect constructability** and long-term performance. I gained hands-on experience with the **slump test**, an essential field tool for ensuring concrete quality, especially in infrastructure projects under **RDA supervision**.

Week 16 – Concrete Works and Slump Test Observations

This week, I participated in quality control procedures related to fresh concrete placement at various road construction elements (e.g., culverts, headwalls). A key focus was on the **Slump Test**, which is critical to ensure proper **workability** and **consistency** of the concrete used on site.

Slump Test – Purpose and Importance

The **Slump Test** is a simple, standardized field test used to evaluate the **workability or consistency** of fresh concrete before it sets. It is vital for:

- Verifying **uniformity across batches**
- Ensuring **suitability for placement and compaction**
- Monitoring the **water-cement ratio**, which directly impacts concrete **strength and durability**

In road construction, particularly under ICTAD and RDA guidelines, this test is mandatory before placing concrete in structural elements.

Note: Excessive water reduces strength. Therefore, maintaining a proper slump value helps control strength indirectly.

Slump Test – Step-by-Step Procedure (As Per Site Practice)

1. Setup

- Place the **slump cone** on a clean, level surface.
- Step on the footholds to keep the cone stable.

2. Filling the Cone

- Fill in **3 equal layers** of concrete.
- **Tamp each layer 25 times** with the steel rod in a uniform way.

3. Levelling

- Strike off excess concrete at the top with the tamping rod.
- Clean away spillage around the base.

4. Lifting the Cone

- Lift the cone **vertically and steadily**.
- Place it upside down near the slumped concrete.

5. Measuring the Slump

- Lay the steel rod across the top of the upturned cone.
- Measure the **vertical distance** from the rod to the highest point of the slumped concrete.
- Record the slump value to the nearest **5 mm**.

Types of Slump and Their Meaning

Slump Type

Appearance

Interpretation

True Slump	Mix settles evenly, retains shape	Good cohesion and workability
Zero Slump	No subsidence	Very dry mix, suitable for low-workability uses (e.g., road base)
Shear Slump	Cone leans to one side	Mix has excess water or poor cohesion
Collapse Slump	Mix collapses completely	Water-cement ratio too high; mix must be rejected



Site Observations

- Slump tests were performed **before concrete placement** to ensure each batch met the design specification (usually **75–100 mm**).
 - **True slump** was most common, indicating good quality mixes.
 - **Shear slump** occasionally occurred and was corrected by adjusting the water or admixture dosage.
 - Batches showing **collapse slump** were not allowed for use and returned or modified.
-



Conclusion

The Slump Test remains a **quick and effective field test** to assess concrete workability. Through this week's activities, I learned how concrete quality is managed practically on-site and how **slump test results guide field decisions**, ensuring compliance with structural and durability standards under RDA/ICTAD guidelines.

Bitumen Stability Test (Marshall Method)

Comprehensive Guide to Bitumen Stability Test (Marshall Method)

Introduction

The **Bitumen Stability Test**, commonly known as the **Marshall Stability Test**, is used to determine the **strength and plastic flow characteristics** of bituminous mixtures. It plays a critical role in evaluating mix design for asphalt pavements by assessing the **maximum load** a compacted specimen can withstand before failure. This ensures road surfaces are durable, stable, and resistant to deformation under traffic loads.

Purpose of the Test

- Determine the **Marshall Stability** (maximum load) and **Flow Value** (deformation).
 - Identify the **Optimum Bitumen Content (OBC)**.
 - Evaluate the strength and flexibility of bituminous mixtures.
 - Ensure the mix meets specification requirements for structural performance.
-

Equipment Required

- Marshall stability testing machine (with proving ring/load cell & flow meter)
 - Specimen mould (101.6 mm dia. × 63.5 mm height)
 - Compaction hammer (4.5 kg, 457 mm drop height)
 - Bitumen heating oven
 - Water bath (maintained at $60^{\circ}\text{C} \pm 1^{\circ}\text{C}$)
 - Balance (accuracy 0.1 g)
 - Mixing tray and mixing tools
 - Thermometer, filter paper
-

Test Procedure

1. Sample Preparation

- Heat aggregates and bitumen to mixing temperature (~150–160°C).
- Mix thoroughly and place into a preheated Marshall mould.
- Compact with **75 blows per side** (or 50 for light traffic roads).
- Cool and extrude specimen.

2. Conditioning

- Place specimen in a **water bath at 60°C** for 30–40 minutes.

3. Testing

- Place the specimen in the Marshall testing machine.
- Apply load at 50 mm/min until failure.
- Record:
 - **Stability** = Maximum load (kN or kg)
 - **Flow** = Deformation at peak load (measured in 0.25 mm units)

Calculations

1. **Marshall Stability** = Maximum load at failure (kN or kg)
2. **Marshall Flow** = Total deformation (mm)
3. Plot graphs to evaluate:
 - Bitumen content vs. stability
 - Bitumen content vs. flow
 - Bitumen content vs. density and voids

From these graphs, determine **Optimum Bitumen Content (OBC)** where:

- Stability is at or near maximum
- Flow is within range
- Air voids meet target specification

Acceptance Criteria (Typical Values)

Property	Light Traffic	Medium Traffic	Heavy Traffic
Stability (kg)	> 500	> 750	> 1000
Flow (mm)	2 – 4	2 – 4	2 – 4
Air Voids (%)	3 – 5	3 – 5	3 – 5
Voids in Mineral Agg.	13 – 15	14 – 16	15 – 17

Reporting Format

A standard Marshall Test report should include:

- Sample ID, mix design, date of test
- Bitumen content, compaction details
- Stability and Flow values
- Bulk density, % air voids, VMA, VFB
- Graphs for mix property trends
- Optimum Bitumen Content (OBC)
- Remarks on compliance with project specs

Precautions

- Ensure uniform heating and mixing of materials.
- Compaction should be consistent.

- Use water bath at exact 60°C to maintain consistency.
 - Calibrate testing machine and flow meter regularly.
-

Conclusion

The Marshall Stability Test is a critical quality control and design tool in bituminous mix design. It ensures that asphalt mixes provide adequate load-bearing strength and deformation resistance. This test guides pavement engineers in selecting the best binder content and aggregate combination for long-lasting road performance.

This guide supports lab engineers, QA/QC staff, and road construction professionals in effectively conducting and interpreting Marshall stability tests.

Concrete Mix Methods and Mix Design

Week 17 – Concrete Mix Methods and Mix Design

This week, I studied and compared different concrete mix methods used in road and structural works under the project, especially focusing on mix design principles and the use of **HSR guidelines**, **Nominal Mix**, and **Ready Mix Concrete (RMC)**.

1. Highway Schedule of Rates (HSR) – Guideline Reference

The **HSR (Highway Schedule of Rates)** is a guideline issued by RDA/ICTAD that defines standardized material quantities and proportions for various grades of concrete used in road and bridge construction. These mix ratios are often used in **nominal mix** design when a detailed lab-based mix design is not required.

Example from HSR:

- **Grade 25 Concrete (M25)**
Cement: 1
Fine Aggregate (Sand): 1.5
Coarse Aggregate: 3
Water: ~0.5 (by weight of cement)

✓ HSR ensures consistency, cost estimation, and ease of construction, especially in low-risk or non-critical elements.

2. Nominal Mix Concrete

A **Nominal Mix** is a fixed-proportion concrete mix typically used for general construction when high precision or performance is not required.

Grade	Typical Ratio (C : FA : CA)
M15	1 : 2 : 4
M20	1 : 1.5 : 3
M25	1 : 1 : 2

 **Characteristics:**

- Based on **volume batching**
 - Does not account for material-specific properties (e.g., moisture, absorption)
 - Easy to prepare and used for **blinding layers, non-critical structures, and site-made concrete**
-

3. Concrete Mix Design (by Lab Trial)

For critical structures (e.g., headwalls, culverts), **design mix concrete** is used. This is proportioned by weight and optimized through lab testing to achieve:

- Target compressive strength (e.g., 25 MPa for M25)
- Required workability (e.g., slump 75–100 mm)
- Durability under exposure conditions

Parameters Considered:

- **Water-Cement Ratio** (low = higher strength)
- **Aggregate Grading**
- **Moisture Content**
- **Specific Gravity of materials**
- **Use of admixtures** (e.g., water reducers, retarders)

 Design mixes ensure strength, quality, and durability — especially important for RDA-supervised road works.

4. Ready Mix Concrete (RMC)

Ready Mix Concrete is prepared at a batching plant and delivered to site in transit mixers. It provides:

- **Consistency across batches**

- **Time savings**
- **Better control over mix design**
- **Compliance with RDA quality standards**

At site, **slump tests** are conducted before placement, and **cube samples** are taken for strength testing at 7 & 28 days.

✓ **Site Observations:**

- RMC was used in culvert and headwall concreting operations.
- Concrete was verified via slump tests and cube mould casting for later compressive strength testing.
- Mix certificates and delivery slips were checked for each batch.

✓ **Conclusion**

This week provided comprehensive insight into concrete mix practices used on-site. I learned how **HSR-based nominal mixes** are applied for general use, while **design mix and RMC** are chosen for structurally critical elements. Understanding mix design parameters helped me appreciate the importance of material quality and proportioning in achieving durable concrete structures.

Bitumen Work and Surface Preparation

Week 19 – Bitumen Work and Surface Preparation

This week, I was actively involved in observing and assisting with **bitumen-related activities**, particularly during the **tack coat application** and **bitumen tray testing**. Bitumen application is a critical part of flexible pavement construction and must follow **ICTAD specifications** for quality and performance.

Role of Bitumen in Road Construction

Bitumen acts as a **binding agent** between pavement layers and enhances:

- **Adhesion** between the old surface and the new asphalt layer
 - **Waterproofing** and protection against moisture damage
 - **Load transfer** across pavement layers
-

1. Bitumen Tray Test (Tack Coat Rate Check)

The **tray test** is used to verify the **application rate of bitumen** (tack coat) before asphalt is laid.

Procedure:

1. Place a pre-weighed **metal tray** (known weight) on the prepared surface before bitumen spraying.
2. Allow the spray to cover the tray area.
3. Collect and reweigh the tray to determine the amount of bitumen sprayed (by weight difference).
4. Calculate the **application rate (kg/m²)** and compare with specifications.

Standard Application Rates (per ICTAD / RDA):

Surface Type	Typical Tack Coat Rate
Granular Base (e.g., ABC)	0.6 – 0.9 kg/m ²

Existing Bituminous Surface 0.3 – 0.6 kg/m²

✅ Results must be within the specified range to ensure proper bonding and prevent slippage or delamination of the asphalt layer.

2. Bitumen Temperature Handling

Bitumen is heated and maintained at specific temperatures to ensure proper flow and adhesion.

Process Stage	Target Temperature
Heating in boiler	150 – 180°C
Spraying temperature	140 – 160°C
Minimum surface temp before spraying	≥ 25°C

🔥 Overheating can cause **bitumen degradation** (burning), while underheating leads to **poor sprayability and bonding**.

3. Surface Preparation for Bitumen Application

Before applying bitumen tack coat:

- The base surface (ABC or existing asphalt) must be **clean, dry, and dust-free**
- Any loose material or standing water must be removed
- Proper **sweep and inspection** is carried out by the contractor and verified by the Consultant/RE

📌 Observation:

In one instance, the consultant rejected bitumen spraying because the surface had fine dust particles, highlighting the importance of proper preparation.

4. QA Documentation

- All bitumen batches are documented with **delivery slips and temperature logs**
 - **Tray test results** are filed with chainage references
 - QA records include **type of bitumen used (e.g., 60/70 penetration grade)** and application area
-

Conclusion

Bitumen work requires precision and strict quality control to ensure layer bonding and pavement durability. Through this week's involvement, I gained valuable experience in **tray testing, temperature monitoring, and site inspection procedures** as per **ICTAD and RDA standards**. These practices are essential for achieving long-lasting, high-performance asphalt layers.

1. Prime Coat

Purpose:

- Applied to **unbound granular surfaces** (e.g., **Aggregate Base Course - ABC**)
- **Penetrates into the base layer** to bind loose particles
- Prepares the surface for the first bituminous layer (usually a binder or asphalt layer)

Material Used:

- **Low-viscosity bitumen cutbacks or emulsions**
 - Example: MC-30, MC-70 (medium-curing cutbacks)

Function:

- Acts as a **sealant and binder** for granular materials
- Improves adhesion of the first asphalt layer
- Controls dust and moisture penetration

Application Rate:

- Typically **0.7 – 1.2 L/m²** depending on base type and porosity
-

2. Bitumen (General Term)

Definition:

- Bitumen is a **viscous, black, hydrocarbon binder** derived from crude oil refining
- Used in various forms like:
 - **Penetration grade bitumen** (e.g., 60/70)
 - **Cutback bitumen** (bitumen + solvent)
 - **Bitumen emulsions** (bitumen + water + emulsifier)

Uses in Road Construction:

- Binder in asphalt mixes
- Prime coat and tack coat
- Seal coats and waterproofing

Grades:

- 60/70, 80/100 (based on penetration value in tenths of mm)

 So, **bitumen** is the base material used to create **prime coats, tack coats, and asphalt mixes**.

3. Tack Coat

Purpose:

- Applied between **two bituminous layers** (e.g., old asphalt surface and new asphalt overlay)
- Promotes **bonding between asphalt layers**
- Prevents slippage and layer separation

Material Used:

- Bitumen emulsion or diluted bitumen
 - Example: **Cationic Emulsion (RS-1, SS-1)**
 - Can also use penetration grade bitumen (diluted with kerosene)

Application Rate:

- Typically **0.3 – 0.6 kg/m²** for bitumen-to-bitumen layers

Summary Table:

Feature	Prime Coat	Bitumen (General)	Tack Coat
Applied On	Unbound granular base (e.g., ABC)	Raw binder used in all applications	Bituminous surface or binder course
Purpose	Seal and bind loose base particles	Acts as a binder	Promote bonding between asphalt layers
Material	Cutback/emulsified bitumen	60/70 penetration, cutbacks, emulsions	Emulsion or diluted penetration bitumen
Application Rate	0.7 – 1.2 L/m ²	–	0.3 – 0.6 kg/m ²

Timing	Before first asphalt layer	Used in multiple stages	Between asphalt layers
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 Remember:

- **Prime Coat** → **Unbound base**
- **Tack Coat** → **Between bituminous layers**
- **Bitumen** → **Raw binder used to make both**

Prime Coat vs. Bitumen vs. Tack Coat

Prime Coat

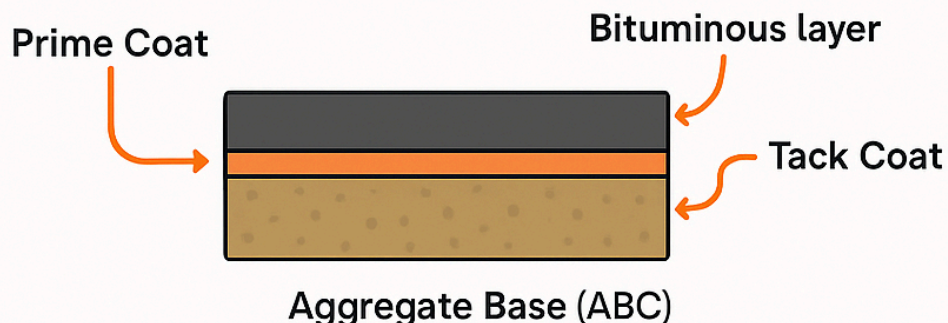
- ✓ Applied to unbound granular surfaces (e.g. ABC)
- ✓ Used as a sealant and binder for the base layer

Bitumen

- ✓ A viscous hydrocarbon binder derived from crude oil
- ✓ Used as a binder in asphalt mixes, prime coats, and tack coats

Tack Coat

- ✓ Applied between bituminous layers
- ✓ Promotes bonding between asphalt layers



Asphalt Mix Design, Batching, and Testing

Week 20 – Asphalt Mix Design, Batching, and Testing

This week, I gained practical exposure to the process of **Asphalt Mix Design validation**, **batching at the asphalt plant**, and **field sampling** for laboratory testing. Ensuring that the asphalt mixture meets specified quality standards is critical for long-term pavement performance and durability.

1. Asphalt Mix Design – Preparation and Submission

- The initial **asphalt mix design** is prepared by the contractor or certified laboratory based on:
 - Aggregate grading
 - Bitumen content
 - Voids in mineral aggregate (VMA)
 - Marshall Stability and Flow values
- The proposed design is submitted to the **Resident Engineer (RE)** and **RDA** for preliminary review before plant production begins.

Typical Mix Parameters Checked:

Parameter	Acceptance Range (per ICTAD)
Bitumen Content (%)	4.5 – 5.5%
Marshall Stability (kN)	≥ 9.0
Flow Value (mm)	2 – 4
Air Voids (%)	3 – 5
VMA (%)	14 – 18

2. Asphalt Plant Batching Process

- Once the design is approved, the **asphalt plant (batch or drum type)** is set to produce the mix using controlled input from:

- Aggregates (coarse + fine)
- Filler (e.g., cement or lime dust)
- Bitumen (temperature-controlled)
- Each component is weighed and fed precisely according to the mix formula.

Observations:

- Plant operator adjusted the bitumen pump and aggregate gate settings based on lab proportions.
 - Proper heating temperature was maintained:
 - Aggregates: 160–170°C
 - Bitumen: 150–160°C
 - Final mix: ~155–165°C
-

3. Plant Trial and Sample Collection

Before allowing full-scale production for road laying, a **trial batch** is prepared and tested.

- Samples were collected from:
 - **Hot Bin** (for gradation check)
 - **Mixer Outlet** (for extraction and Marshall tests)
 - These samples were immediately transported to the lab for confirmation testing.
-

4. Asphalt Extraction Test

The **Extraction Test** checks if the **bitumen content** in the produced mix matches the design.

Test Method:

- A sample of asphalt mix is weighed and placed in the extraction device (usually a centrifuge or filter system).
- **Solvent (e.g., trichloroethylene)** is used to dissolve the bitumen.
- The extracted aggregates are washed, dried, and reweighed.
- Bitumen content is calculated by difference.



Acceptable Range (per ICTAD):

- **Bitumen content deviation:** $\pm 0.3\%$ from the design mix



If results fall outside the acceptable range, the mix is adjusted and retested before use in paving.

Sieve Analysis for Asphalt Aggregates

Week 21 – Sieve Analysis for Asphalt Aggregates

This week, I participated in conducting **Sieve Analysis** on aggregates used in asphalt production. The purpose of this test is to ensure that aggregate gradation meets the specified limits in the approved **Asphalt Mix Design** and complies with **ICTAD Section 1602** standards.

Purpose of Sieve Analysis

Sieve Analysis determines the **particle size distribution** of aggregates used in asphalt mixtures. Proper gradation is crucial for:

- Achieving **target density and compaction**
 - Maintaining **mix stability and durability**
 - Ensuring **smooth surface finish** and **void control** in the final asphalt layer
-

Equipment Used

- Standard sieve set (coarse and fine sieves: 37.5 mm – 0.075 mm)
 - Mechanical sieve shaker or hand shaking method
 - Oven (105–110°C) for drying
 - Balance (0.1 g accuracy)
 - Brush and pan
-

Test Procedure (as per ASTM C136 / BS 812-103.1)

1. Sample Preparation

- Aggregates were taken from hot bin samples at the asphalt plant.

- Oven-dried to remove moisture and ensure accuracy.

2. Weighing and Sieving

- A specific quantity of sample (~500–1000 g) was placed in the top sieve.
- Sieves were stacked from coarse (top) to fine (bottom), including a pan.
- The sample was sieved for 10–15 minutes.

3. Weight Calculation

- Material retained on each sieve was weighed.
- Percent passing and percent retained were calculated.

Typical Sieve Sizes for Asphalt Mix Gradation

Sieve Size (mm)	Example Spec Limits (Dense-Graded Asphalt)
19.0	100%
13.2	85–100%
9.5	70–90%
4.75	50–70%
2.36	35–55%
0.600	20–35%
0.075	4–10%

Note: Exact limits vary depending on mix type (e.g., wearing course, binder course, etc.)

Observations

- The tested sample conformed well to the **approved gradation envelope** from the mix design.

- Slight deviation in **finer particles** was observed in one trial, leading to an adjustment in **filler feed rate** at the plant.
- The gradation results were plotted on semi-log graph paper to visually confirm conformity with target curves.

Conclusion

Sieve Analysis is a key test in asphalt quality control. It ensures that aggregate grading supports **good compaction, stability, and long-term durability** of the pavement. This week improved my practical understanding of how **aggregate gradation directly influences asphalt performance**, and how it is continuously monitored and corrected during production.

Marshall Stability and Flow Test

Week 22 – Marshall Stability and Flow Test

This week, I was involved in performing the **Marshall Stability and Flow Test**, which is a standard laboratory procedure used to determine the **load-bearing capacity and plastic deformation characteristics** of asphalt mixtures. The test is essential for verifying whether the asphalt produced at the plant conforms to the approved **mix design** and meets **ICTAD Section 1602** requirements.

Purpose of the Marshall Test

- Evaluate the **strength (stability)** of the asphalt mix
 - Measure **deformation (flow value)** under loading
 - Ensure the mix has adequate **stiffness and flexibility**
 - Validate production against the **design mix performance criteria**
-

Equipment Used

- Marshall compaction machine (50 or 75 blows per face)
 - Mould assembly (diameter 101.6 mm, height 63.5 mm)
 - Water bath (maintained at 60°C)
 - Marshall testing frame with load dial and flow gauge
-

Test Procedure (As per ASTM D6927 / ICTAD Spec)

1. Sample Preparation

- Asphalt mix was collected from the plant or site.
- Cylindrical specimens were compacted with **75 blows per face** using the Marshall hammer.

2. Curing

- Specimens were kept for 24 hours at room temperature.
- Then immersed in a water bath at **60°C for 30–40 minutes** before testing.

3. Testing

- The specimen was placed in the Marshall testing machine.
- Load was applied at a rate of **50 mm/min** until the specimen failed.
- Maximum load = **Marshall Stability**
- Deformation (in mm) at peak load = **Flow Value**

Standard Acceptance Criteria (ICTAD / RDA Typical Specs)

Parameter	Target Range
Marshall Stability	≥ 9.0 kN (minimum)
Flow Value	2.0 – 4.0 mm
Air Voids	3.0 – 5.0%
VMA (Voids in Mineral Aggregate)	14 – 18%

Site Observations and Results

- The tested asphalt mix (Dense Graded Wearing Course) achieved:
 - **Stability:** 11.2 kN
 - **Flow:** 3.1 mm
→  Within acceptable range
- Each batch was tested during plant trials and pre-laying validations.
- Specimens that failed to meet minimum stability were traced back to **bitumen content or gradation issues**, prompting minor plant adjustments.

Asphalt Laying, Trial, and Core Cutting

Week 23 – Asphalt Laying, Trial, and Core Cutting

This week, I observed and participated in key on-site operations related to **asphalt laying**, including the **trial laying process**, **pre-laying sample checks**, and **core cutting** for quality verification after compaction. These steps ensure that the final pavement meets the structural and functional standards set by the **Road Development Authority (RDA)** and **ICTAD Specifications**.

1. Asphalt Laying Process

The asphalt laying began at selected road sections after proper base layer preparation (ABC layer), surface cleaning, and bitumen tack coat application.

Key Site Activities:

- Asphalt was transported from the approved plant in **insulated trucks** to maintain temperature.
 - **Paver machine** laid the hot mix to the specified thickness and camber.
 - **Steel wheel and pneumatic rollers** compacted the mix while temperature was within the specified rolling range (135–160°C).
 - The surface finish, joint alignment, and compaction were continuously monitored by the site engineer and consultant.
-

2. Laying Trial and Pre-Laying Checks

Before full-scale production laying, an **asphalt laying trial** was conducted to:

- Verify the workability of the mix under site conditions.
- Confirm the effectiveness of the plant batch settings.
- Conduct on-site **sample collection** for:
 - **Extraction Test** (bitumen content)
 - **Sieve Analysis** (gradation)

- **Marshall Test** (stability and flow)

✅ The results were compared to the approved asphalt mix design. Once the trial batch passed all parameters, the site was cleared for production-scale laying.

3. Core Cutting for Asphalt Quality Verification

After laying and final compaction, **core samples** were extracted to assess in-situ quality.

Procedure:

- Cores were taken from random chainages using a **core drilling machine**.
- The extracted cylinders were measured and tested for:
 - **Actual thickness**
 - **Density (Bulk Specific Gravity)**
 - **Air voids and compaction level**

Acceptance Criteria (Typical):

Parameter	Requirement (RDA/ICTAD)
Thickness Tolerance	±10 mm from design
Density	≥ 95% of Marshall Density
Voids	Within mix design limits

📌 If values fall below acceptable ranges, re-compaction or overlay may be required (none observed this week).

✅ Conclusion

This week's activities strengthened my understanding of **on-site asphalt paving operations**, including laying trials, real-time quality monitoring, and **post-laying core testing**. These checks are crucial for achieving a smooth, well-compacted, and durable asphalt surface in compliance with RDA project standards.

Tab 44

There are **two major types of asphalt plants**, based on the production method:

1. Batch Mix Asphalt Plant

- **Working Principle:** Produces asphalt in batches — each batch is separately weighed and mixed.
 - **Key Features:**
 - Precise control over mix design and quality.
 - Suitable for projects with varied specifications.
 - Higher flexibility, but lower output than continuous plants.
 - **Main Components:**
 - Cold aggregate feeder
 - Dryer drum
 - Screening unit
 - Hot bins
 - Weighing unit
 - Pugmill mixer
 - Bitumen and filler storage
 - **Best Used For:** Urban roads, airports, and projects requiring high precision and variety in mix designs.
-

2. Drum Mix Asphalt Plant (Continuous Type)

- **Working Principle:** Mixes the aggregate and bitumen continuously in a rotating drum.
- **Key Features:**

- High production capacity.
 - Lower cost and simpler design.
 - Less precise than batch plants, but faster.
 - **Main Components:**
 - Cold aggregate feeder
 - Conveyor belt
 - Drying and mixing drum
 - Bitumen supply system
 - Storage silo
 - **Best Used For:** High-volume rural roads and highway projects where mix uniformity is acceptable.
-

Optional Classification:

You may also see plants categorized as:

- **Mobile Asphalt Plant** – Easily moved from one site to another.
 - **Fixed (Stationary) Asphalt Plant** – Installed in one location for long-term use.
-

Asphalt plants are essential for producing the asphalt mixtures used in road construction. Here's a structured overview of the **major types of asphalt plants**, categorized by mixing method, mobility, and functionality:

By Mixing Method

Type	Description	Best For
Batch Mix Plant	Produces asphalt in discrete batches with precise control over mix composition.	Highways, airports, urban roads requiring strict quality standards.
Drum Mix Plant	Continuous production where aggregates and bitumen are mixed in a rotating drum.	Large-scale projects needing consistent output.
Parallel Flow Drum Plant	Aggregates and hot gases flow in the same direction.	Simple operation and maintenance.
Counterflow Drum Plant	Aggregates and gases flow in opposite directions for better heat transfer.	Efficient drying and reduced emissions.
Double Barrel Drum Plant	Combines parallel and counterflow features for enhanced RAP usage.	Projects with recycled asphalt requirements.

By Mobility

Type	Description	Best For
Stationary Plant	Permanently installed; high capacity and stability.	Long-term, large-scale infrastructure projects.
Mobile Plant	Trailer-mounted or modular; easy to relocate.	Temporary or remote job sites.

Semi-Stationary Plant	Installed on trailers; reassembled at new sites.	Medium-term projects with moderate relocation needs.
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 By Functionality

Type	Description	Best For
Traditional Plant	Standard design without advanced environmental controls.	Budget-conscious projects.
Eco-Friendly Plant	Includes dust filters, enclosed systems, and emission controls.	Projects with environmental regulations.
Recycling Plant (RAP)	Uses reclaimed asphalt pavement to reduce material costs and emissions.	Sustainable construction projects.
Hybrid Plant	Combines batch, drum, and continuous features.	Versatile, high-efficiency projects.

 Other Classifications

- **By Capacity:** Small (<40 t/h), Medium (40–120 t/h), Large (>120 t/h)
- **By Fuel Type:** Diesel, Gas, Coal
- **By Asphalt Type Produced:** Hot Mix, Warm Mix, Cold Mix